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(12) **United States Patent**
Ahn

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(45) **Date of Patent:** **Aug. 12, 2025**

(54) **CONTAINER CAP AND CONTAINER
COMBINED WITH SAME**

USPC 215/316, 253, 254, 256, 270–271,
215/329–331, 343; 220/86.1; 206/219,
206/432, 497; 222/147

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See application file for complete search history.

(72) Inventor: **Jin Hee Ahn**, Yongin-si (KR)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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B65D 41/04 (2006.01)

(52) **U.S. Cl.**
CPC **B65D 41/0428** (2013.01); **B65D 41/0414**
(2013.01); **B65D 2251/20** (2013.01)

(58) **Field of Classification Search**
CPC B65D 41/0428; B65D 41/0414; B65D
41/0421; B65D 41/3442; B65D 41/0407;
B65D 41/0485; B65D 43/0225; B65D
23/00; B65D 51/18; B65D 53/02; B65D
2251/20

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Primary Examiner — Orlando E Aviles

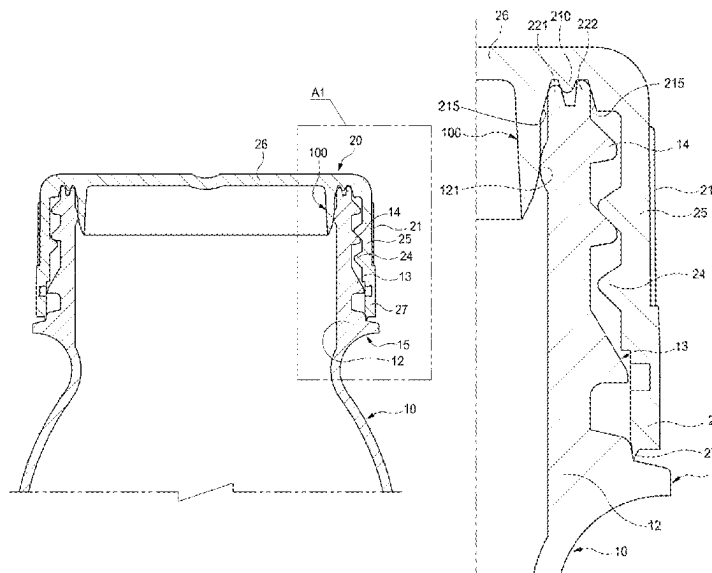
Assistant Examiner — Brijesh V. Patel

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(57) **ABSTRACT**

A container cap which is combined with a container con-
taining a beverage or the like, and a container combined with
the container cap. The container cap is combined with a
container inlet of a container body. A plurality of close
contact portions in contact with the container inlet are
formed, and the close contact portions form a closed curve
enclosing the container inlet so as to block inflow of external
air from the outside into the container body.

2 Claims, 55 Drawing Sheets



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FIG. 1

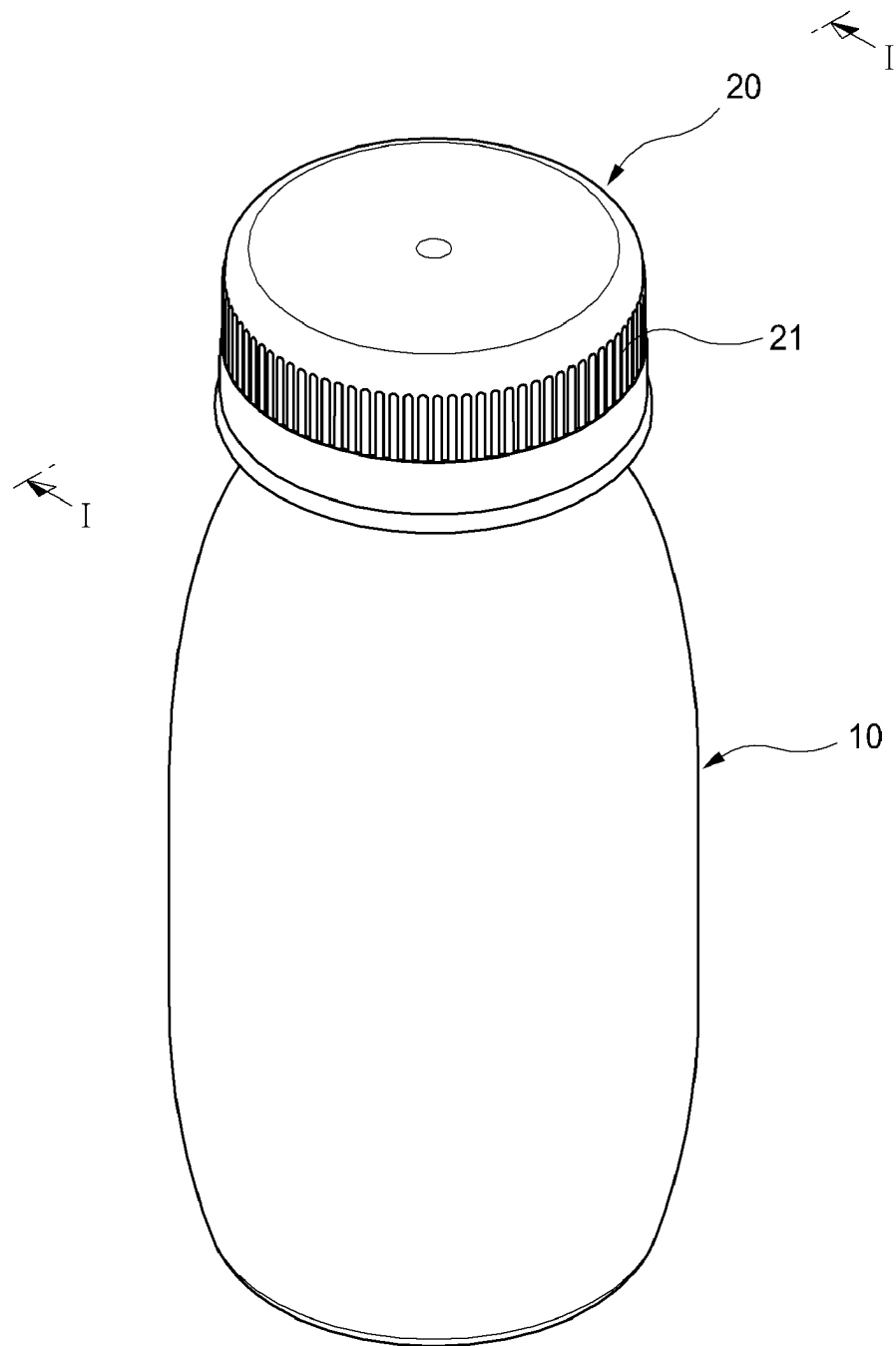


FIG. 2A

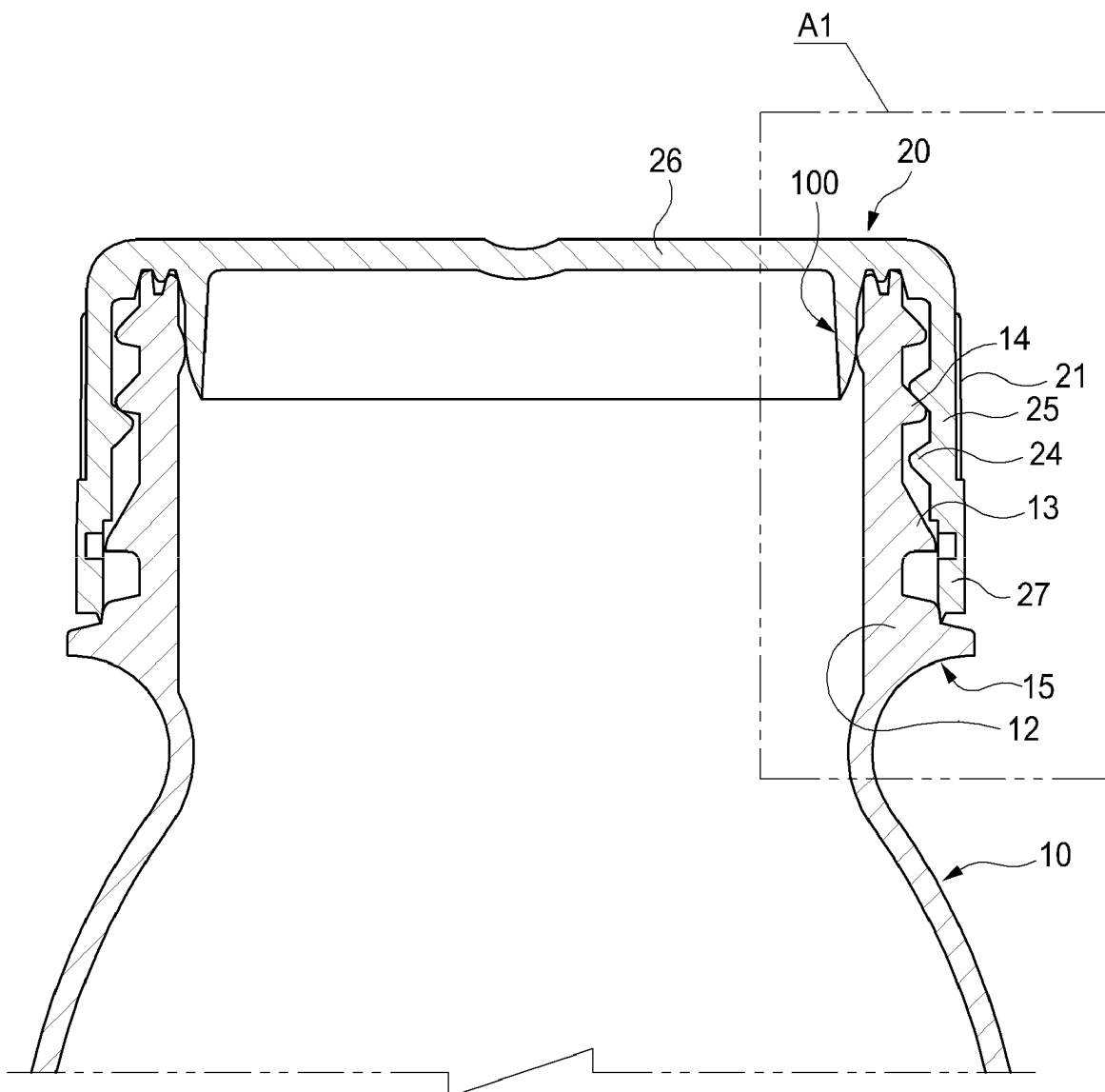


FIG. 2B

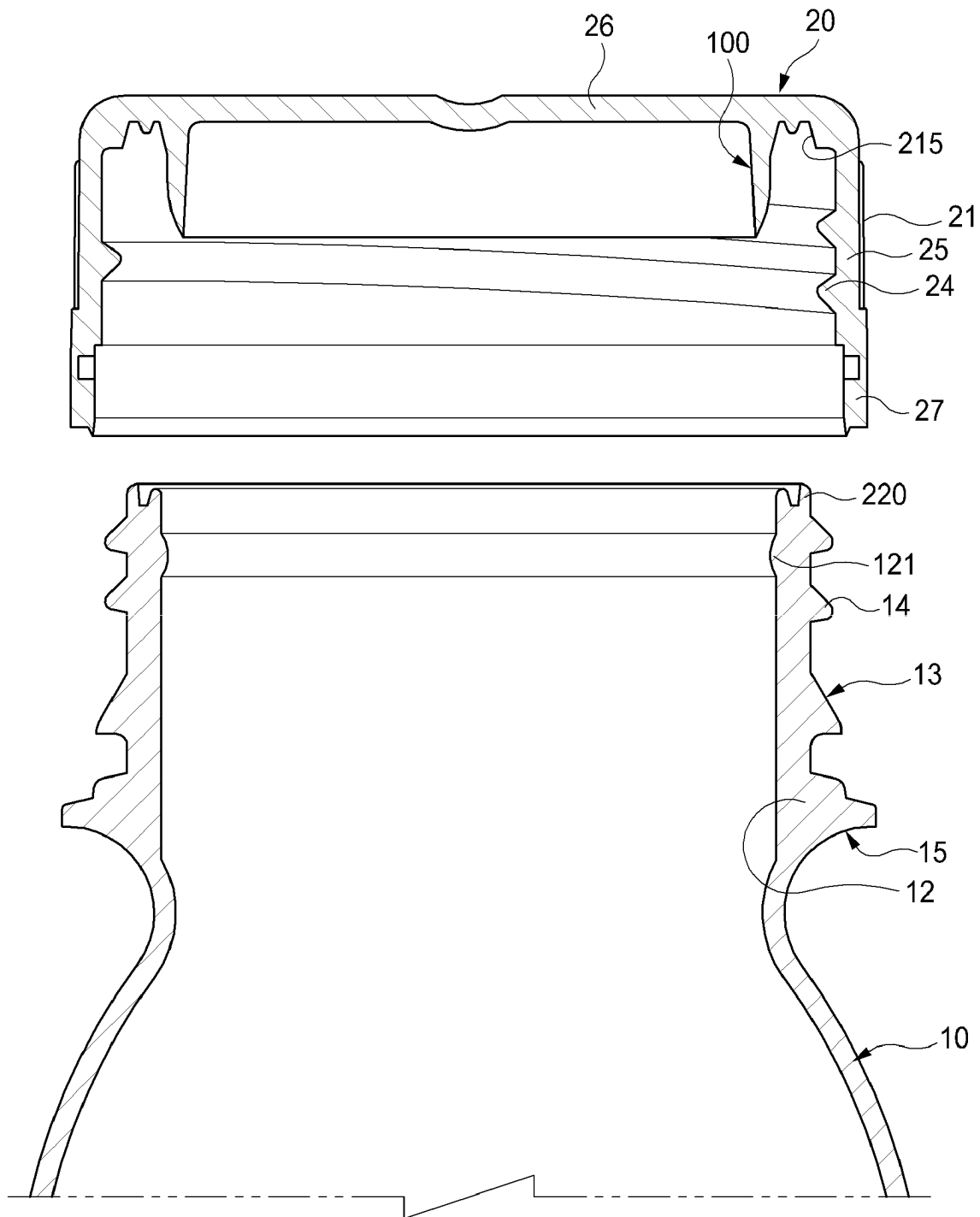


FIG. 3

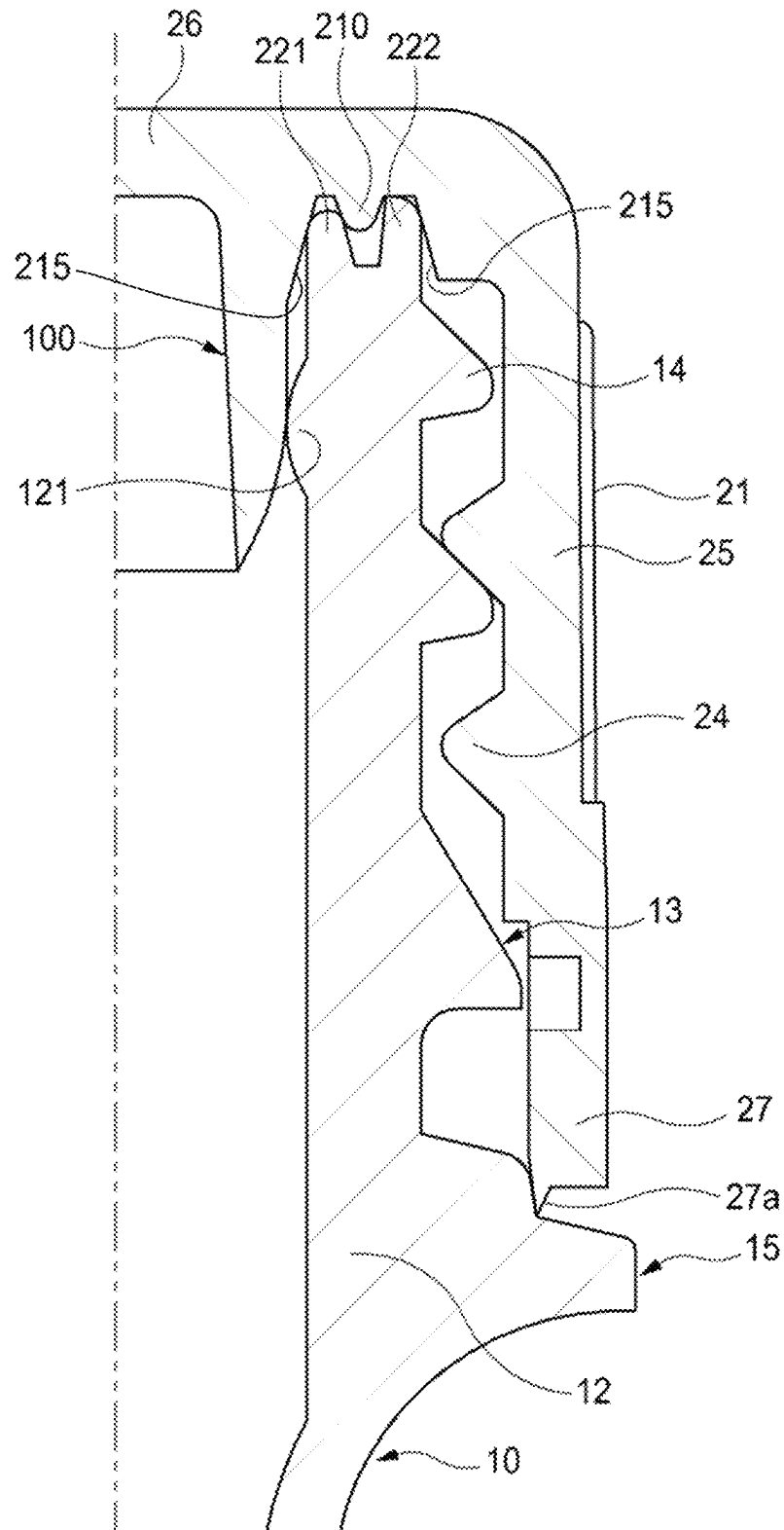


FIG. 4A

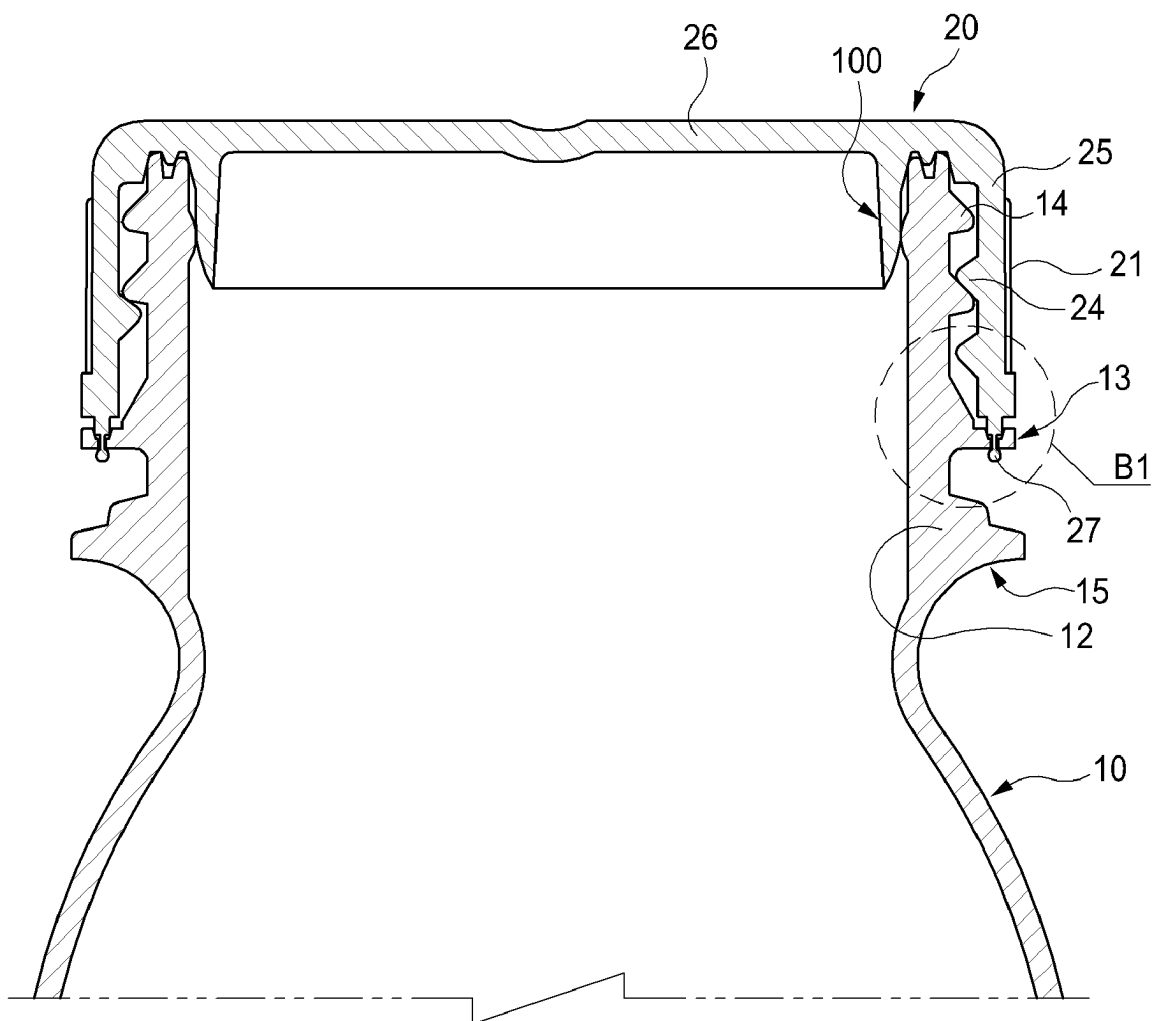


FIG. 4B

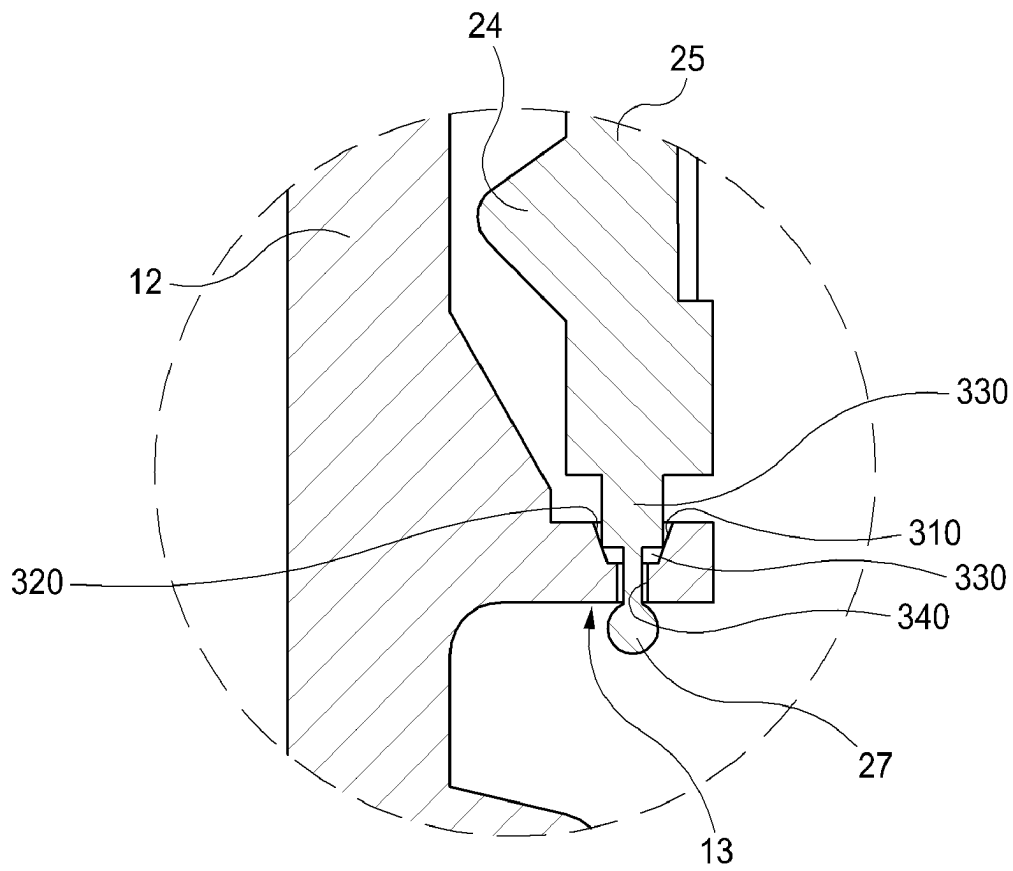


FIG. 5A

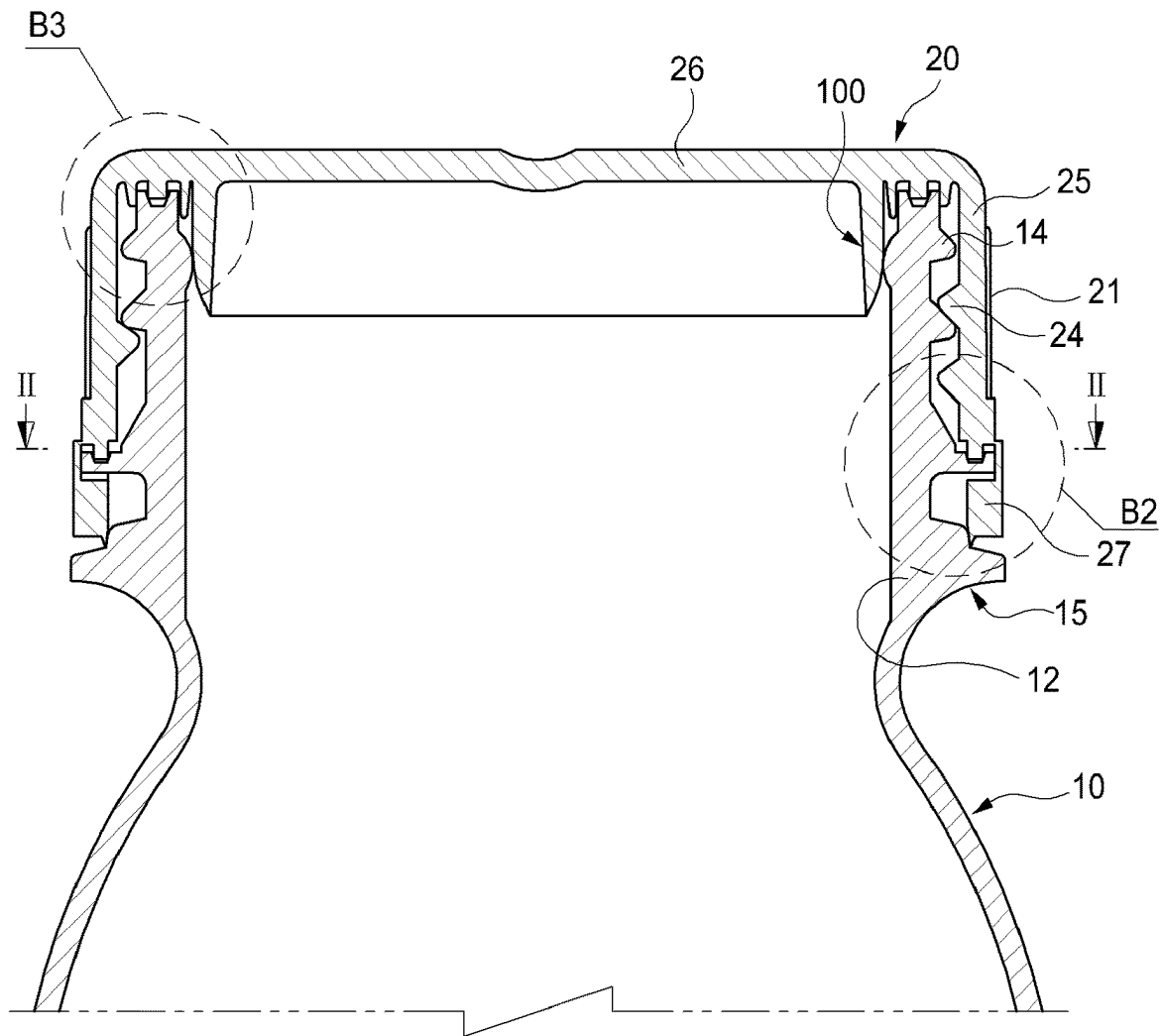


FIG. 5B

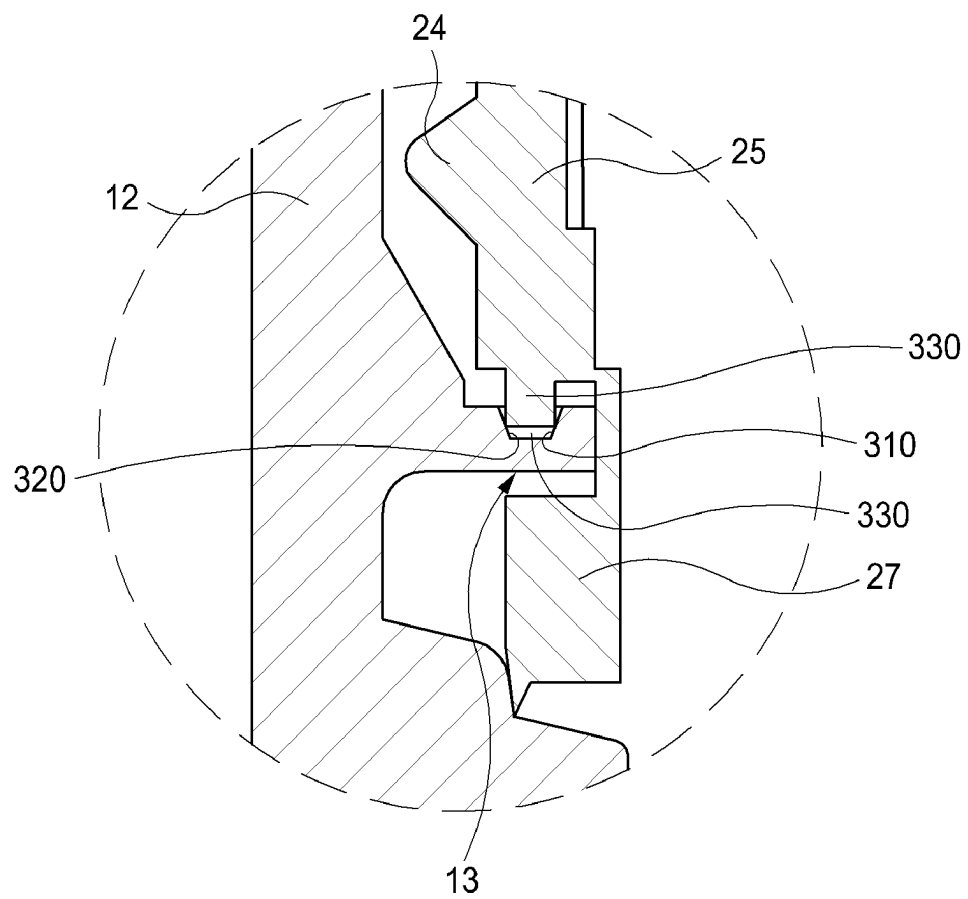


FIG. 5C

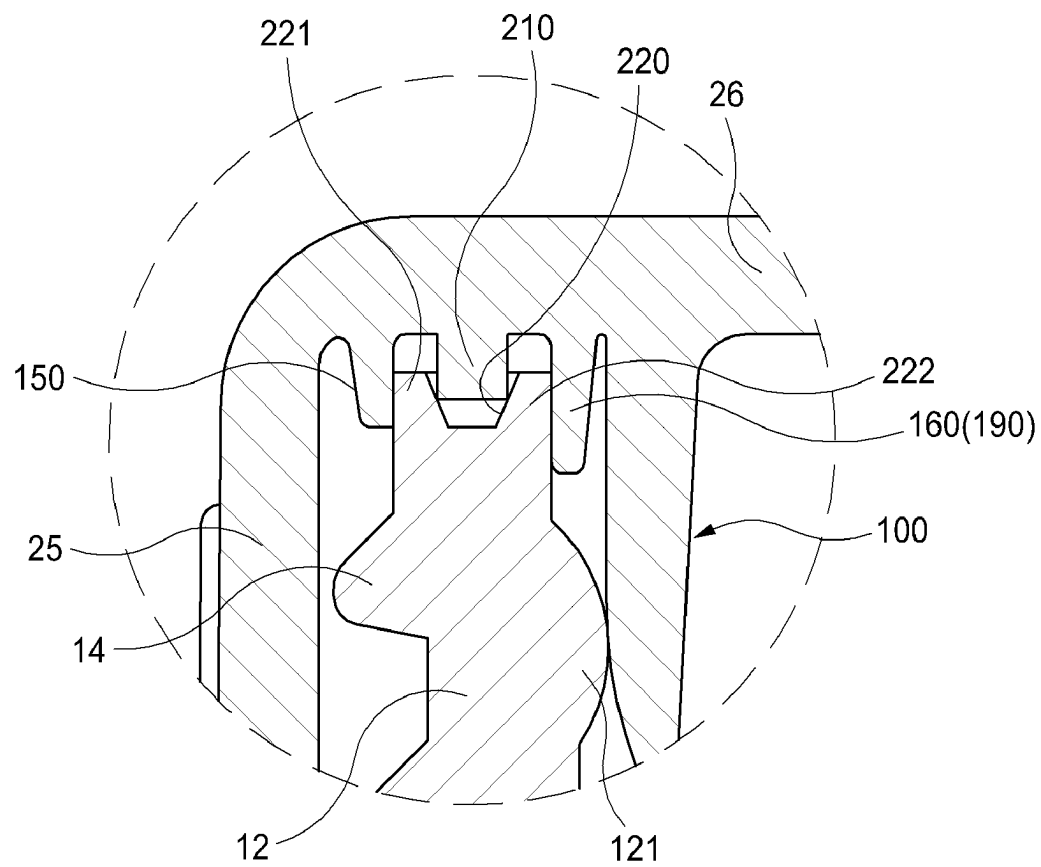


FIG. 6A

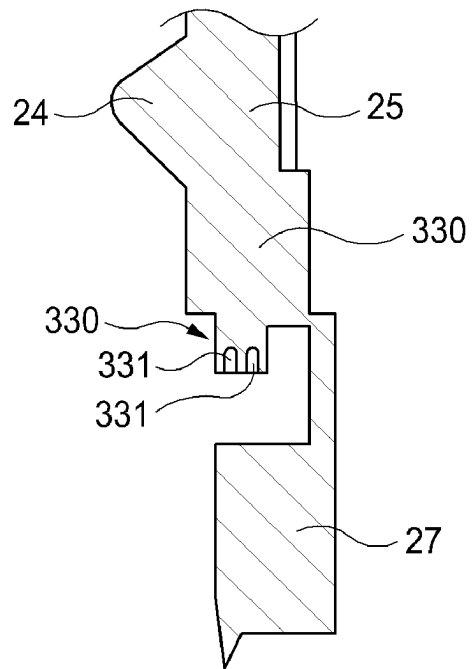


FIG. 6B

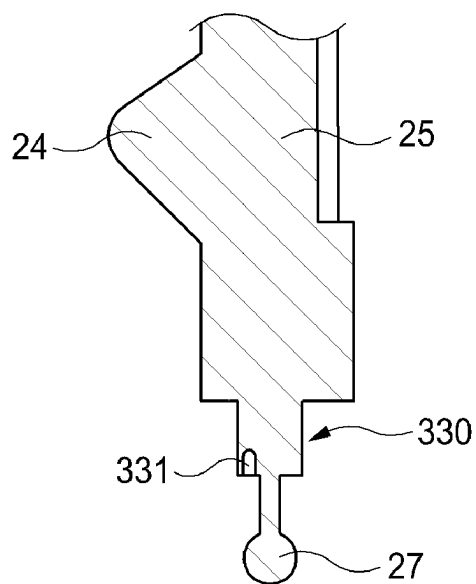


FIG. 7

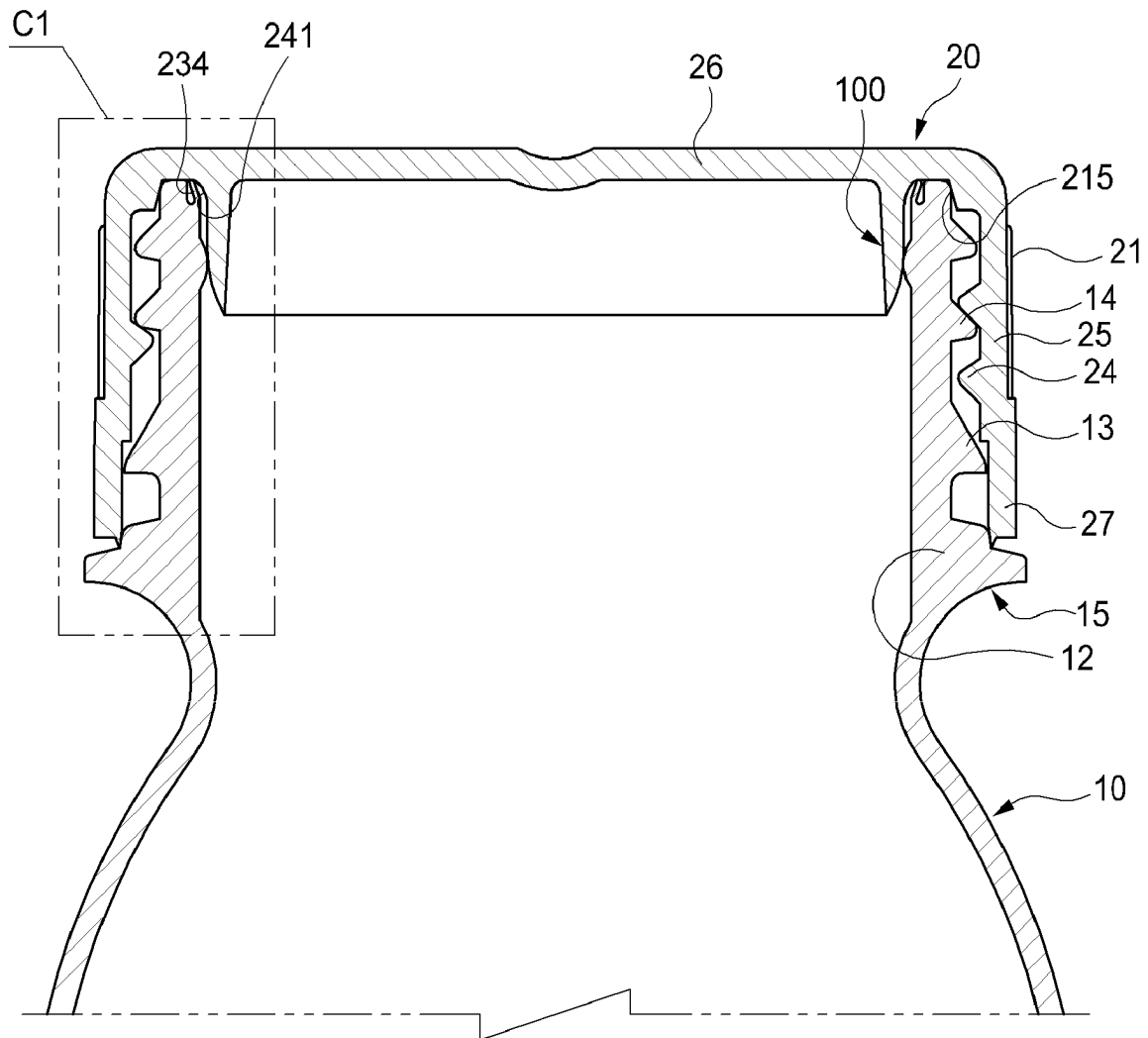


FIG. 8A

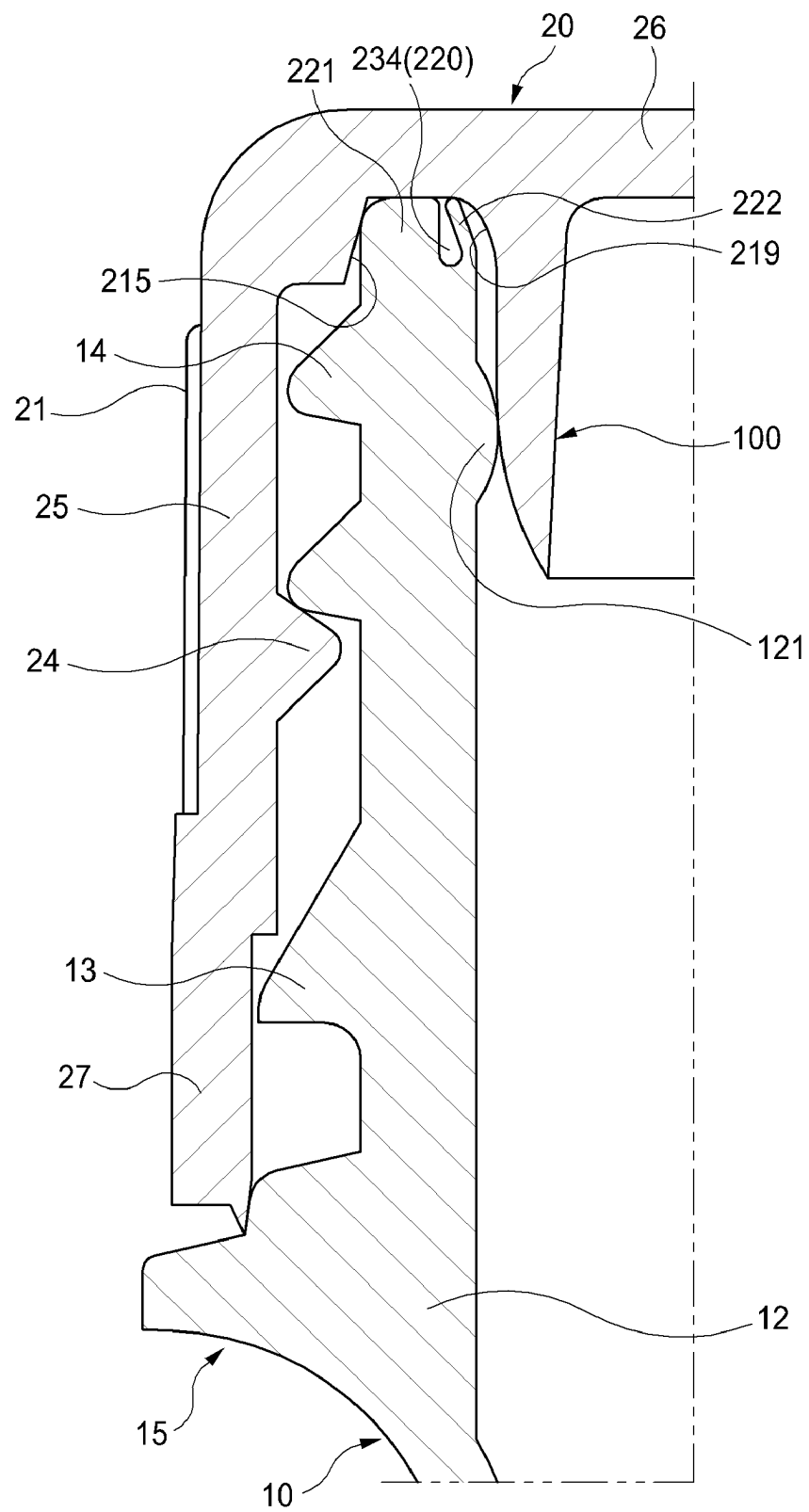


FIG. 8B

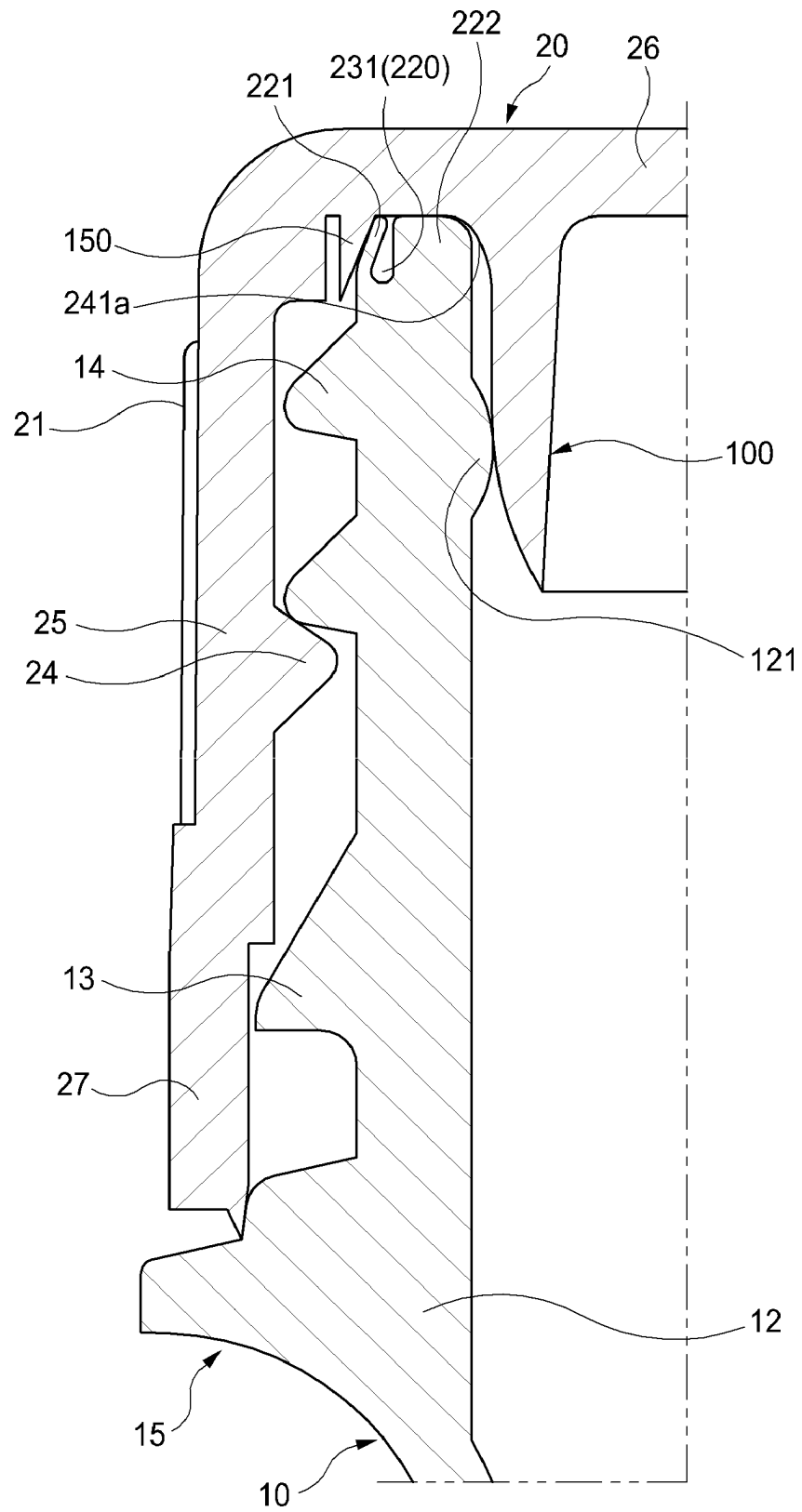


FIG. 8C

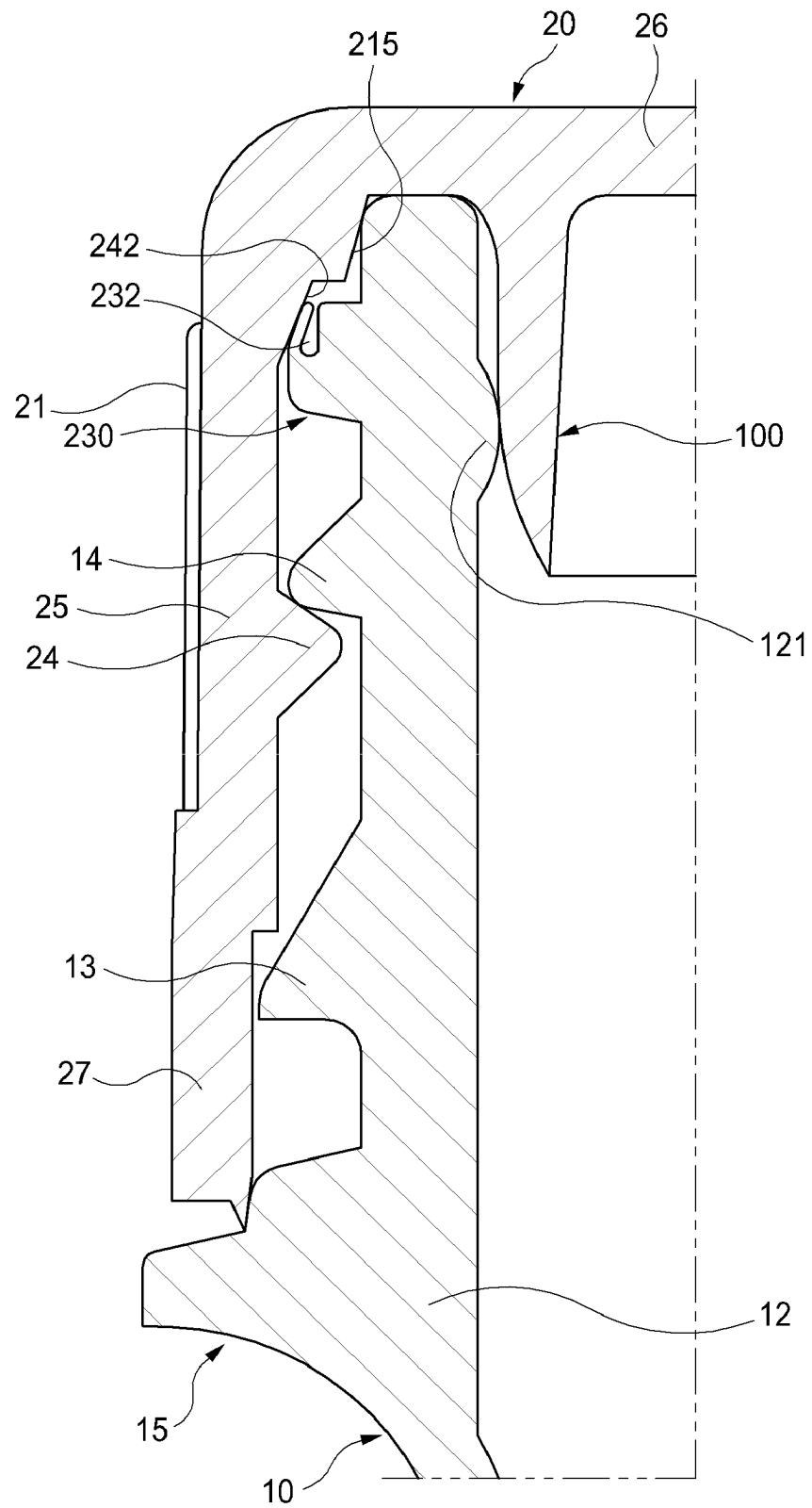


FIG. 8D

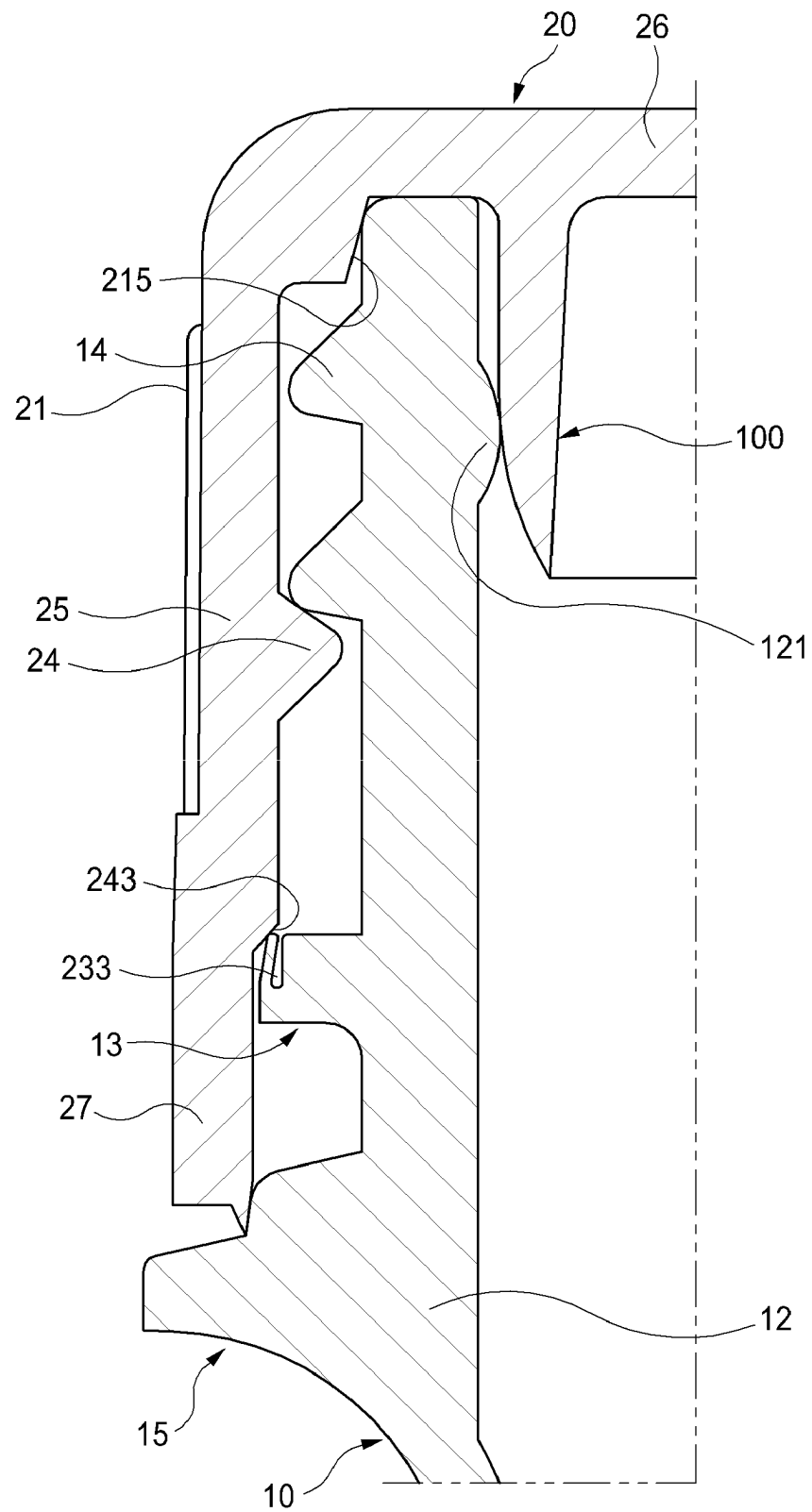


FIG. 9

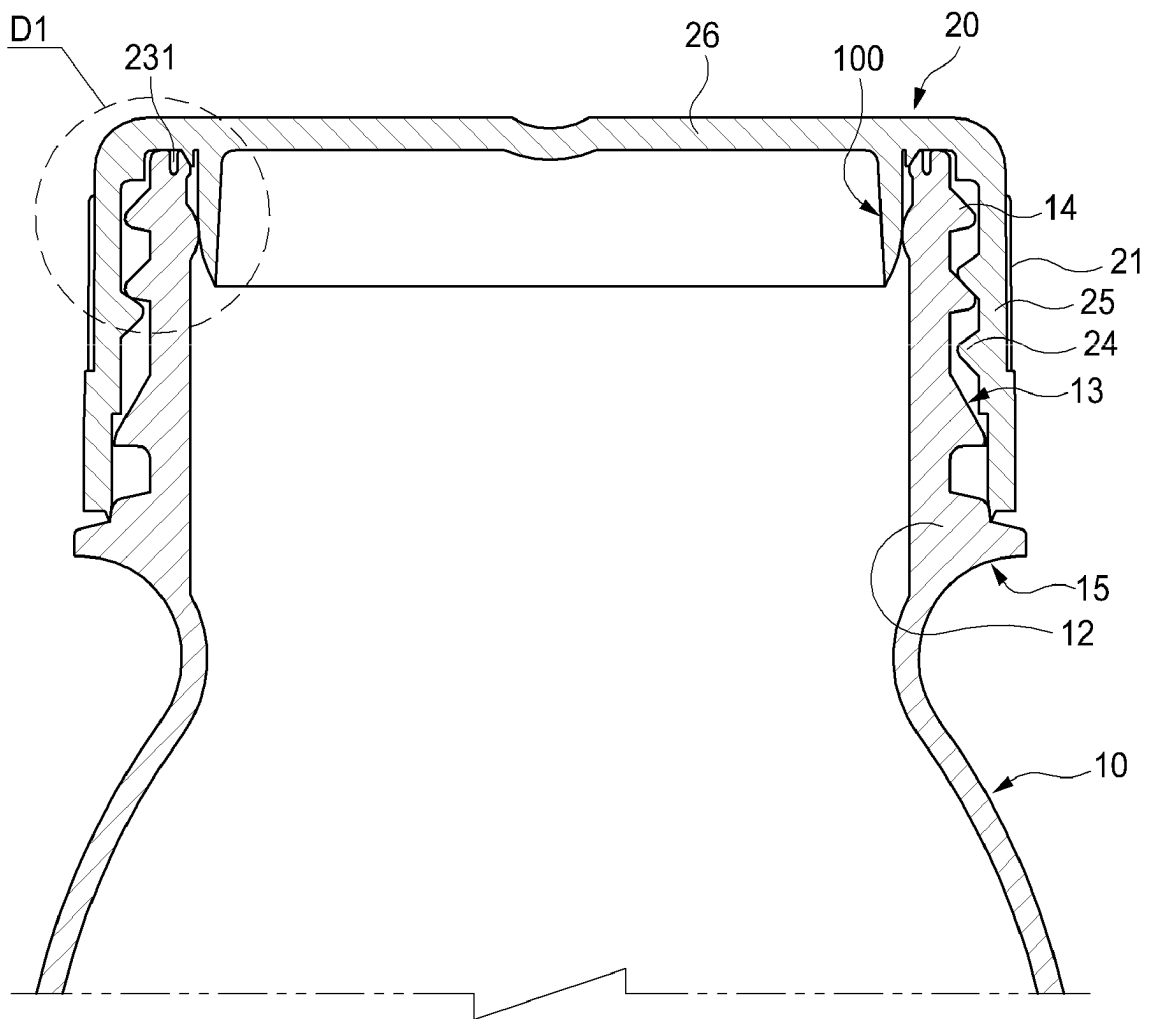


FIG. 10A

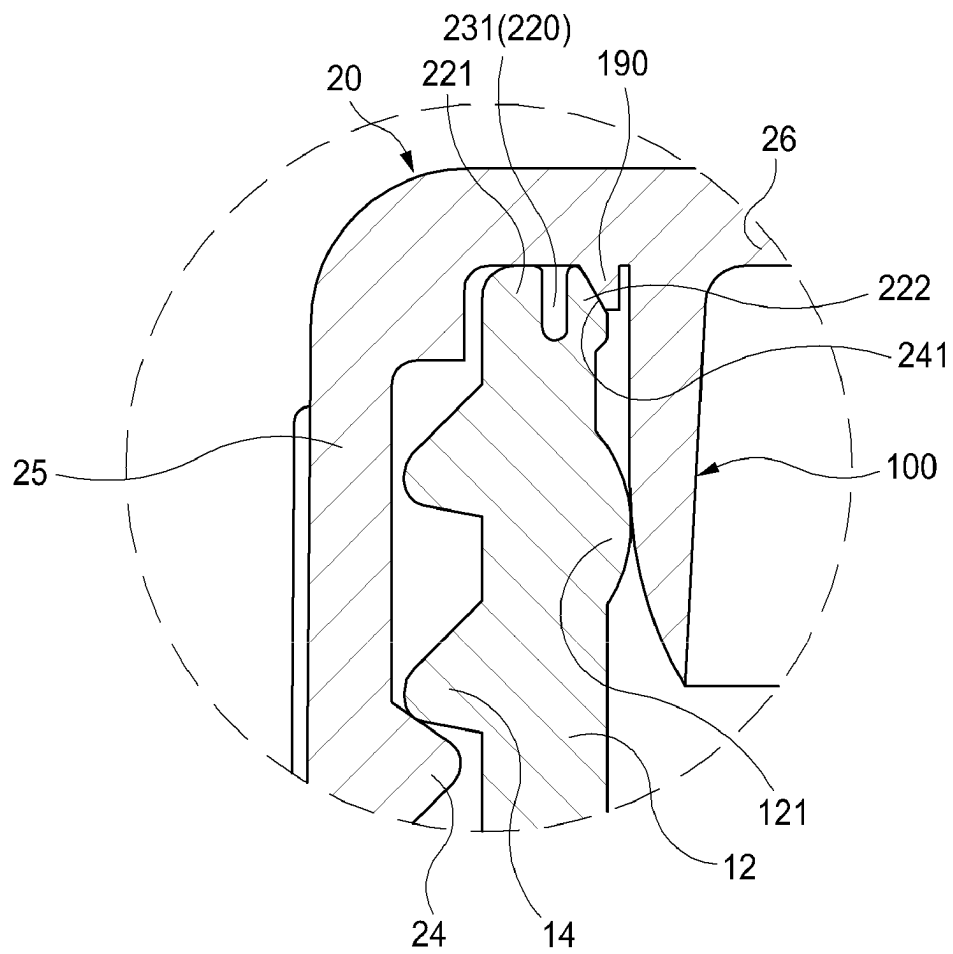


FIG. 10B

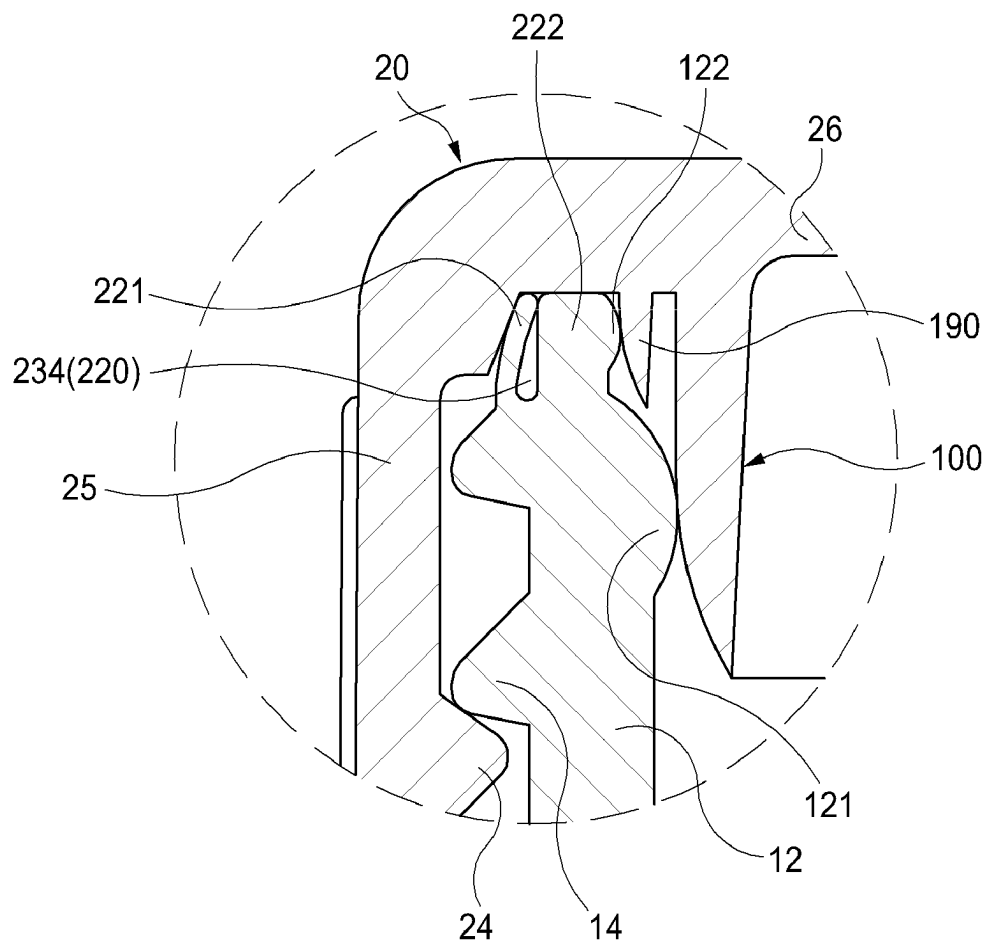


FIG. 10C

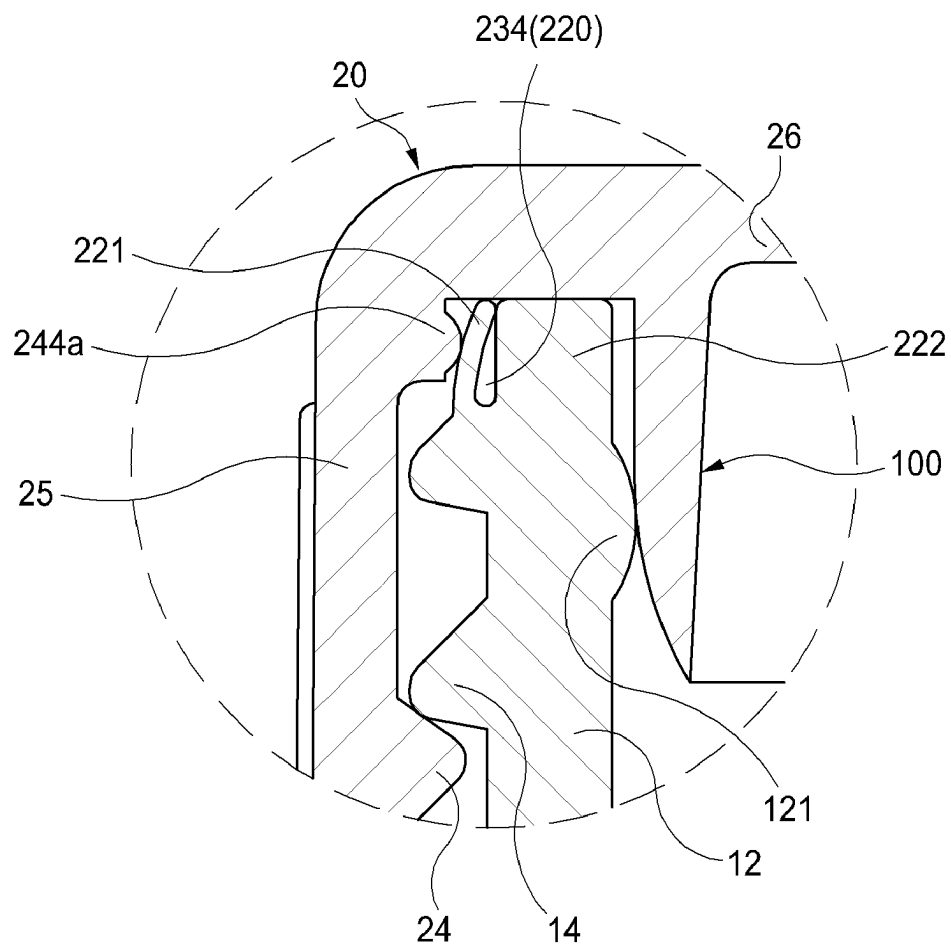


FIG. 10D

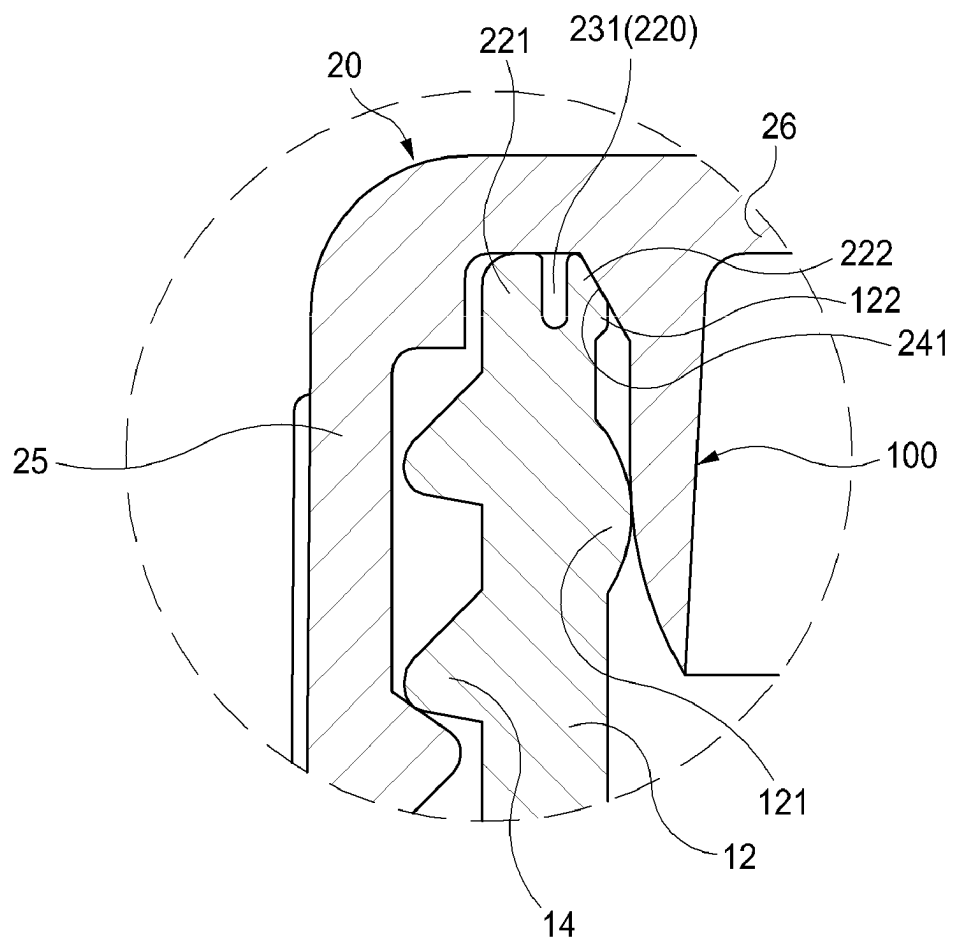


FIG. 11A

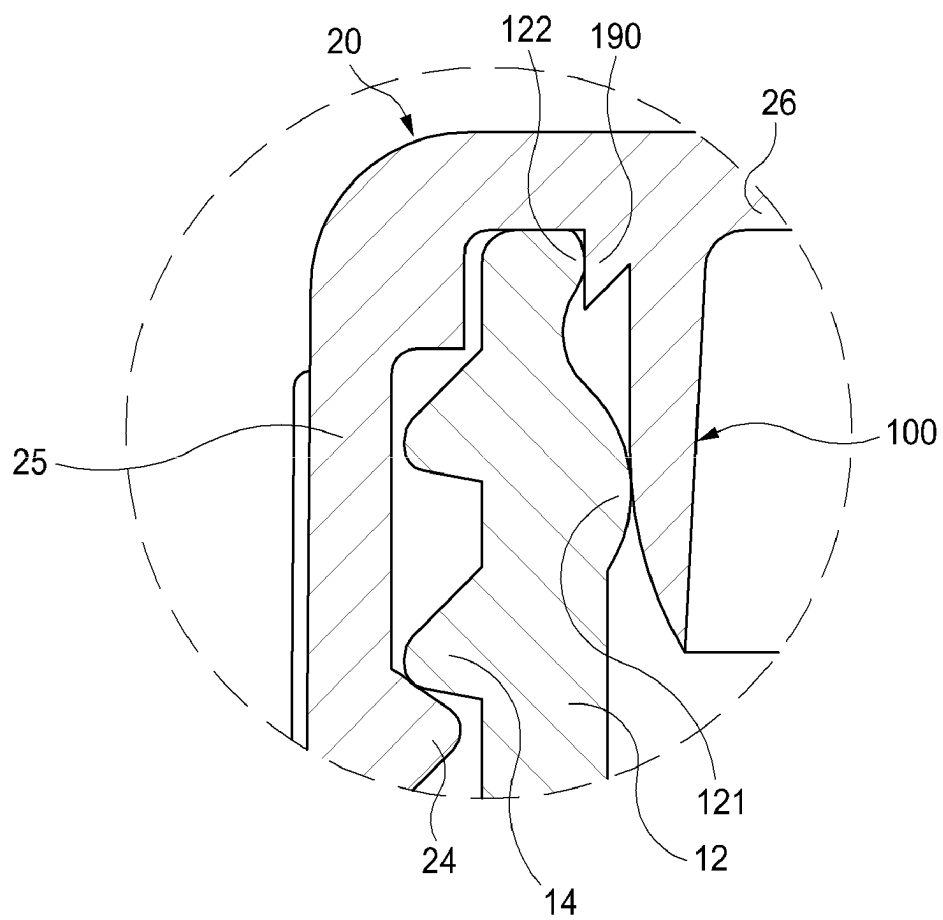


FIG. 11B

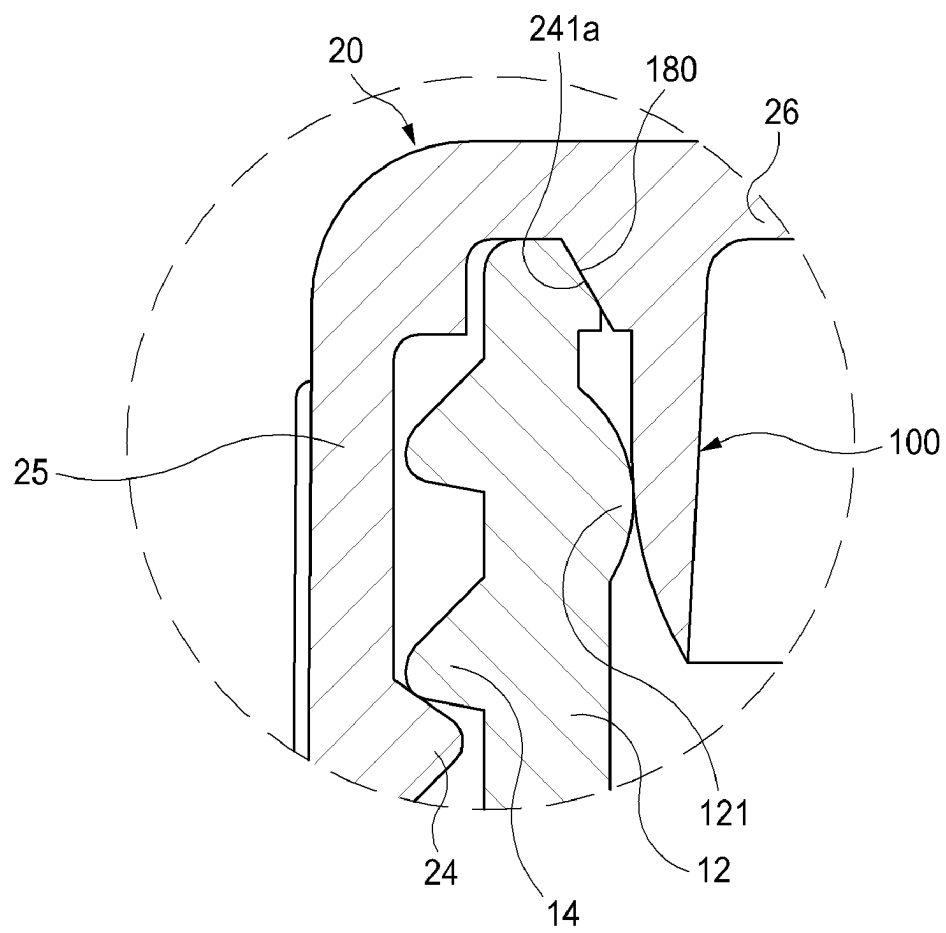


FIG. 11C

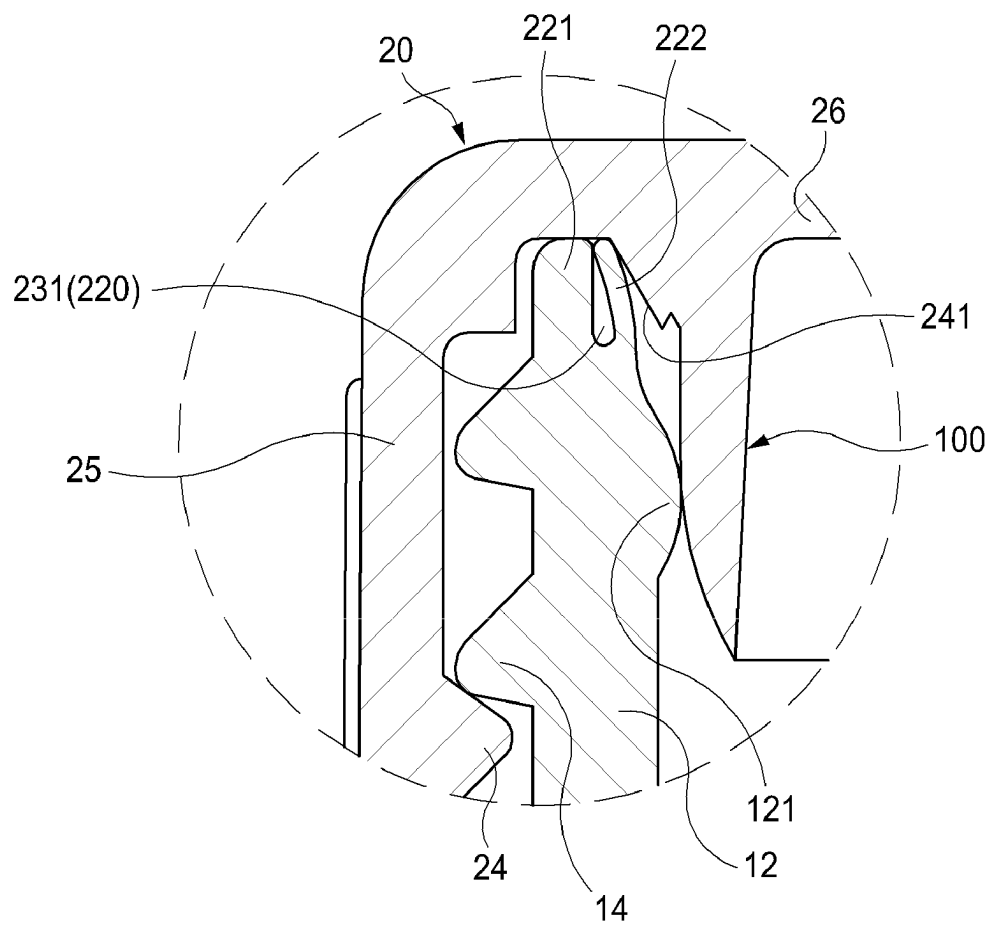


FIG. 11D

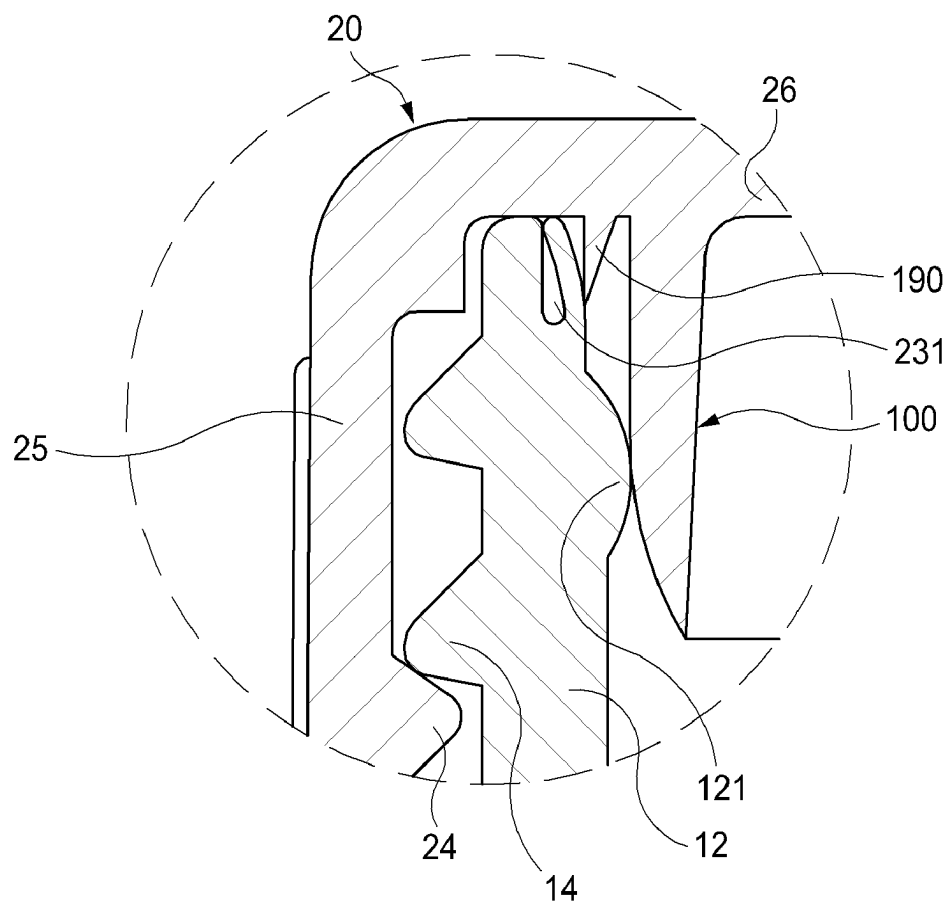


FIG. 11E

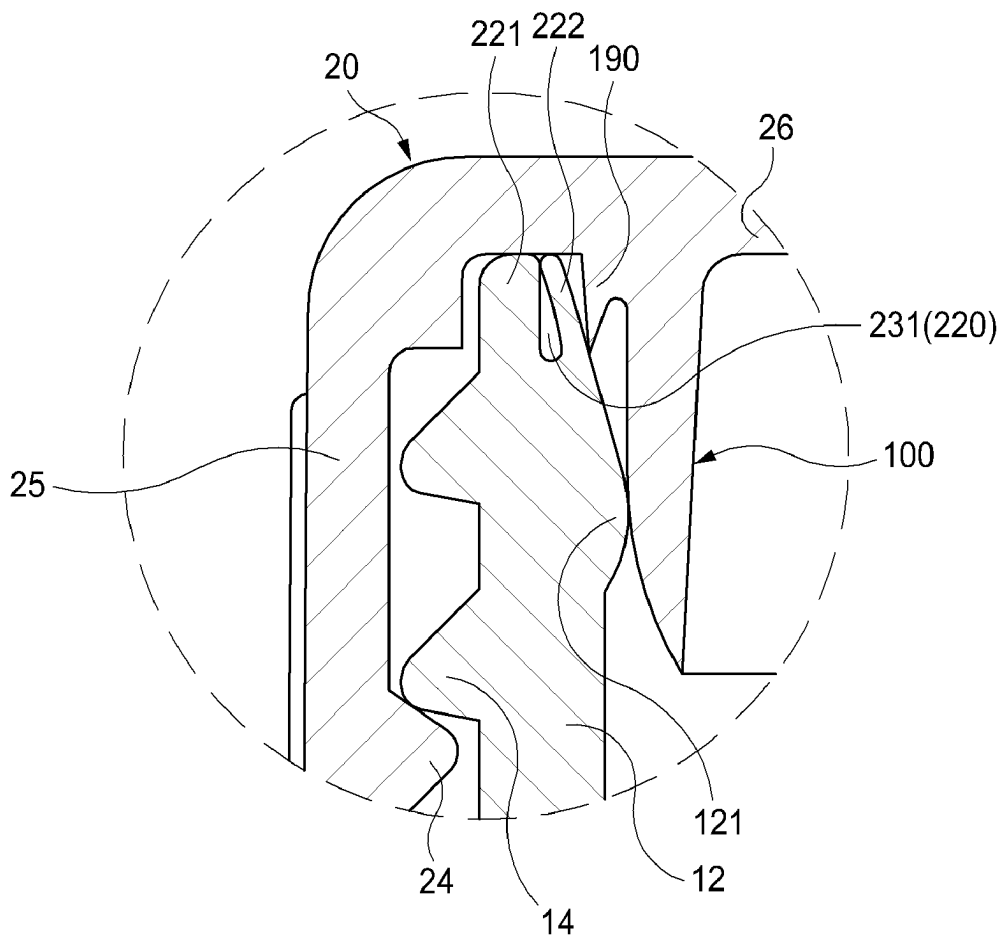


FIG. 12

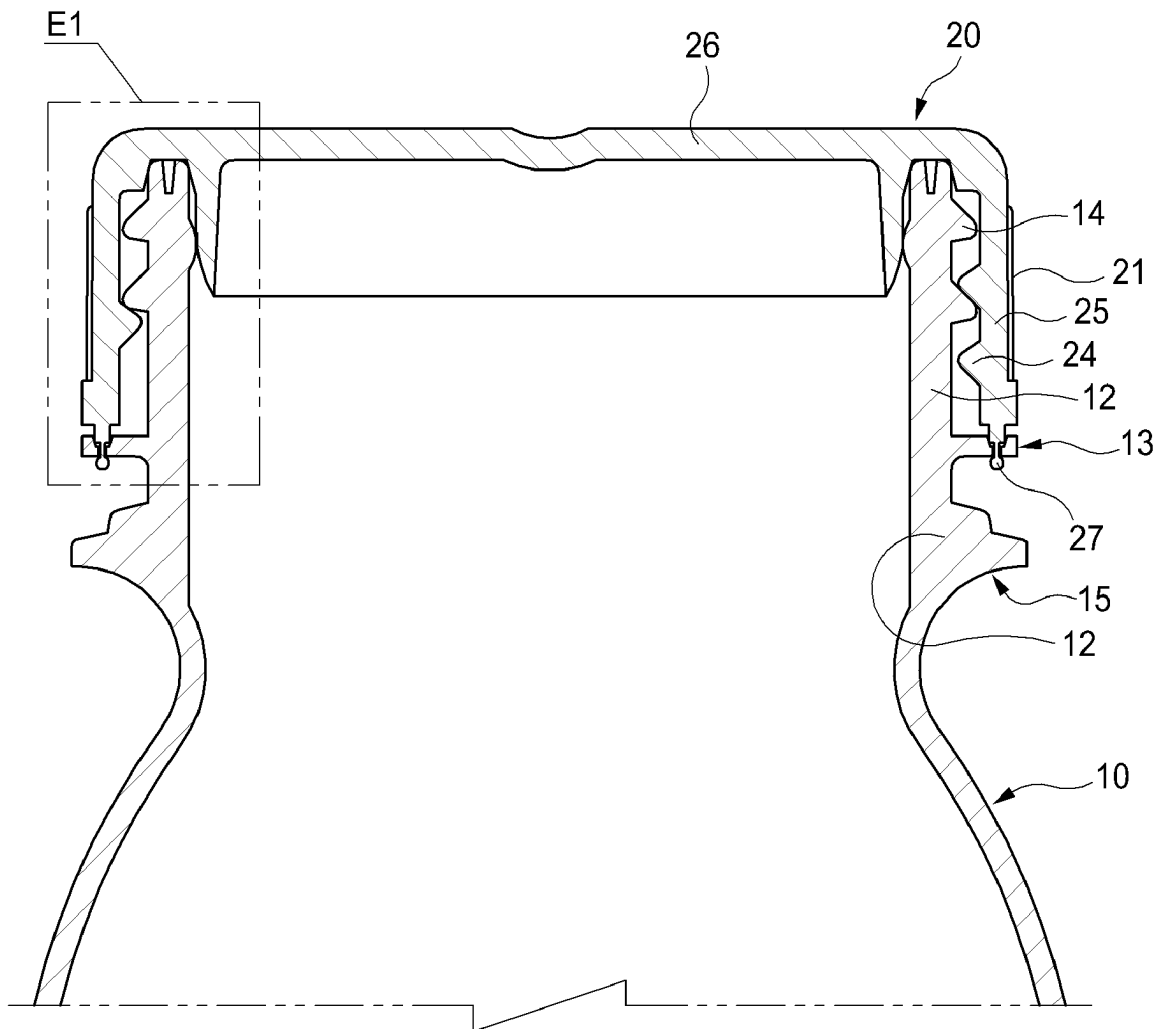


FIG. 13

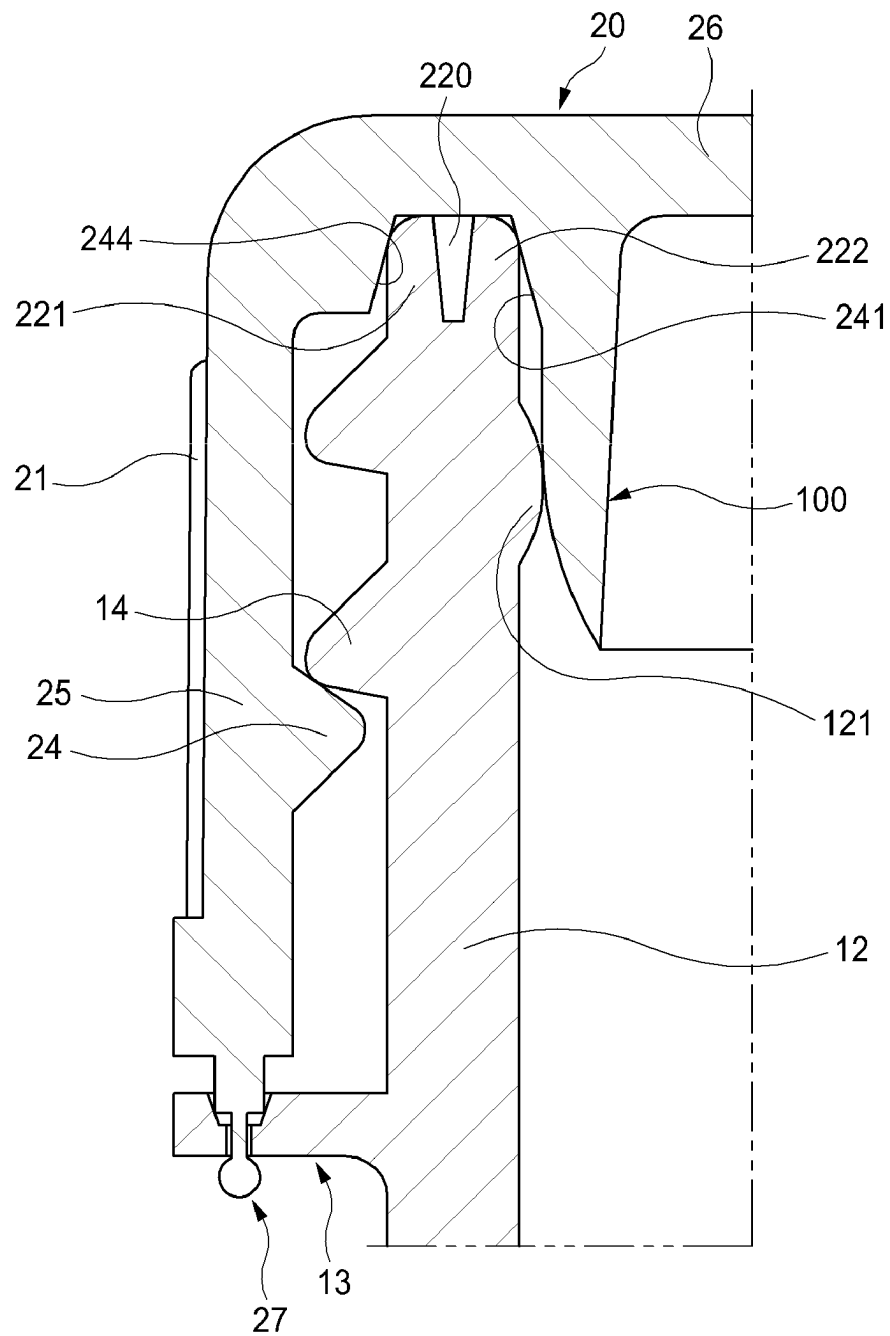


FIG. 14

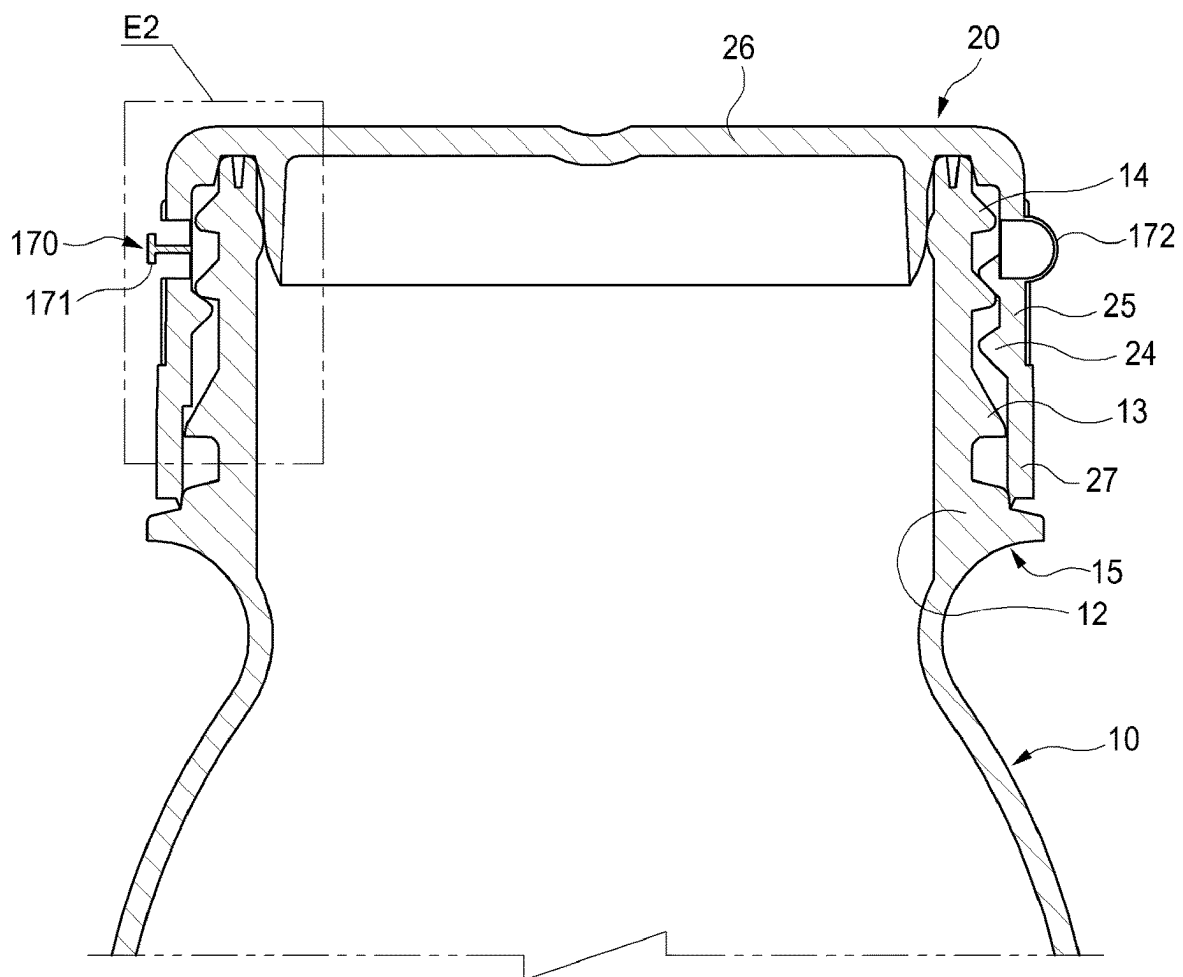


FIG. 15

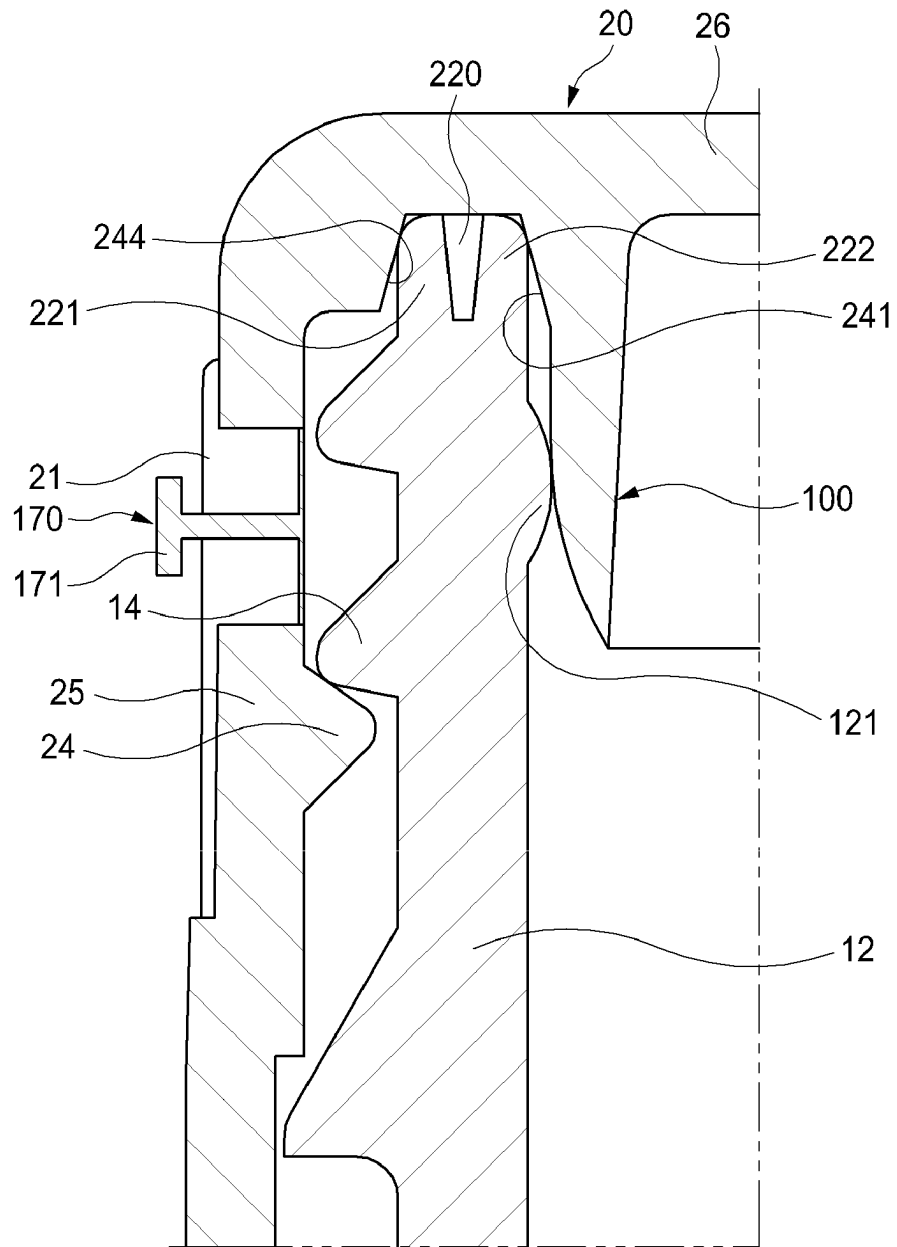


FIG. 16

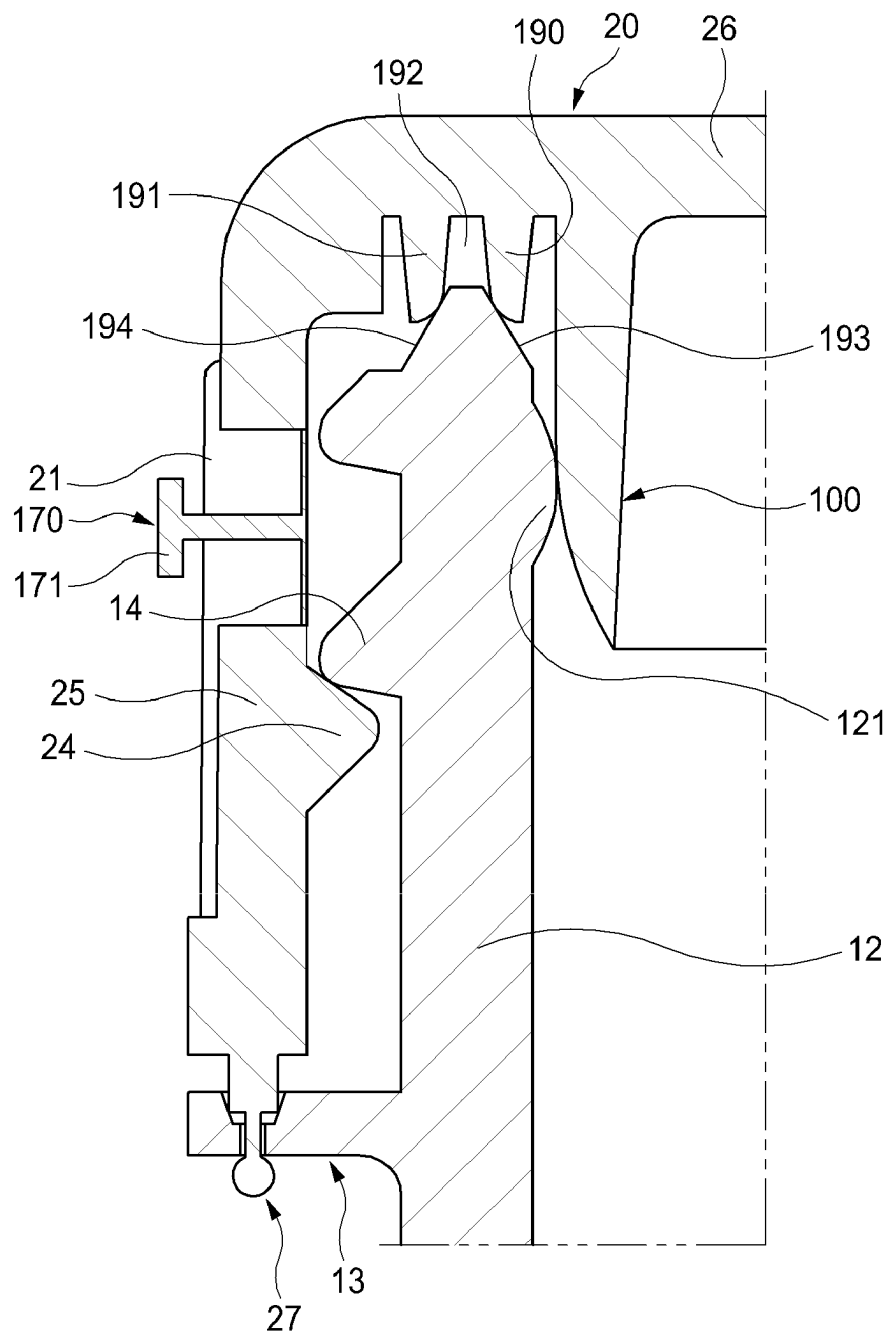


FIG. 17

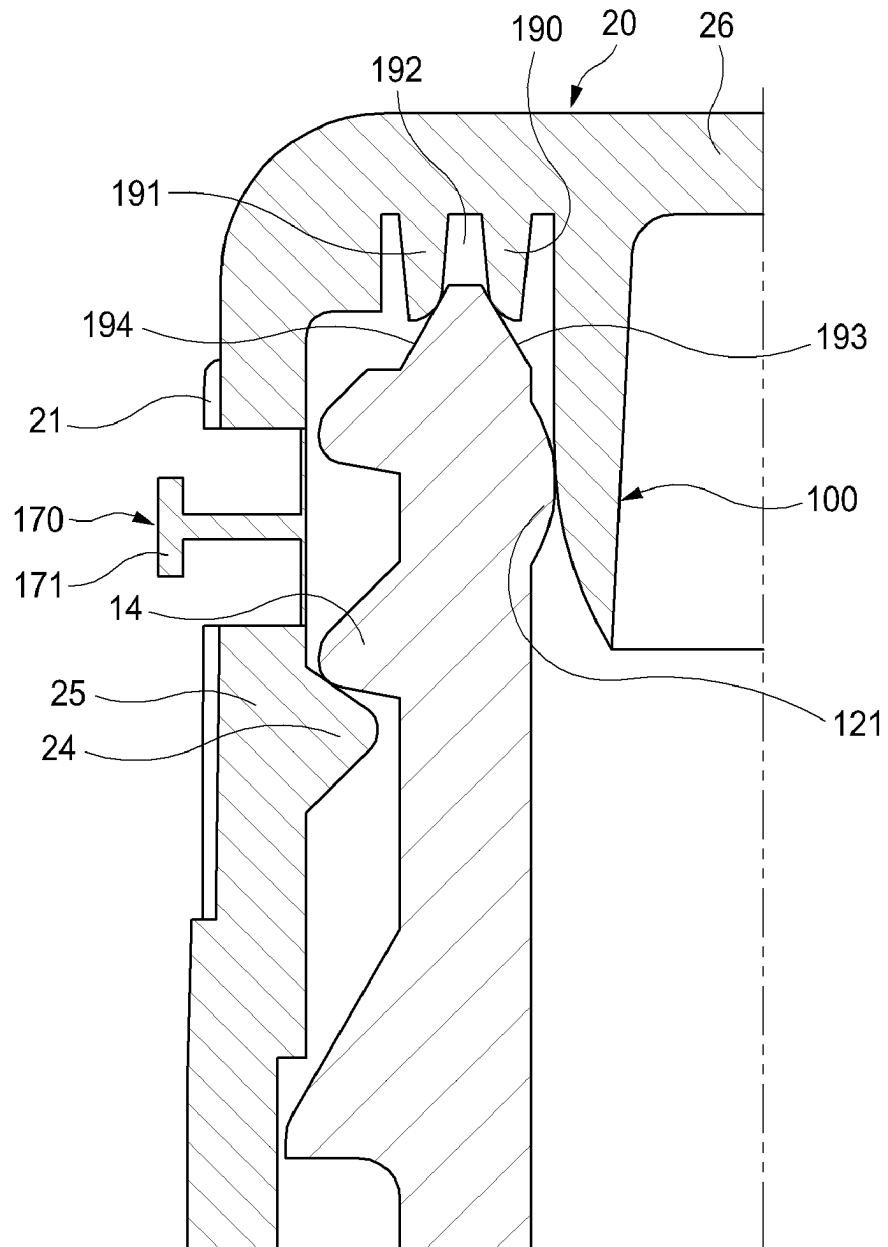


FIG. 18A

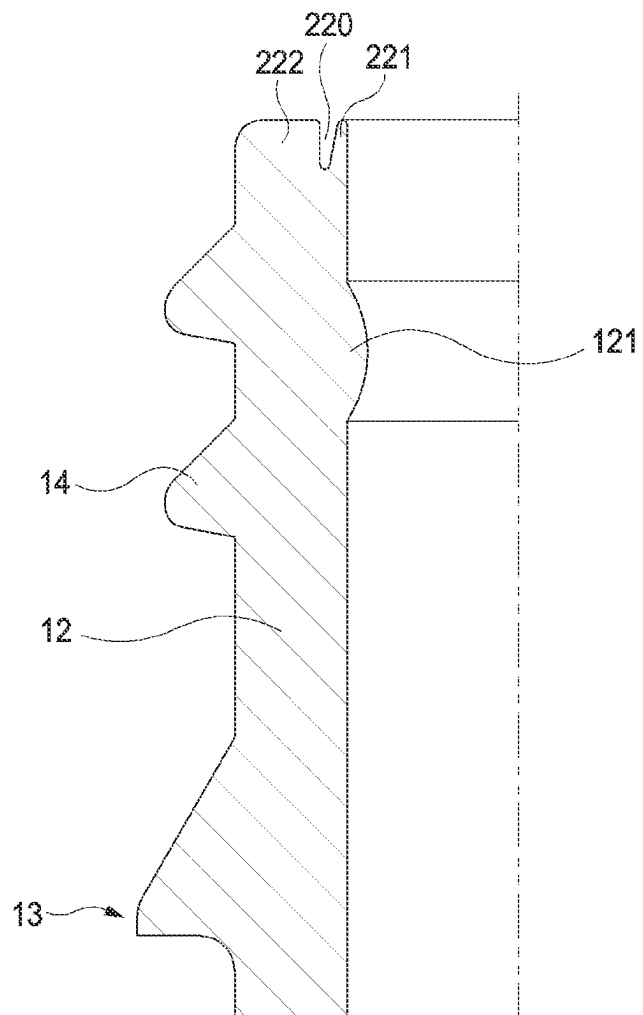


FIG. 18B

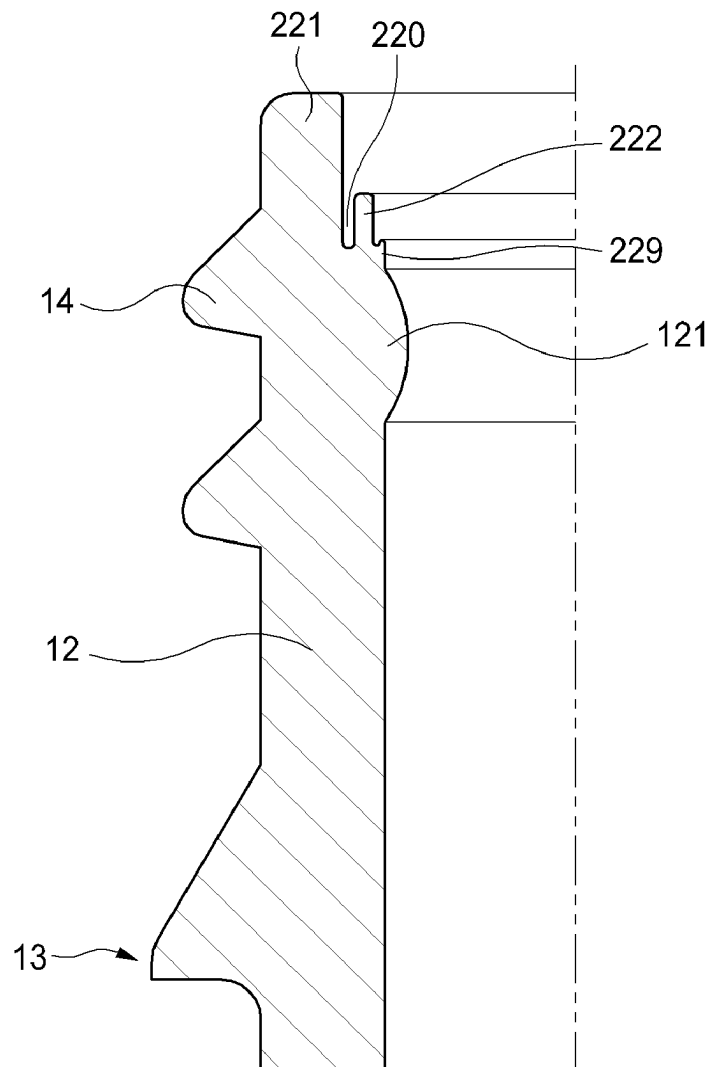


FIG. 18C

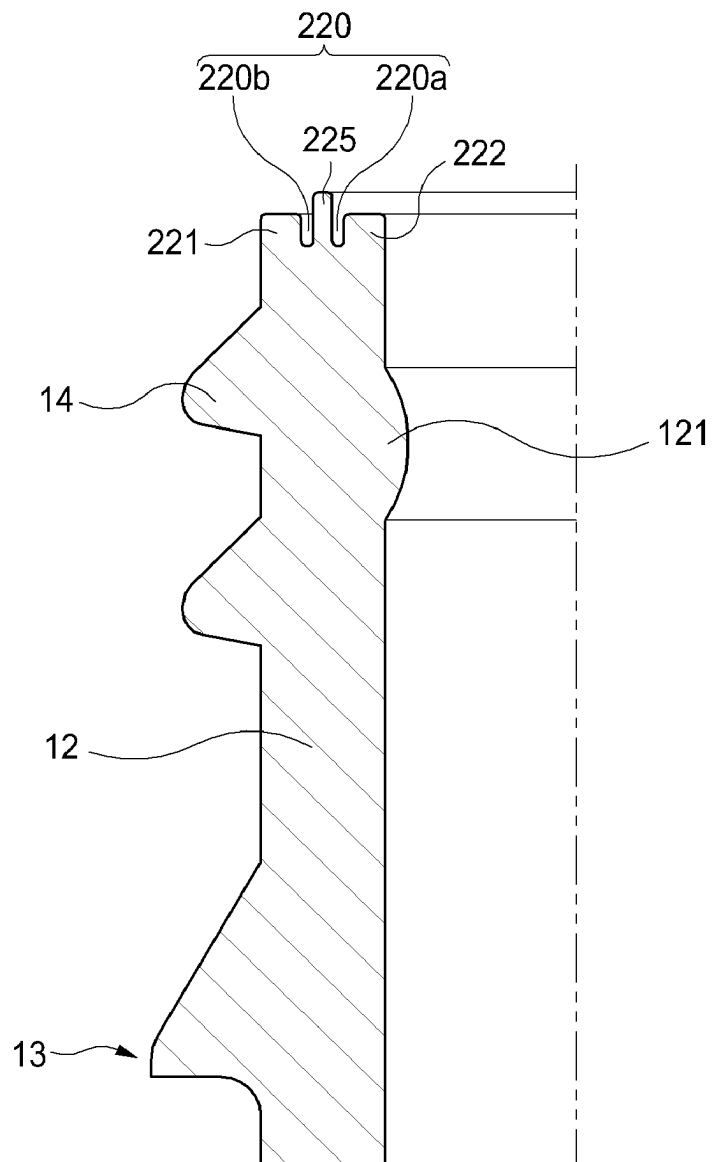


FIG. 18D

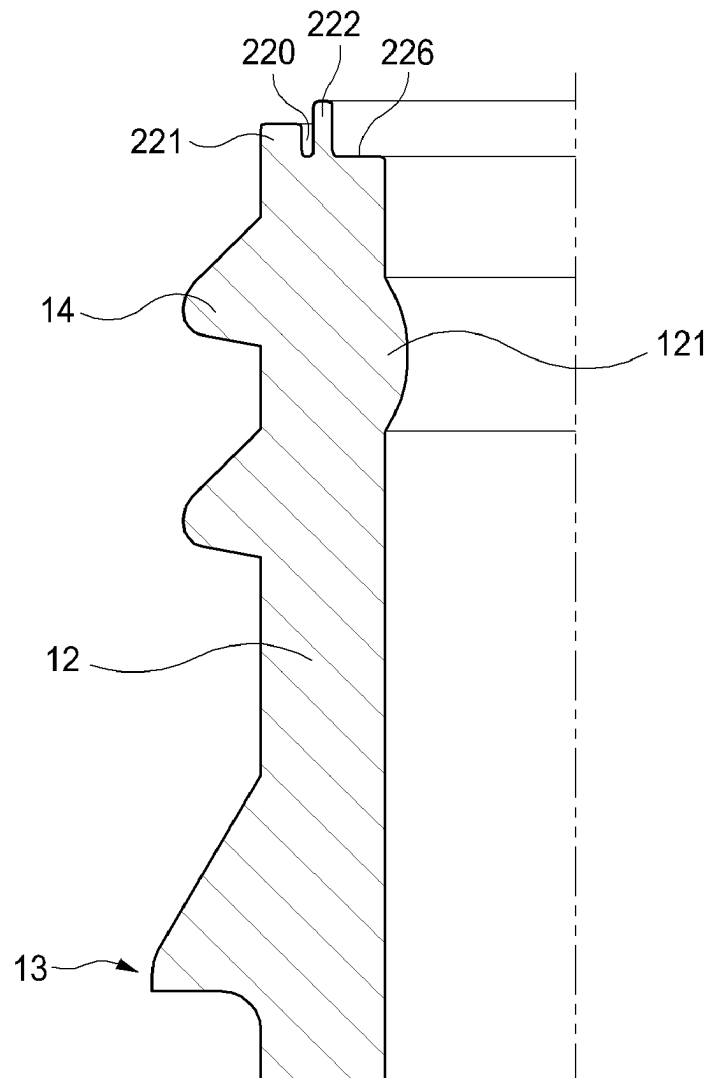


FIG. 18E

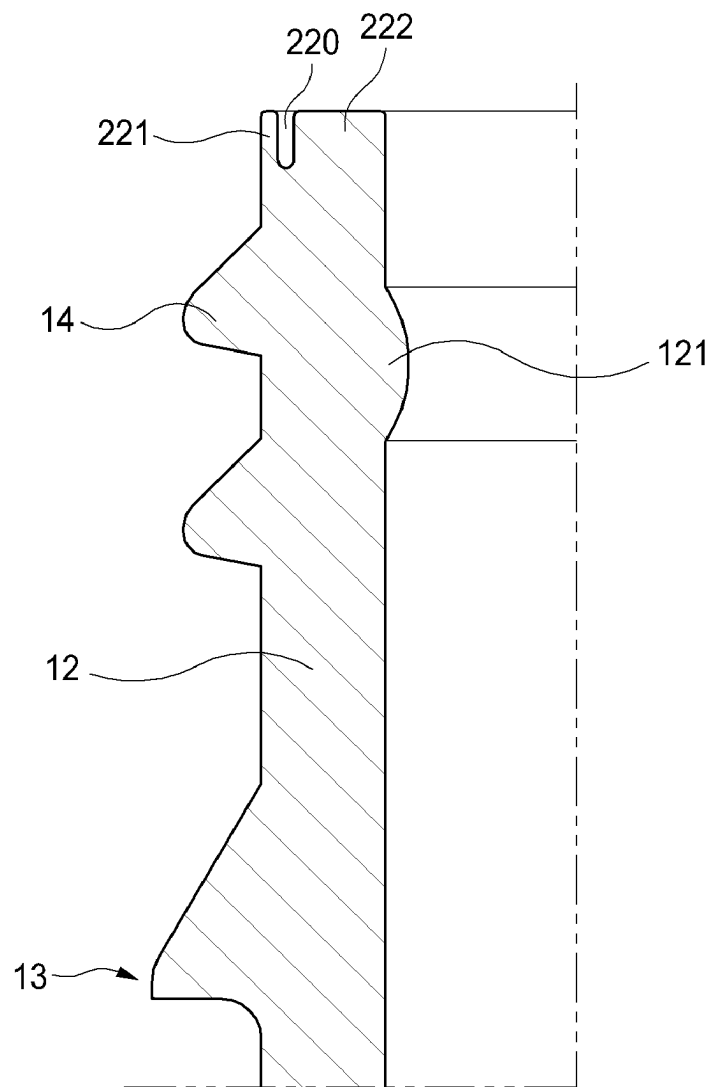


FIG. 18F

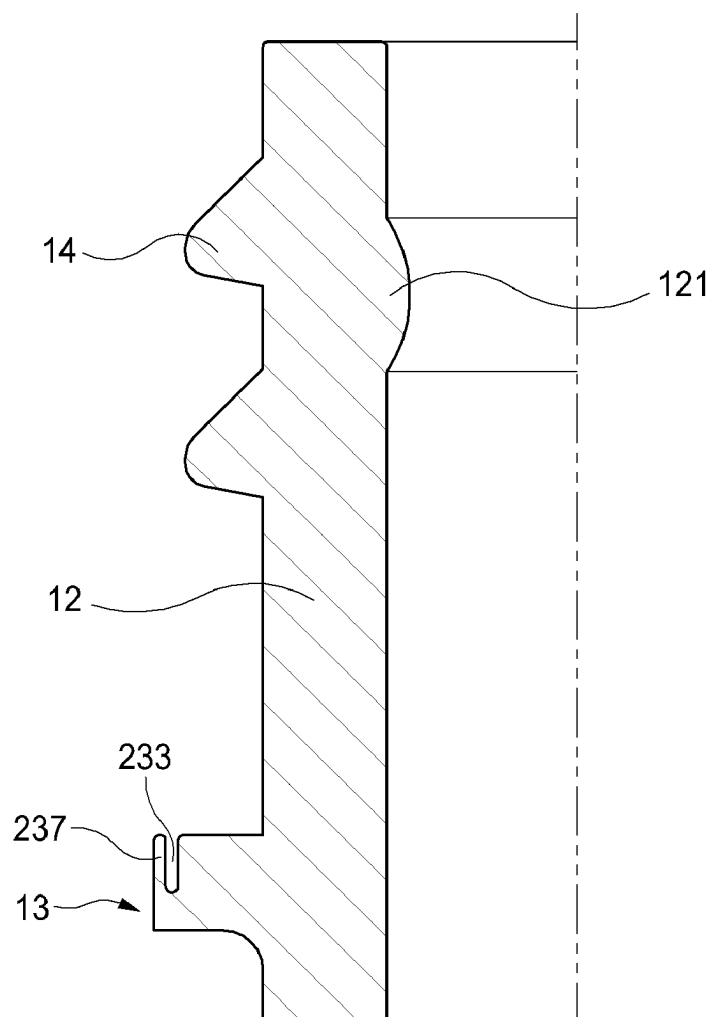


FIG. 18G

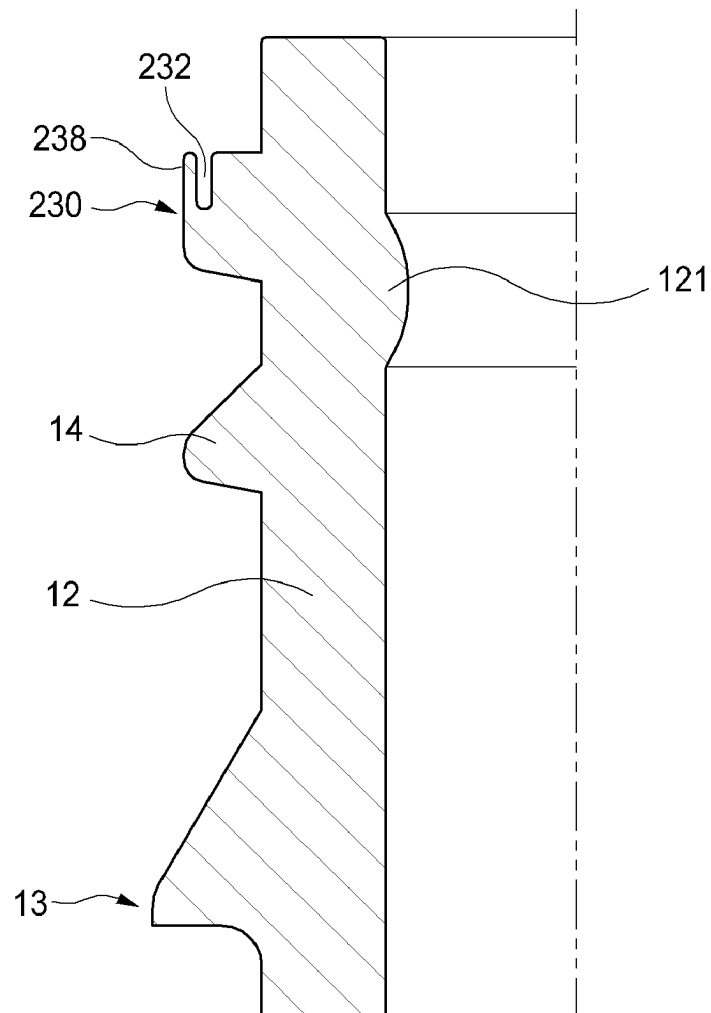


FIG. 18H

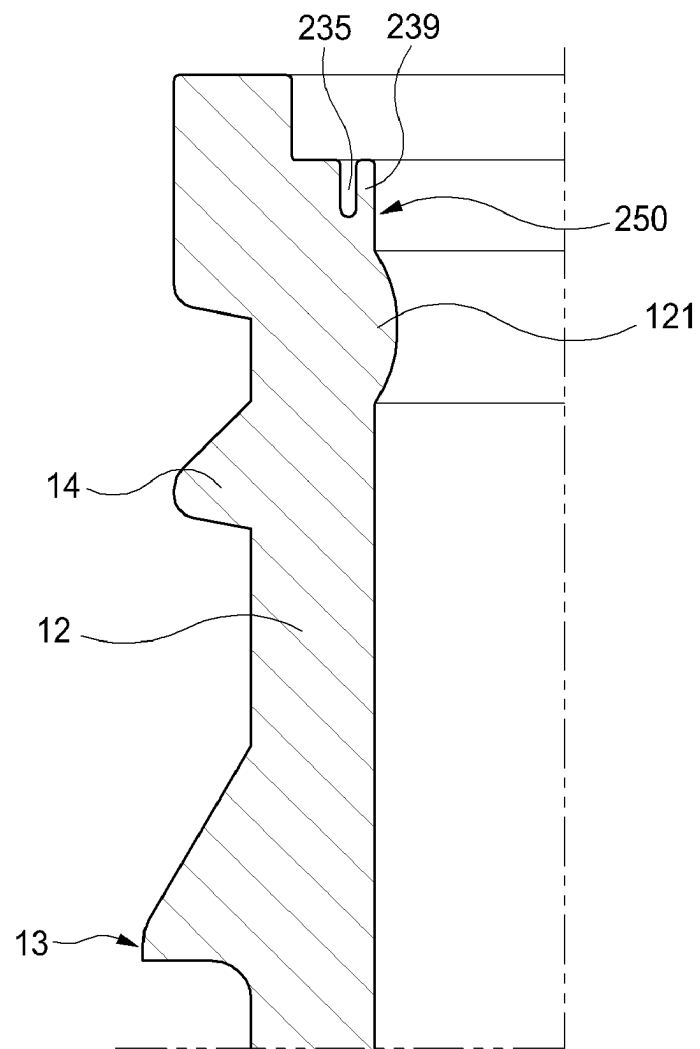


FIG. 19A

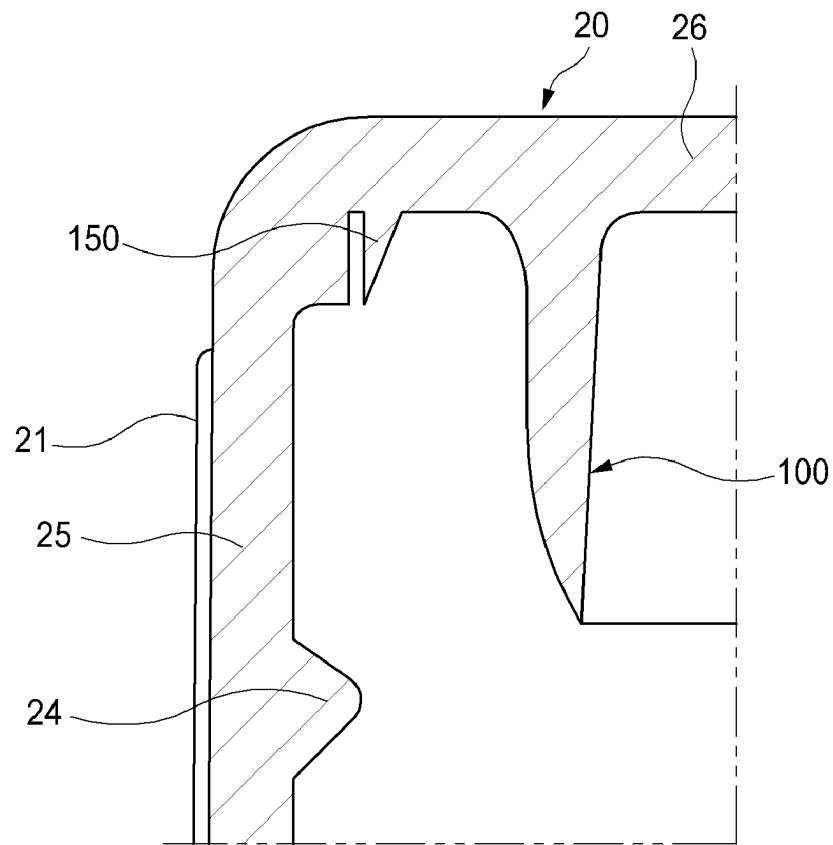


FIG. 19B

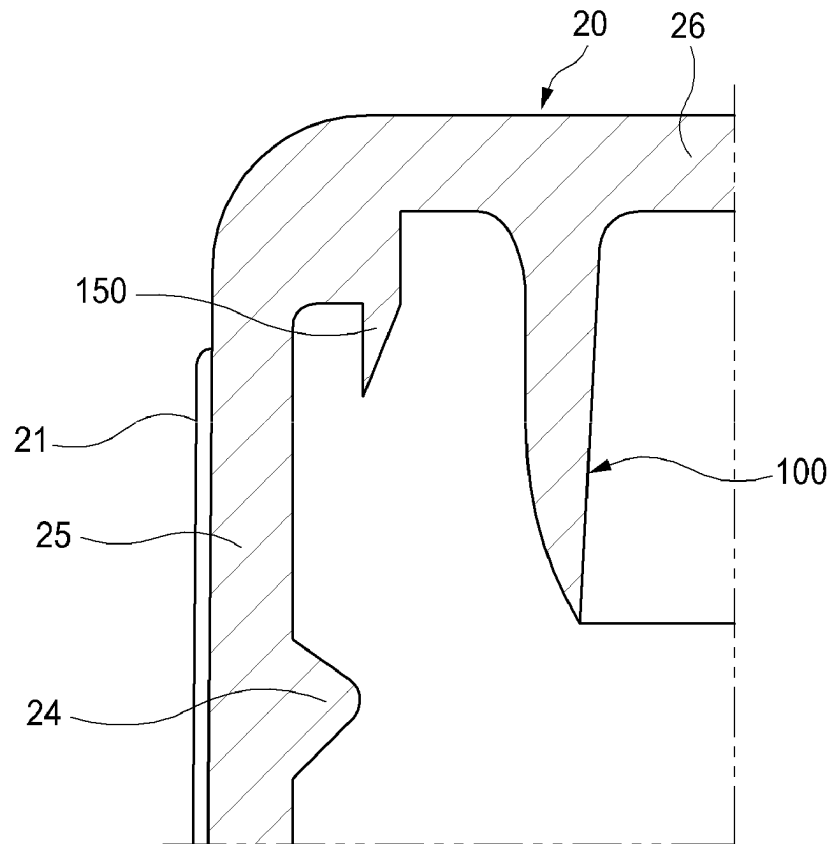


FIG. 19C

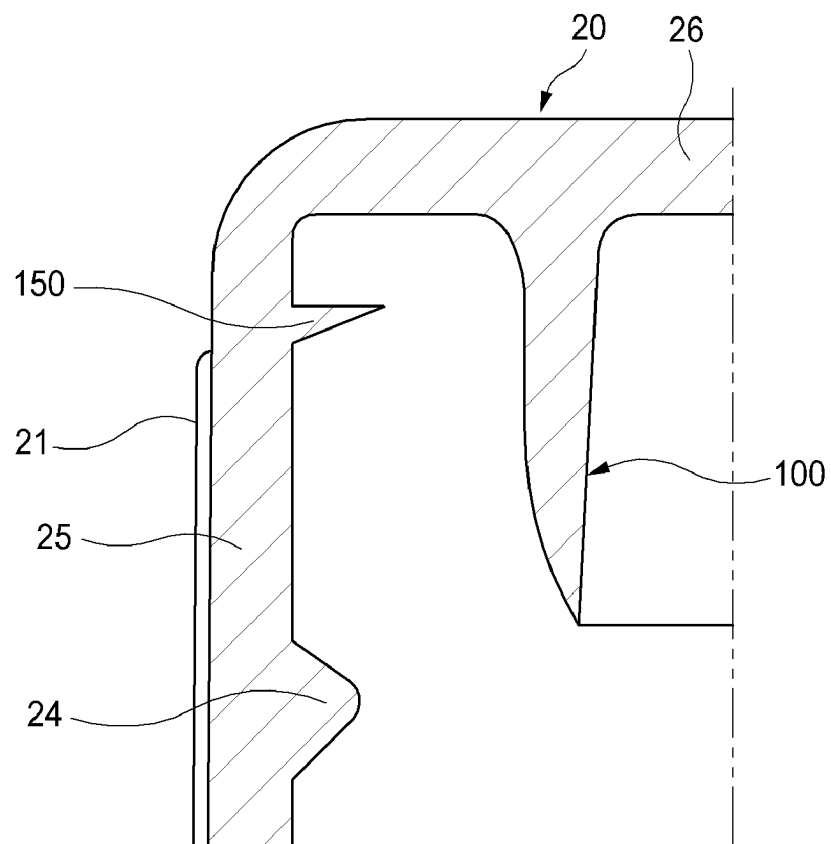


FIG. 20A

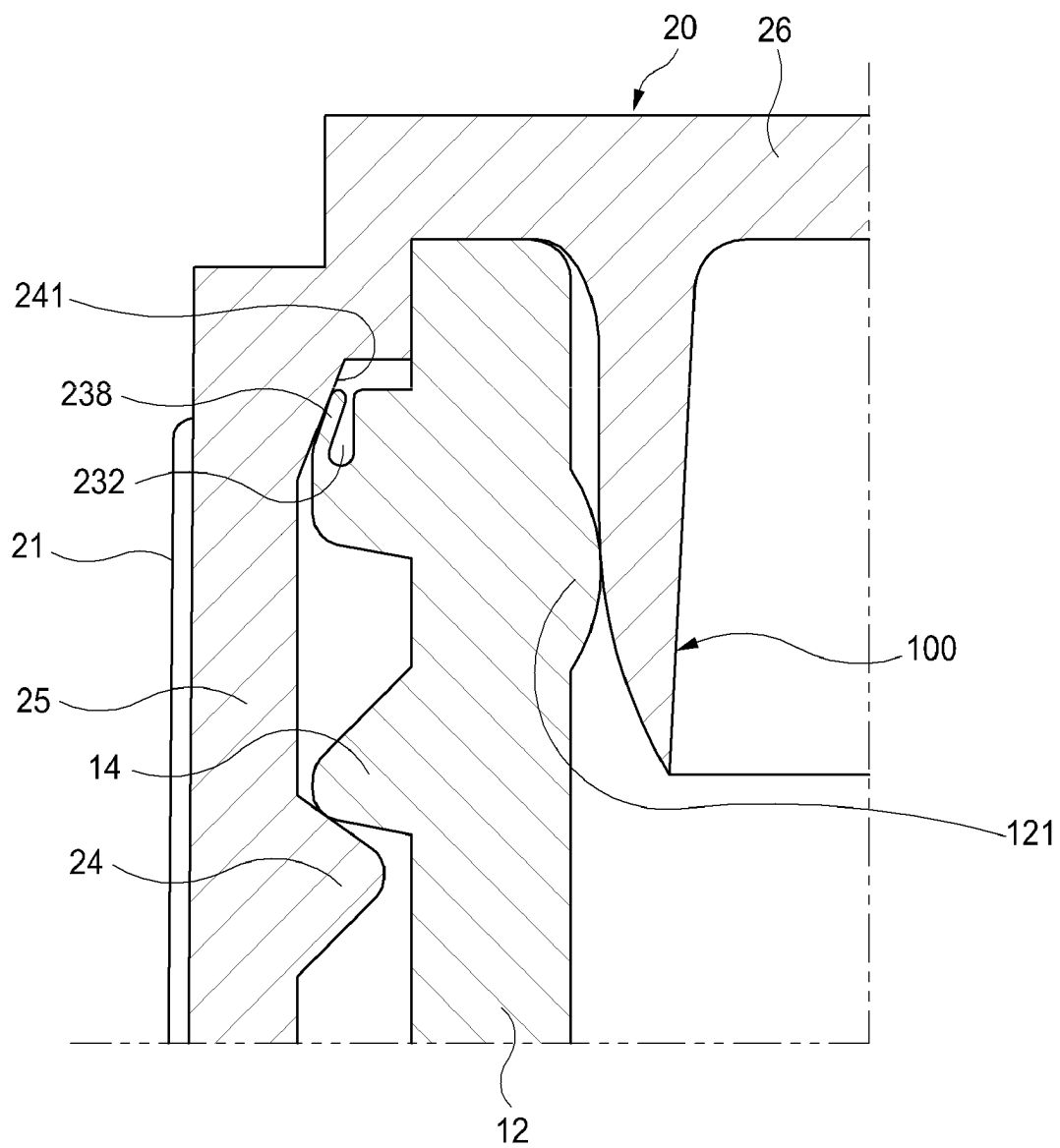


FIG. 20B

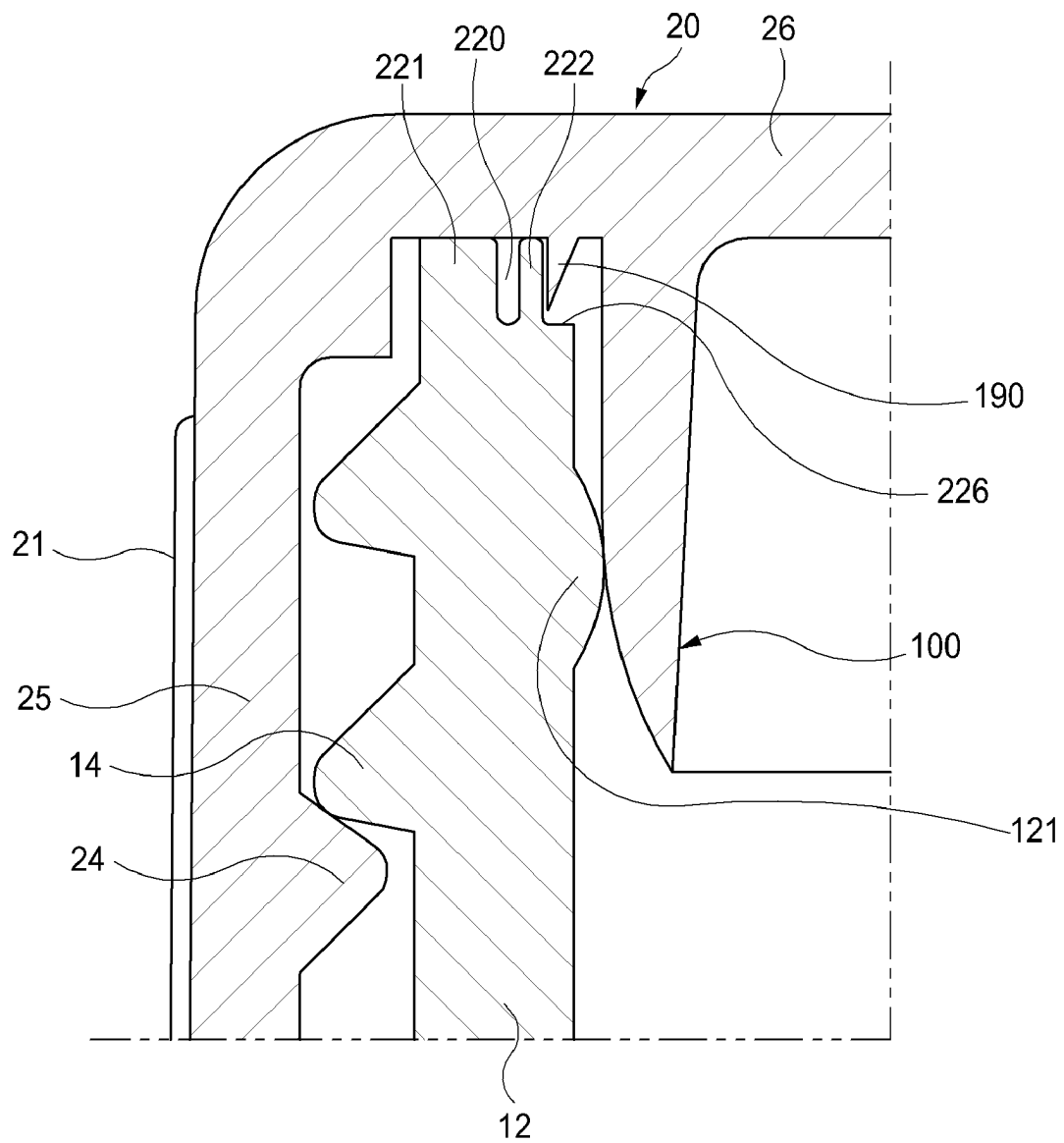


FIG. 20C

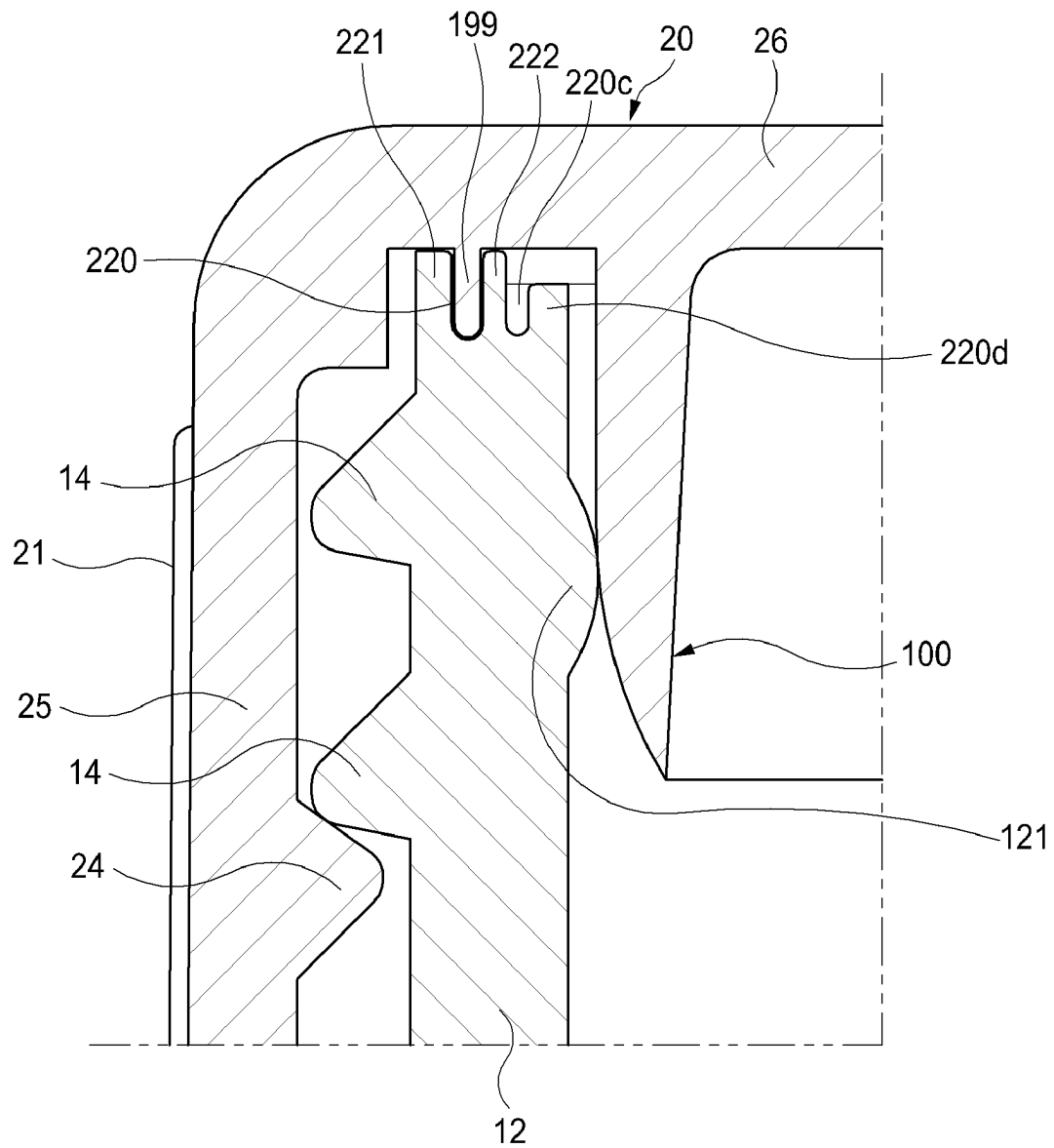


FIG. 20D

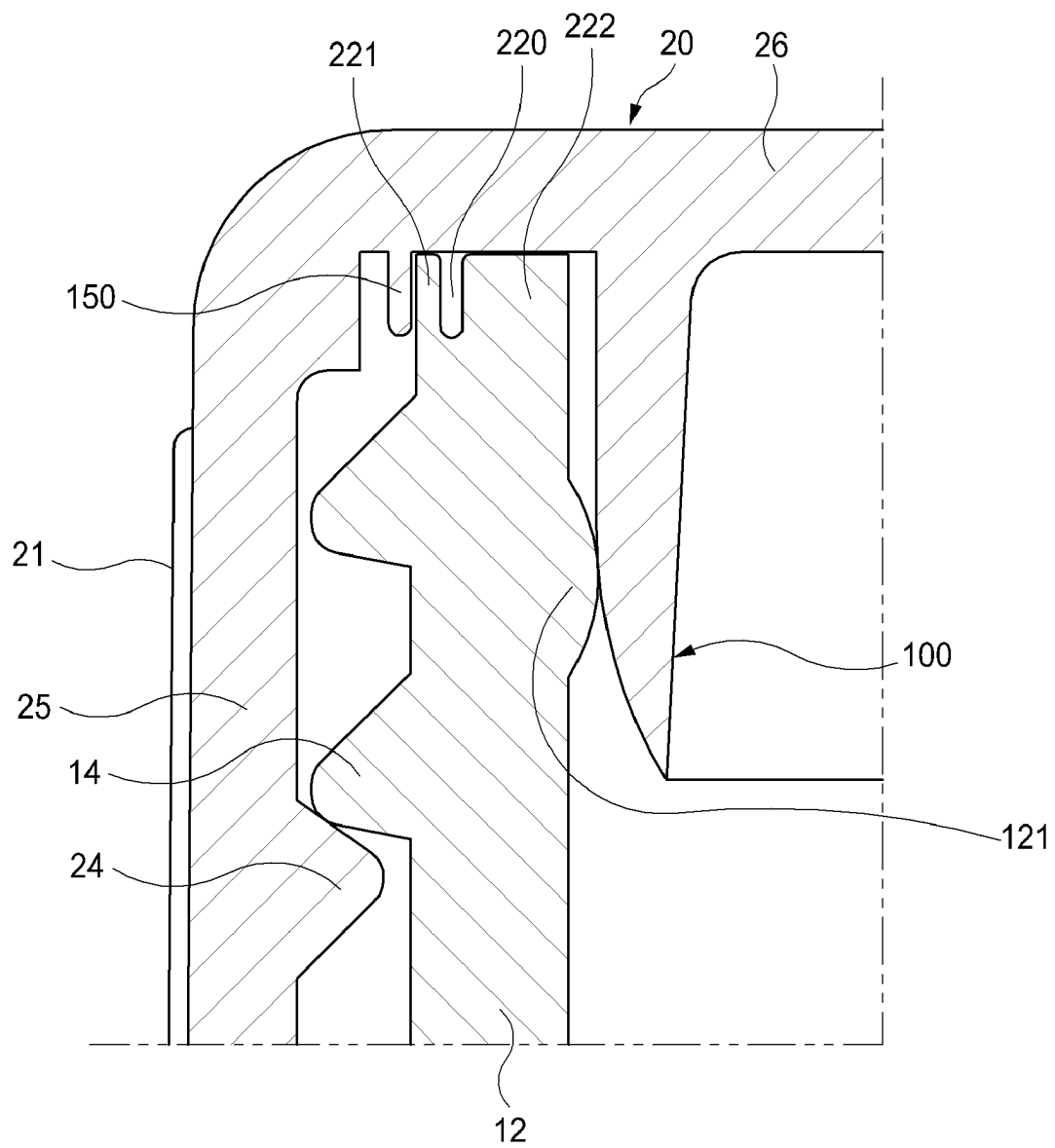


FIG. 21A

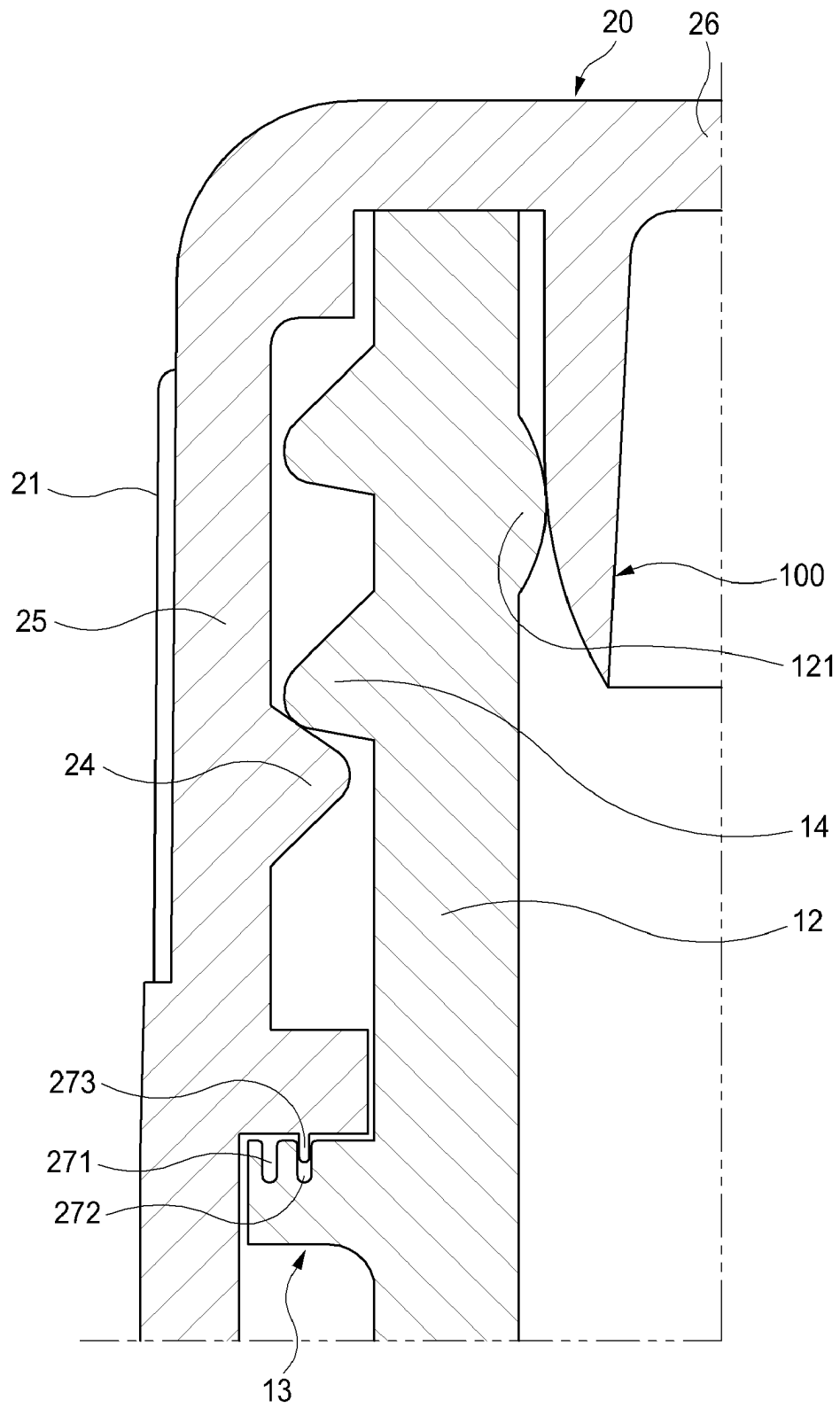


FIG. 21B

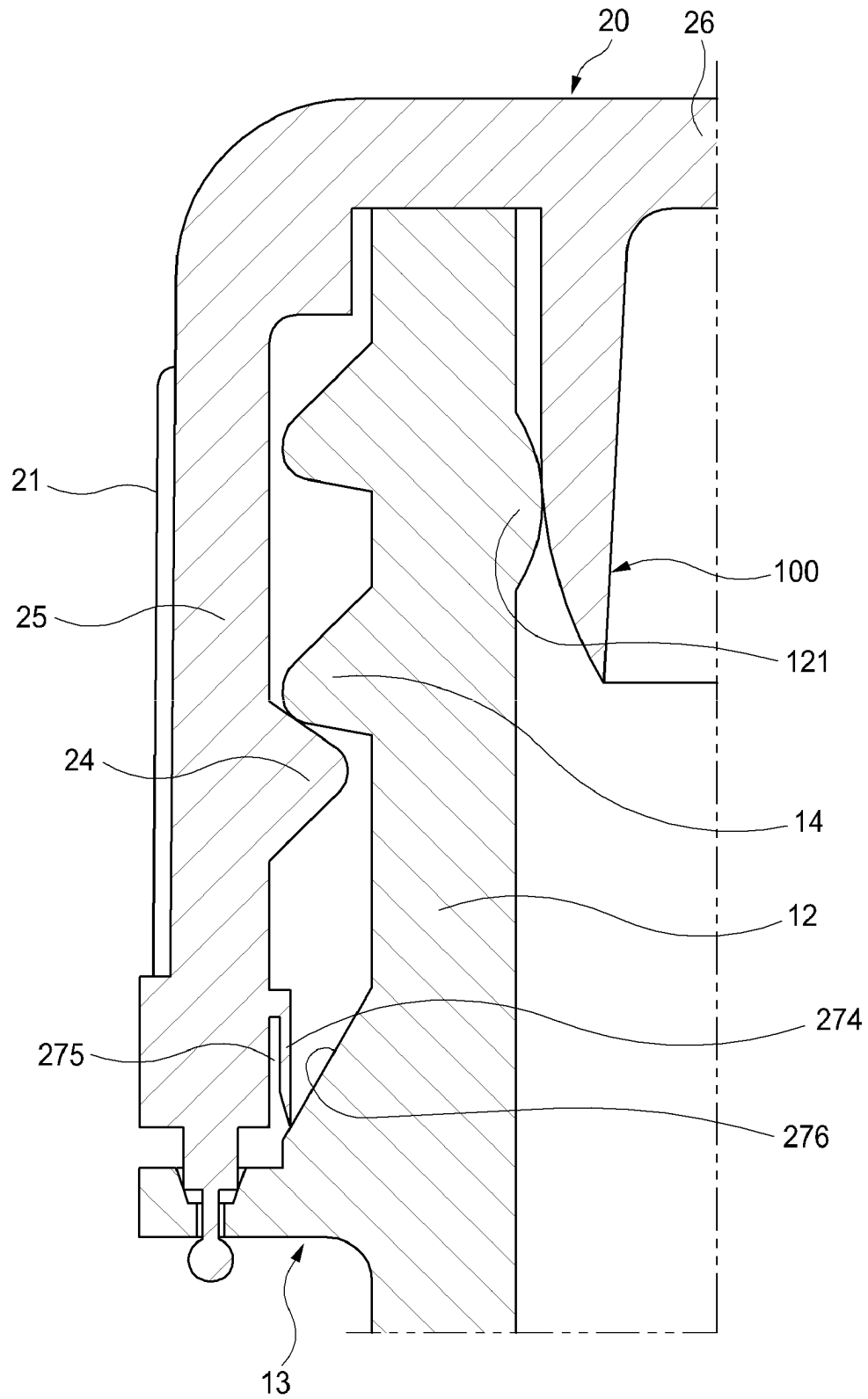


FIG. 21C

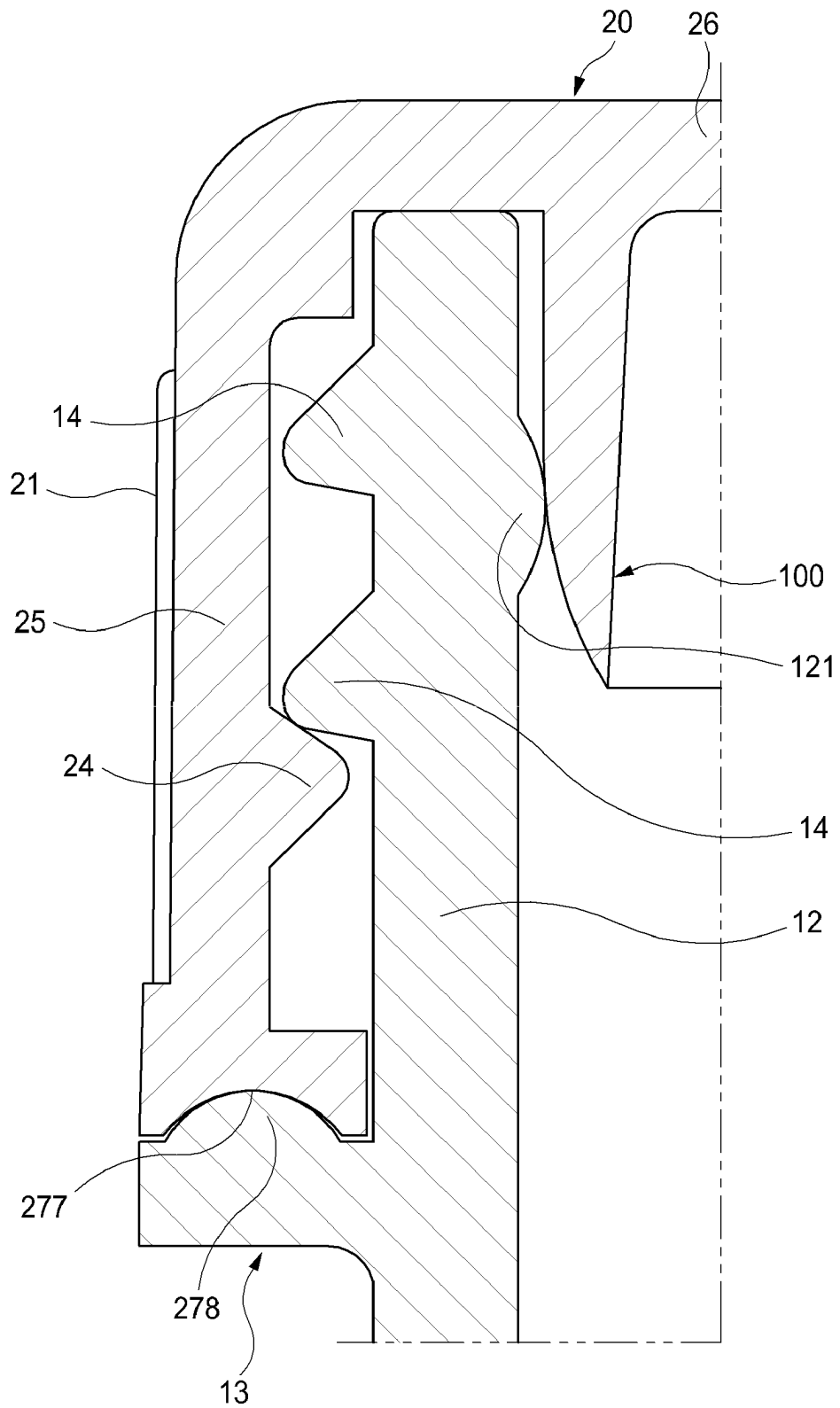


FIG. 22

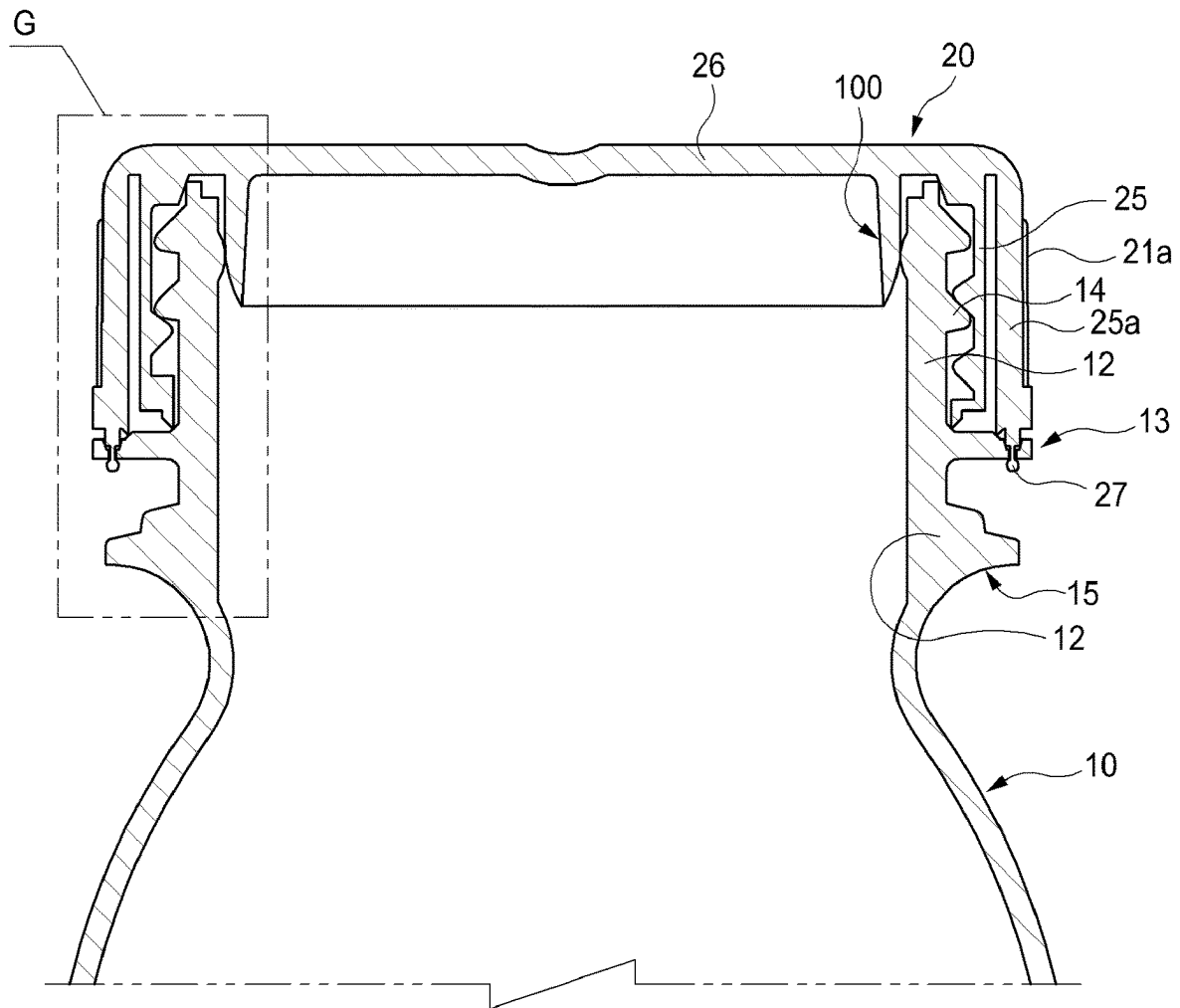


FIG. 23

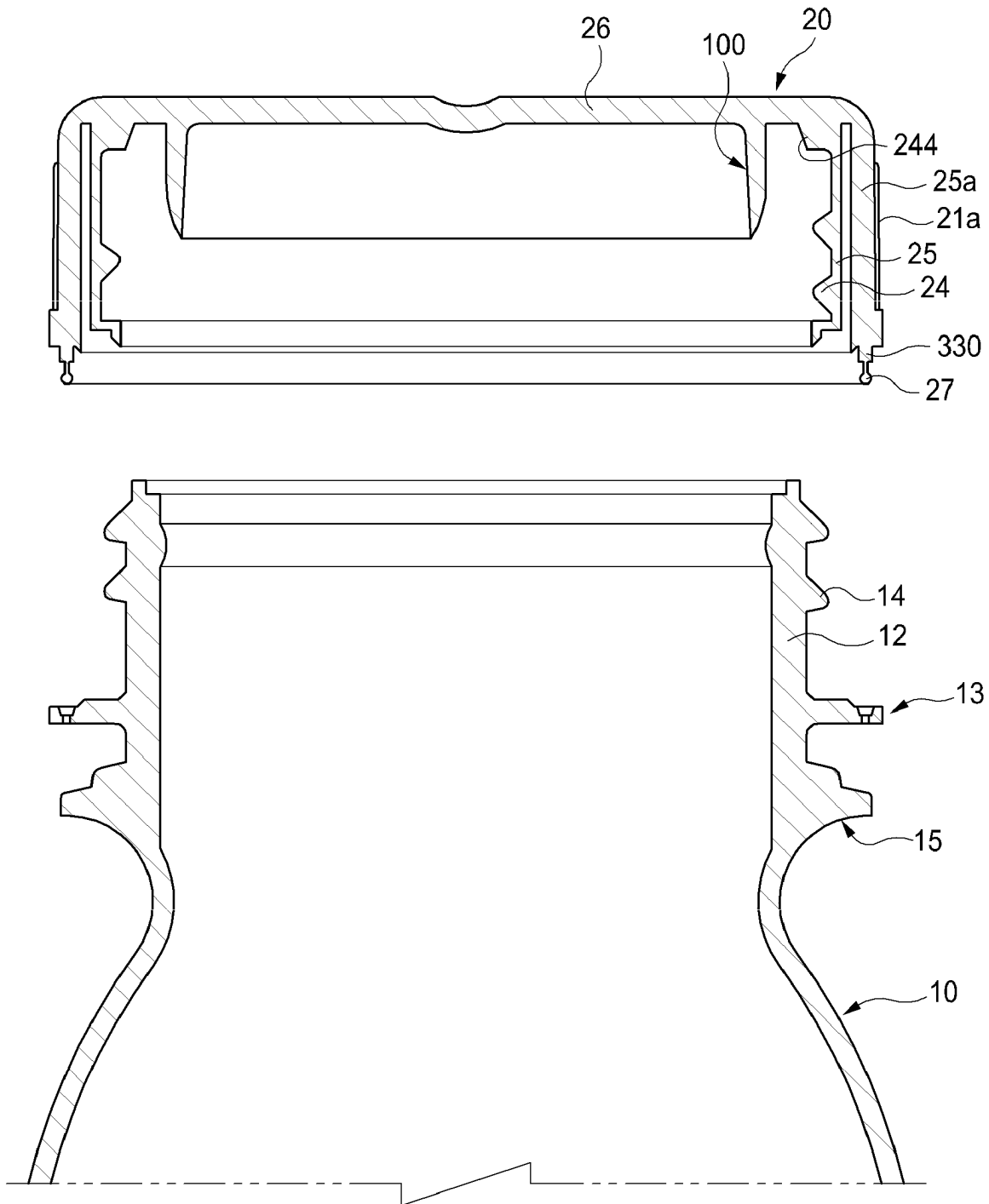


FIG. 24

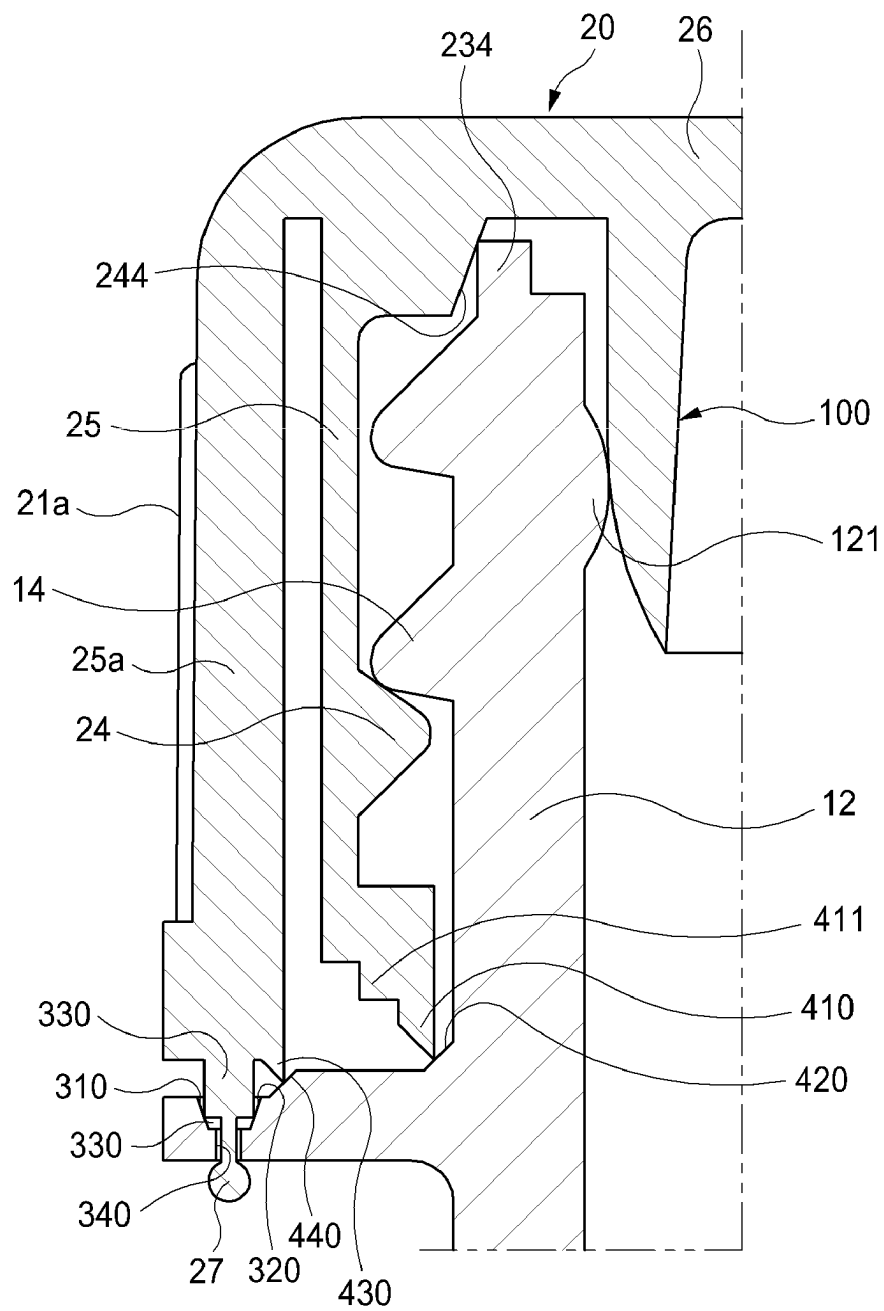


FIG. 25

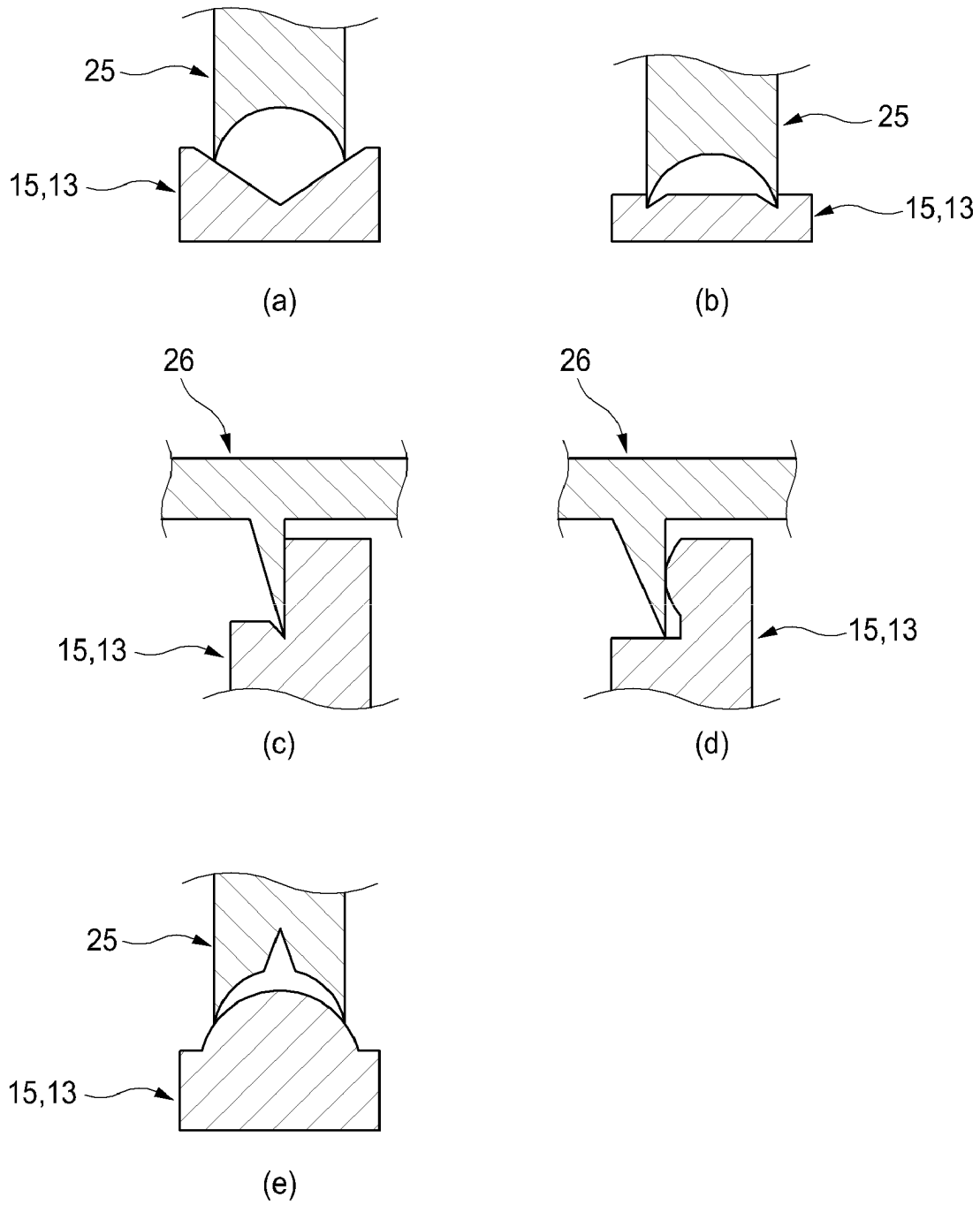


FIG. 26A

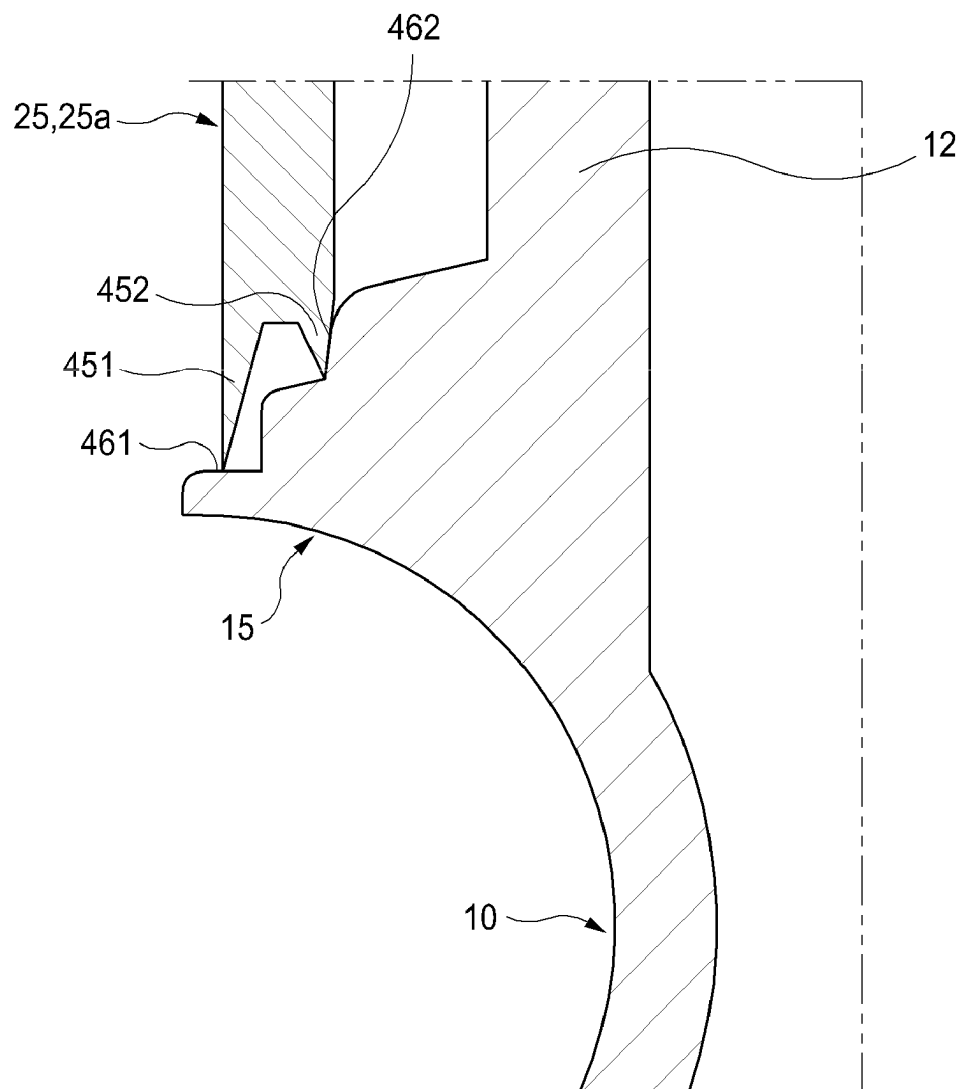
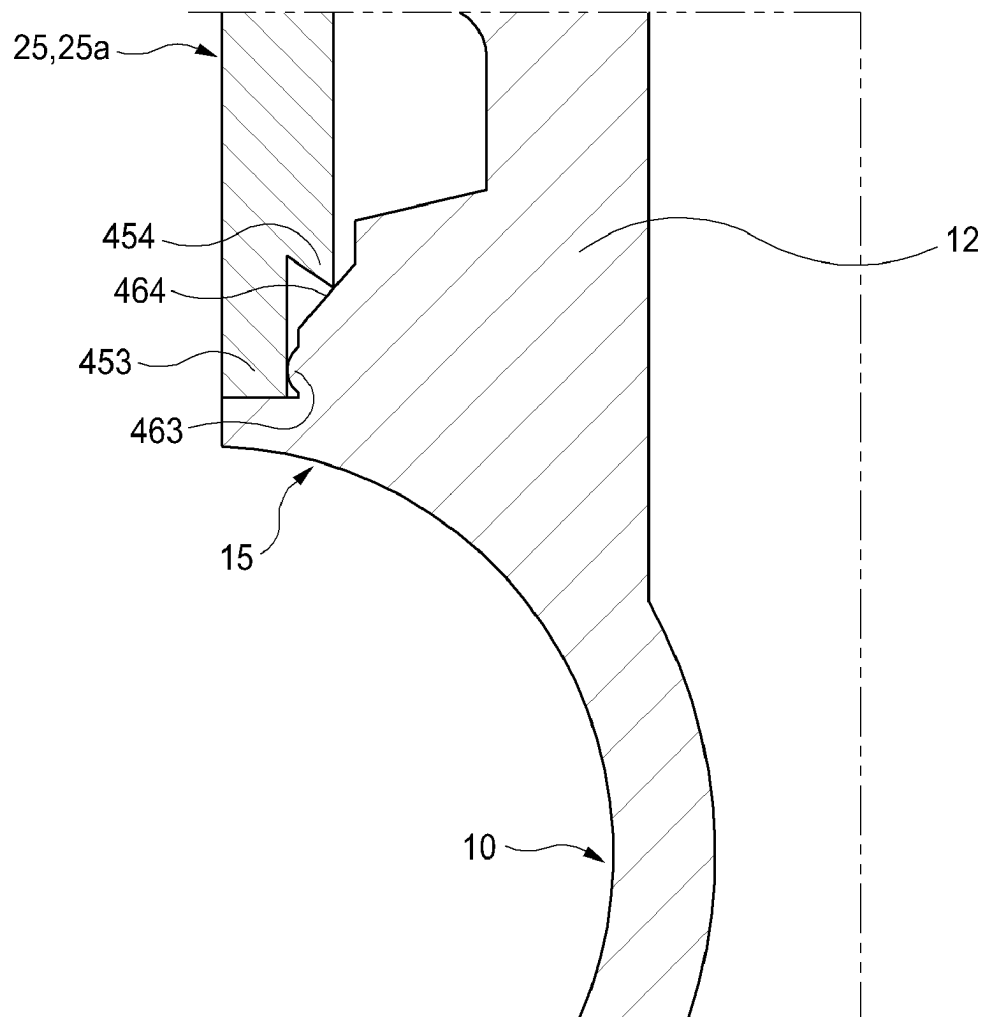


FIG. 26B



1

CONTAINER CAP AND CONTAINER COMBINED WITH SAME

TECHNICAL FIELD

The present invention relates to a container cap and a container combined with same, and more particularly, to a container cap combined with a container containing a beverage or the like and a container combined with the container cap.

BACKGROUND ART

A plastic container includes a container containing contents such as a beverage and a container cap combined with a container inlet.

However, the container cap combined with the container inlet may form a minute gap due to a tolerance with the container inlet. Through the gap, the contents such as a beverage may be leaked to the outside, or moisture and bacteria may be introduced from the outside into the container to contaminated or deteriorated.

In order to prevent above described limitations, typically, 1) a sealing member is mounted to a top surface of the container inlet, 2) an inner ring is formed on an inner bottom surface of the container cap, or 3) a packing is mounted to an inner space of the container cap.

However, although the 1) is the safest method, the method of the 1) may increase a manufacturing cost by adding a process of installing the sealing member to the top surface of the container inlet and may not be recycled because a different material is combined.

The method of 2) has a limitation in completely blocking moisture, external air, or bacteria because the tolerance is generated between an inner wall or an outer wall of the container and an inner space of the container cap when the container cap is combined with the container inlet.

DISCLOSURE OF THE INVENTION

Technical Problem

The present invention provides a container cap having a structure combined with a container inlet to completely block moisture, external air, or bacteria from being introduced into a container and a container combined with same to solve the above-described limitations.

Technical Solution

The present invention is disclosed to achieve the objects of the invention. The present invention will solve the above-described limitations by deforming structures of various portions that are combined to have a blocking function when the container inlet is combined with the container cap.

In order to achieve the objects, the present invention provides a container cap **20** which is combined with a container inlet **12** of a container body **10**, wherein a plurality of close contact portions in contact with the container inlet **12** are formed, and the close contact portions form a closed curve enclosing the container inlet **12** so as to block inflow of external air from the outside into the container body **10**.

According to the present invention, a container includes: a container body **10** containing contents and having a container inlet **12**; and a container cap **20** combined with the container inlet **12**. Here, a plurality of close contact portions in close contact with each other are formed on at least one

2

of the container cap **20** and the container inlet **12**, and the close contact portions form a closed curve enclosing the container inlet **12** so as to block inflow of external air from the outside into the container body **10**.

The container may include: a container body **10** containing contents and having a container inlet **12**; and a container cap **20** combined with the container inlet **12**. Here, the container cap **20** may include: a body part **26** in close contact with an upper end of the container inlet **12**; a sidewall part **25** which extends downward from the body part **26** and in which a female screw part **24** that is screw-coupled with a male screw part **14** formed on an outer circumferential surface of the container inlet **12** is formed; and an inner ring **100** protruding from a bottom surface of the body part **26** and inserted to an inner circumferential surface of the container inlet **12**.

At least one inner circumferential protruding part **121** that protrudes toward the inner ring **100** to contact an outer circumferential surface of the inner ring **100** when the container cap **20** is combined with the container inlet **12** may be formed on an inner circumferential surface of the container inlet **12**.

At least one inner ring protruding part that protrudes toward an inner circumferential surface of the container cap **20** to contact the inner circumferential surface of the container cap **20** when the container cap **20** is combined with the container inlet **12** may be formed on the container cap **20**.

A middle ring **210** inserted to a recessed groove **220** having a circular shape formed on the upper end of the container inlet **12** and formed on the bottom surface of the body part **26** to form at least one close contact portion when the container cap **20** is combined with the container inlet **12** may be formed on the container cap **20**.

An inner upper end **221** may be lower than an outer upper end **222** based on the recessed groove **220** in the upper end of the container inlet **12**.

At least one of an inner upper end **221** and an outer upper end **222** of the container inlet **12** may have a curved outer circumferential surface, and an inclined surface **215** in contact with the curved outer circumferential surface when the container cap **20** is combined with the container inlet **12** may be formed on the container cap **20**.

Advantageous Effects

The container cap and the container combined with the same according to the present invention have an advantage in that the close contact portions contacting the container inlet form the closed curve, and the plurality of close contact portions are combined with the container inlet to completely block moisture, external air, and bacteria from being introduced into the container.

Also, the container cap and the container combined with the same according to the present invention have an advantage in that as the close contact portions contacting the container inlet form the closed curve, and the plurality of close contact portions completely block moisture, external air, and bacteria, the sealing force of the container is improved to secure safety of the contents contained in the container and diversify the kinds of the contents.

Also, the container cap and the container combined with the same according to the present invention have an advantage in that as the close contact portions contacting the container inlet form the closed curve, and the plurality of close contact portions completely block moisture, external air, and bacteria, the sealing member for sealing the container is not necessary to remarkably reduce the manufac-

turing costs of the product including the container and contribute to the environment.

Also, the container cap and the container combined with the same according to the present invention have an advantage in that the close contact portions contacting the container inlet form the closed curve, the plurality of close contact portions completely block moisture, external air, and bacteria, and the container cap is maximally recycled as the container and the container cap are made of the single material such as synthetic resin materials (e.g., PE, PP, PET).

Furthermore, the container cap and the container combined with the same according to the present invention have an advantage in that as the container and the container cap are made of the single material such as synthetic resin materials, the container cap and all sorts of containers are maximally recycled.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view illustrating a container according to a first embodiment of the present invention.

FIG. 2A is a cross-sectional view taken along line I-I of FIG. 1, and FIG. 2B is a cross-sectional view illustrating a state in which a container cap is separated from the container.

FIG. 3 is an enlarged cross-sectional view illustrating portion A1 of FIG. 2A.

FIG. 4A is a cross-sectional view illustrating a container according to a second embodiment of the present invention, and FIG. 4B is an enlarged cross-sectional view illustrating portion B1 of FIG. 4A.

FIG. 5A is a cross-sectional view illustrating a modified example of the container in FIG. 4A, and FIG. 5B is an enlarged cross-sectional view illustrating portion B2 of FIG. 5A.

FIG. 5C is an enlarged cross-sectional view illustrating portion B3 of FIG. 5A.

FIGS. 6A and 6B are enlarged cross-sectional views respectively illustrating a modified example of a portion in FIG. 4A and a modified example of a portion in FIG. 5B.

FIG. 7 is a cross-sectional view illustrating a container according to a third embodiment of the present invention.

FIG. 8A is an enlarged cross-sectional view illustrating portion C1 of FIG. 7.

FIG. 8B is an enlarged cross-sectional view illustrating a modified example of FIG. 8A.

FIG. 8C is an enlarged cross-sectional view illustrating an example in which a flexibility applying groove is formed in a wall part as a modified example of FIG. 8A.

FIG. 8D is an enlarged cross-sectional view illustrating an example in which a flexibility applying groove is formed in a scut combining projection as a modified example of FIG. 8A.

FIG. 9 is a cross-sectional view illustrating a container according to a fourth embodiment of the present invention.

FIG. 10A is an enlarged cross-sectional view illustrating portion D1 of FIG. 9.

FIG. 10B is an enlarged cross-sectional view illustrating a modified example of FIG. 10A.

FIG. 10C is an enlarged cross-sectional view illustrating a modified example of FIG. 10A.

FIG. 10D is an enlarged cross-sectional view illustrating a modified example of FIG. 10C.

FIGS. 11A to 11E are enlarged cross-sectional views illustrating various embodiments for sealing the container as a modified example of portion D1 of FIG. 9.

FIG. 12 is a cross-sectional view illustrating a container according to a fifth embodiment of the present invention.

FIG. 13 is an enlarged cross-sectional view illustrating portion E1 of FIG. 12.

FIG. 14 is a cross-sectional view illustrating a modified example of the container of FIG. 12.

FIG. 15 is an enlarged cross-sectional view illustrating portion E2 of FIG. 14.

FIG. 16 is an enlarged cross-sectional view illustrating a modified example of FIG. 13.

FIG. 17 is an enlarged cross-sectional view illustrating a modified example of FIG. 15.

FIGS. 18A to 18H are partial cross-sectional views illustrating examples of a structure of a container inlet used for the container according to the present invention.

FIGS. 19A to 19C are partial cross-sectional views illustrating examples of a structure of a container cap used for the container according to the present invention.

FIGS. 20A to 20D are partial cross-sectional views illustrating examples of structures of the container according to the present invention.

FIGS. 21A to 21C are partial cross-sectional views illustrating examples of structures of the container according to the present invention.

FIG. 22 is a cross-sectional view illustrating a container according to a seventh embodiment of the present invention.

FIG. 23 is a cross-sectional view illustrating a state in which the container cap is separated from the container of FIG. 22.

FIG. 24 is an enlarged cross-sectional view illustrating portion G of FIG. 22.

FIG. 25 shows cross-sectional views illustrating examples of contact structures between the container cap and the container inlet in the container according to the present invention.

FIGS. 26A and 26B are partial cross-sectional views illustrating structures of at least one projection ring formed on a lower end of a sidewall part of the container cap and the container inlet contacting the same in the container according to the present invention.

MODE FOR CARRYING OUT THE INVENTION

Hereinafter, a container cap and a container combined with same according to the present invention will be described with reference to the accompanying drawings.

The present invention provides a novel container cap and a container structure, which are capable of preventing external air or bacteria from being introduced into a container or preventing contents from being leaked to the outside.

Specifically, a container according to the present invention includes a container body 10 containing the contents and having a container inlet 12 and a container cap 20 combined with the container inlet 12. Here, at least one of the container cap 20 and the container inlet 12 forms a plurality of close contact portions, and the close contact portions form a closed curve enclosing the container inlet 12 so as to block inflow of external air from the outside into the container body 10.

Here, the container body 10 may contain the contents including powder, tablets, and granules in addition to liquid materials such as a beverage and be made of various materials such as a synthetic resin material and a glass material according to shapes and materials of liquids, powder, or particles.

5

Particularly, the container body **10** may be made of the same material as the container cap **20** such as PP, PE, and PET for combination with the container cap according to the present invention.

The close contact portion, as a portion closely contacting at least one of the container cap **20** and the container inlet **12**, may include all sorts of structures of forming a portion closely contacting the container cap **20** and the container inlet **12**.

Particularly, the close contact portion of the present invention may be combined with the container inlet to completely block moisture, external air, or bacteria from being introduced into the container as the container cap **20** closely contacts the container inlet **12** by a descending force when the container cap **20** is rotated and/or descended to be combined with the container inlet **12** (e.g., one piece or two piece one-touch cap and undercut cap (snap-type cap)).

In the container body **10**, the container inlet **12** that is a portion combined with the container cap **20** has a basic shape of a cylindrical shape, and the rest portion may have various configurations according to a design.

Also, the container inlet **12** may be an outlet formed in the container body **10** so that the contents are discharged to the outside and include various configurations according to a combining structure of the container because the container inlet **12** is coupled with the container cap **20**.

Particularly, since the container cap **20** of main products is combined by rotation, the container inlet **12** may have a partially cylindrical shape. The cylindrical container inlet **12** may be applied to products having a combining structure, in which the container cap descends to be combined with the container inlet **12**, like one-piece one touch cap (also known as Duma cap), two-piece one touch cap in which a cap and a body are integrated by a hinge, and a undercut cap that is combined only with undercut (representative products: *lactobacillus* yogurt products).

The container cap **20** is a component combined with the container inlet **12** of the container body **10**. The container cap **20** has the basic shape of an overall cylindrical shape corresponding to a shape of the container inlet **12** and having an opened side heading toward the container inlet **12**.

For example, the container cap **20** may include: a body part **26** closely contacting an upper end of the container inlet **12**; and a sidewall part **25** extending downward from the body part **26** and combined with the container inlet **12**.

The body part **26** may be a portion closely contacting the upper end of the container inlet **12** and have various shapes according to an outer structure of the container cap **20**, e.g., a circular plate shape.

The container cap **20** may include an inner ring **100**, so called a ring gasket, protruding from a bottom surface of the body part **26** and inserted to an inner circumferential surface of the container inlet **12** to increase a sealing force of the container body **10**.

The inner ring **100**, as a portion protruding from the bottom surface of the body part **26** and inserted to the inner circumferential surface of the container inlet **12** to increase the sealing force of the container body **10**, may have various configurations according to a contact structure with the container inlet **12**.

The sidewall part **25**, as a component extending downward from the body part **26** and combined with the container inlet **12**, may have various configurations according to a combining structure with the container inlet **12**.

For example, the sidewall part **25** may be combined with the container inlet **12** through a screw combining structure, and here, a female screw part **24** screw-combined with a

6

male screw part **14** formed on an outer circumferential surface of the container inlet **12** may be formed.

Also, the sidewall part **25** may be combined with the container inlet **12** by a snap-type combining structure (e.g., one-piece and two-piece one touch cap and undercut cap structure).

Here, as at least one ring-shaped combining projection may be formed on the outer circumferential surface of the container inlet **12**, and a ring-shaped combining groove to which the combining projection formed on the container inlet **12** is inserted is formed in an inner surface of the sidewall part **25**, the container cap **20** may be combined with the container inlet **12**.

The container cap **20** according to the present invention may include a plurality of close contact portions closely contacting the container inlet **12** by relative close contact with the container inlet **12**.

Also, the close contact portions may form a closed curve enclosing the container inlet **12** to block inflow of external air from the outside into the container body **10** and have various configurations according to the formation position, formation structure, and number of the close contact portions.

A scut part **27** caught by and combined with a scut combining projection **13** formed on the container inlet **12** may be integrated with a lower end of the sidewall part **25** of the container cap **20**.

The scut part **27** is a portion integrated with the lower end of the sidewall part **25** to be caught by and combined with the scut combining projection **13** formed on the container inlet **12** and a portion separated from the sidewall part **25** when the container cap **20** is initially opened.

Here, the scut part **27** may have various configurations according to a combining structure with the scut combining projection **13**.

Also, the scut part **27** may be connected with the sidewall part **25** by a plurality of bridges or a cutting line for easy separation from the sidewall part **25**.

A separation preventing projection **15** for preventing the container cap **20** from being unintentionally separated from the container inlet **12** may be additionally formed at a position corresponding to a lower end of the scut part **27**.

The separation preventing projection **15**, as a component formed at the position corresponding to the lower end of the scut part **27** of the container inlet **12** for preventing the container cap **20** from being unintentionally separated from the container inlet **12**, may protrude further than an outer circumferential surface of the sidewall part **25** of the container cap **20**.

Also, a plurality of projections **21** may be formed along a circumferential direction of the container cap **20** so that a user easily rotates the container cap **20**.

Hereinafter, a specific embodiment of the present invention will be described in detail. Also, characteristic parts will be described in detail for convenience of description.

Although the present invention is described for each embodiment, one or more embodiments may be combined as long as the embodiments are not conflicted with each other.

First Embodiment

As illustrated in FIGS. 1 to 3, according to a first embodiment of the present invention, at least one inner circumferential protruding part **121** protruding toward the inner ring **100** to contact an outer circumferential surface of

the inner ring 100 when the inner ring 100 of the container cap 20 is combined with the container inlet 12 may be formed.

The inner circumferential protruding part 121, as a portion protruding toward the inner ring 100 to contact the outer circumferential surface of the inner ring 100, may have a ring shape in a circumferential direction based on a central axis of a longitudinal direction of the container inlet 12.

Also, the inner circumferential protruding part 121 may have a gentle curved shape so that a cross-section in the longitudinal direction of the container inlet 12 is in surface-contact with the outer circumferential surface of the inner ring 100 instead of line-contact.

In the container cap 20, at least one inner ring protruding part protruding toward an inner circumferential surface of the container cap 20 to contact the inner circumferential surface of the container cap 20 when the container cap 20 is combined with the container inlet 12 may be formed as a component similar to the inner circumferential protruding part 121.

The inner ring protruding part, as a component protruding toward an inner circumferential surface of the container cap 20 to contact the inner circumferential surface of the container cap 20 when the container cap 20 is combined with the container inlet 12, may have a ring shape in a circumferential direction based on the central axis of the longitudinal direction of the container inlet 12.

Also, the inner ring protruding part may have a gentle curved shape so that a cross-section in the longitudinal direction of the container inlet 12 is in surface-contact with the inner circumferential surface of the container inlet 12 instead of line-contact.

In the container cap 20, a middle ring 210 inserted to a recessed groove 220 integrated along a circumference of the upper end of the container inlet 12 when the container cap 20 is combined with the container inlet 12 may be formed on the bottom surface of the body part 26 to increase a sealing force to the container body 10.

The middle ring 210 may have various configurations according to a contact structure with the recessed groove 220 as a ring formed on the bottom surface of the body part 26 and inserted to the recessed groove 220 integrated along the circumference of the upper end of the container inlet 12 when the container cap 20 is combined with the container inlet 12.

For example, the middle ring 210 may have a vertical cross-section having a reverse triangular shape or a reverse trapezoidal shape. Here, an end of the middle ring 210 may have various shapes such as a flat surface or a curved surface.

Also, the recessed groove 220 to which the middle ring 210 is inserted may have a shape corresponding to a cross-sectional shape of the middle ring 210. Here, the cross-sectional shape of the middle ring 210 may have a width that is decreased more than that of the recessed groove 220 so that the middle ring 210 closely contacts the recessed groove 220 when inserted.

Alternatively, two or more recessed groove 220 and two or more middle ring 210 may be formed according to the sealing force.

When the recessed groove 220 is formed, the recessed groove 220 may be formed by two (inner and outer) upper ends 221 and 222 forming the upper end of the container inlet 12.

Here, the two upper ends 221 and 222 may have a structure inclined at a portion in which the recessed groove 220 is formed and having elasticity.

Also, at least one of the two upper ends 221 and 222 may have a curved outer circumferential surface.

Here, as illustrated in FIG. 3, as an inclined surface 215 contacting the curved outer circumferential surface of the upper ends 221 and 222 is formed in the container cap 20, at least one close contact portion may be formed so that the container cap 20 closely contacts the container inlet 12 when the container cap 20 is combined with the container inlet 12.

Also, the upper end of the container inlet 12, i.e., the two upper ends 221 and 222, the inner upper end 221 may be lower than the outer upper end 222 based on the recessed groove 220.

Second Embodiment

As illustrated in FIGS. 4A to 6B, according to a second embodiment of the present invention, a tight sealing contact portion 330 supported by at least one contact support surface 310 and 320 formed on the container inlet 12 may be formed on the container cap 20.

The contact support surface 310 and 320 may include various structures capable of supporting the tight sealing contact portion 330 as a portion supporting the close contact portion 330 when the container cap 20 is combined with the container inlet 12.

For example, as illustrated in FIGS. 4A to 5C, the contact support surface 310 and 320 may be formed on the scut combining projection 13.

Specifically, as illustrated in FIGS. 4B and 5B, a ring-shaped groove 330 that is recessed so that a lower end of the tight sealing contact portion 330 is inserted when the container cap 20 is combined with the container inlet 12 may be formed in the scut combining projection 13, and side surfaces of the groove 330 may form the contact support surfaces 310 and 320.

Specifically, sidewalls of the groove 330, which are inclined in a downward direction to contact a lower end, e.g., a corner, of the tight sealing contact portion 330, may form the contact support surfaces 310 and 320.

Here, the contact support surfaces 310 and 320 may include a first contact support surface 310 disposed at the inside and a second contact support surface 320 disposed at the outside, and when one contact support surface is provided, the second contact support surface 320 disposed at the outside may form the contact support surface.

The scut combining projection 13 may have various shapes according to a combining structure of the scut part 27.

For example, the scut part 27 may extend from the lower end of the tight sealing contact portion 330. Here, the scut part 27 may be combined with the scut combining projection 13 by passing through the groove 330.

As illustrated in FIGS. 4A and 4B, the scut part 27 may include a plurality of scut portions extending in the circumferential direction instead of having a ring structure integrated in the circumferential direction, and a plurality of through-holes 340 may be formed in the groove 330 so that the scut portions are respectively inserted thereto.

For another example, as illustrated in FIGS. 5A and 5B, the scut part 27 may be installed beyond an outside of the scut combining projection 13.

The tight sealing contact portion 330 may have various configurations according to a contact structure of the contact support surface 310 and 320 as a portion formed on the container cap 20 and supported by at least one contact support surface 310 and 320 formed on the container inlet 12.

For example, the tight sealing contact portion **330** may be formed on a lower end of a wall part **25**. The tight sealing contact portion **330** may form the lower end of the wall part **25** or a projection formed at the lower end of the wall part **25** and having a thickness less than a cross-sectional thickness of the wall part in consideration of a connection structure of the scut part **27**.

Here, the tight sealing contact portion **330** may close contact an inclined structure of the contact support surface **310** and **320** when descended to the container inlet **12** by rotation of the container cap **20**.

That is, the tight sealing contact portion **330** is inserted to the groove forming the contact support surface **310** and **320** and closely contacts the inclined structure of the contact support surface **310** and **320** when descended to the container inlet **12** by the rotation of the container cap **20**.

When one pair of contact support surfaces **310** and **320** are provided, the tight sealing contact portion **330** may have a thickness greater than a minimum width of the one pair of contact support surfaces **310** and **320**.

Also, in a cross-section of the lower end of the tight sealing contact portion **330**, each corner may have the right angle.

The tight sealing contact portion **330** may have increased flexibility to increase a sealing effect when closely contacts the contact support surfaces **310** and **320**.

To this end, as illustrated in FIGS. **6A** and **6B**, at least one deformed groove **160** adjacent to a contact portion contacting the contact support surfaces **310** and **320** may be formed in the tight sealing contact portion **330**.

The deformed groove **160** has a ring shape formed along the circumferential direction at a position adjacent to the contact portion contacting the contact support surfaces **310** and **320**. The deformed groove **160** may be deformed when the contact portion contacting the contact support surfaces **310** and **320** contacts the contact support surfaces **310** and **320** to apply flexibility and increase the sealing effect.

An upper end of the container inlet **12** may have the same or similar structure as that of the first embodiment.

As illustrated in FIG. **5C**, at least one contact ring **150** and **160** may be formed on the upper end of the container inlet **12** by extending from the lower end of the body part **26** to closely contact the upper end of the container inlet **12** in addition to or separately from the same or similar structure as that of the first embodiment.

The contact ring **150** and **160** may have a ring shape as a portion extending from the lower end of the body part **26** to closely contact the upper end of the container inlet **12**.

Particularly, the contact ring **150** and **160** may include an inner ring **150** and an outer ring **160** so that the upper end of the container inlet **12** is inserted therebetween.

Third Embodiment

In terms of sealing between the container cap **20** and the container inlet **12**, the sealing force of the container may be remarkably improved by applying flexibility to portions contacting each other.

In the container according to a third embodiment of the present invention, at least one flexibility applying groove **231**, **232**, **233**, and **234** is formed in the container inlet **12**, and a pressing contact part **241**, **242**, **243**, and **215** may be formed in the container cap **20** at a position corresponding to the flexibility applying groove **231**, **232**, **233**, and **234**.

The flexibility applying groove **231**, **232**, **233**, and **234** may be formed in any portion at which the container cap **20** and the container inlet **12** contact each other as a groove

formed to improve the sealing force of the container by applying the flexibility to the portion at which the container cap **20** and the container inlet **12** contact each other.

For example, as illustrated in FIGS. **7** to **8b**, the flexibility applying groove **231** may be formed in the upper end of the container inlet **12**.

The flexibility applying groove **231** formed in the upper end of the container inlet **12** may be as same as the recessed groove **220** of the first embodiment as a ring-shaped groove formed in the upper end of the container inlet **12**.

Particularly, the flexibility applying groove **231** formed in the upper end of the container inlet **12** may be formed adjacent to the inner circumferential surface (refer to FIGS. **7** and **8A**) or the outer circumferential surface (refer to FIG. **8B**) of the upper end of the container inlet **12**, so as to be deformed when contacts the container cap **20**.

For another example, as illustrated in FIGS. **7** to **8A**, when the flexibility applying groove **234** formed in the upper end of the container inlet **12** is formed adjacent to the inner circumferential surface of the upper end of the container inlet **12**, the container cap **20** may press and contact a portion at which the flexibility applying groove **234** is formed by the pressing contact part **215** when the container cap **20** is combined with the container inlet **12** as the inclined surface **215** is formed at a position corresponding to the flexibility applying groove **234**.

As illustrated in FIG. **8B**, when the flexibility applying groove **231** formed in the upper end of the container inlet **12** is formed adjacent to the outer circumferential surface of the upper end of the container inlet **12**, the container cap **20** may press and contact a portion at which the flexibility applying groove **231** is formed by the pressing contact part **241** when the container cap **20** is combined with the container inlet **12** as the inclined surface **241** is formed at a position corresponding to the flexibility applying groove **231**.

As illustrated in FIG. **8B**, when the pressing contact part **241** is formed in the container cap **20**, the pressing contact part **241** may be formed on the outer ring **160** of the second embodiment in addition to the simple inclined surface.

For another example, as illustrated in FIG. **8C**, the flexibility applying groove **232** may be formed in the outer circumferential surface of the container inlet **12**.

Here, a ring-shaped flexibility applying projection **230** may be formed on the outer circumferential surface of the container inlet **12** to form the flexibility applying groove **232** pressed by the container cap **20**, and the flexibility applying groove **232** may be recessed from an upper side to a lower side of the flexibility applying projection **230**.

In the container cap **20**, the pressing contact part **242** may be formed on the inner circumferential surface of the container cap **20** in correspondence to the flexibility applying groove **232**.

The pressing contact part **242** may be an inclined surface that is inclined upward to press a portion in which the flexibility applying groove **232** is formed in the flexibility applying projection **230** by descent of the container cap **20** when the container cap **20** is combined with the container inlet **12**.

For another example, as illustrated in FIG. **8D**, the flexibility applying groove **233** may be formed in the outer circumferential surface, particularly the scut combining projection **13**, of the container inlet **12**.

Here, the flexibility applying groove **233** pressed by the container cap **20** may be recessed from an upper side to a lower side of the scut combining projection **13**.

11

Also, the pressing contact part **243** may be formed on the wall part **25** of the container cap **20** in correspondence to the flexibility applying groove **233**.

The pressing contact part **243** may be an inclined surface that is inclined upward to press a portion in which the flexibility applying groove **13** is formed in the scut combining projection **13** by the descent of the container cap **20** when the container cap **20** is combined with the container inlet **12**.

As described above, the pressing contact part **241**, **242**, **243**, and **215** may include various structures capable of pressing the portion in which the flexibility applying groove **231**, **232**, **233**, and **234** is formed as a portion pressing and contacting the flexibility applying groove at a position corresponding to the flexibility applying groove **231**, **232**, **233**, and **234** formed in the container inlet **12**.

Specifically, the pressing contact part **241**, **242**, **243**, and **215** may be formed as an inclined surface in consideration that the container cap **20** rotates and/or is descended with respect to the container inlet **12** and the pressing contact part presses the portion in which the flexibility applying groove **231**, **232**, **233**, and **234** is formed by using a force applied by the descent.

However, the pressing contact part **241**, **242**, **243**, and **215** may have various structures such as a projection in addition to the inclined surface, which are capable of pressing the portion in which the flexibility applying groove **231**, **232**, **233**, and **234** is formed by the force applied by the descent.

Fourth Embodiment

The present invention may include various embodiments of intentionally increasing the portion closely contacting each other when the container cap **20** is combined with the container inlet **12** to improve the sealing force of the container.

As illustrated in FIGS. **9** and **10A**, the container according to a fourth embodiment of the present invention may have a configuration similar to that in FIGS. **7** and **8A**.

Specifically, the container cap **20** may further include an auxiliary inner ring **190** between the inner ring **100** and the upper end of the container inlet **12** in comparison with the configuration in FIGS. **7** and **8A**.

The auxiliary inner ring **190** may protrude downward from the container cap **20**, particularly the bottom surface of the body part **26**, between the inner ring **100** and the upper end of the container inlet **12**.

Here, the auxiliary inner ring **190** may have a vertical length less than that of the inner ring **100** to prevent interference with the inner ring **100**.

Also, the auxiliary inner ring **190** may form a surface inclined toward the upper end of the container inlet **12**, and the upper end of the container inlet **12** may also form an inclined surface at a portion contacting the auxiliary inner ring **190** as the pressing contact part **241**.

In the container inlet **12**, an inner circumferential protruding part **121** contacting the outer circumferential surface of the inner ring **100** may be formed as described in the first embodiment.

The container having the above-described structure may increase the contact portion between the container cap **20** and the container inlet **12** to improve the sealing force of the container.

As illustrated in FIG. **10B**, as a modified example of FIG. **10A**, the container according to the fourth embodiment of the present invention may form an inclined surface as the pressing contact part **241** for pressing a portion at which the

12

flexibility applying groove **234** is formed (a portion of the inner ring **100** and the auxiliary inner ring **190** are integrated instead of including the auxiliary inner ring **190**).

In the embodiment in FIGS. **10A** and **10B**, the flexibility applying groove **234** or the recessed groove **220** may be formed by an inner upper end **222** and an outer upper end **221** as illustrated in FIG. **3**, and the inner upper end **222** and the outer upper end **221** may have the same height or different heights.

Particularly, the inner upper end **222** may have a height less than that of the outer upper end **221** in FIGS. **10A** and **10B**.

As a modified example of FIG. **10A**, the container according to the fourth embodiment of the present invention may include the flexibility applying groove **234** formed adjacent to the outer circumferential surface of the upper end of the container inlet **12** as illustrated in FIG. **10C**.

Here, the container cap **20** may form an inclined surface **215** as the pressing contact part **215** to press the portion in which the flexibility applying groove **234** is formed.

In FIG. **10C**, the outer upper end may have a height less than that of the inner upper end **222**.

Also, as illustrated in FIG. **10D**, as a modified example of FIG. **10B**, the container according to the fourth embodiment of the present invention may include a projection **244a** formed on the container cap **20** as the pressing contact part for pressing the portion in which the flexibility applying groove **234** is formed.

In order to improve the sealing force of the container, the contact portion between the container cap **20** and the upper end of the container inlet **12** may be relatively increased. Particularly, the plurality of contact portions may be formed along a path from the outside to the inside of the container instead of being connected to each other.

As a first example illustrated in FIG. **11A**, an auxiliary inner ring **190** closely contacting the upper end of the container inlet **12** may be formed on the container cap **20** in addition to the inner ring **100** and the inner circumferential protruding part **121** contacting the outer circumferential surface of the inner ring **100** of the first embodiment.

The auxiliary inner ring **190** may protrude downward from the container cap **20**, particularly the bottom surface of the body part **26**, between the inner ring **100** and the upper end of the container inlet **12**.

Here, the auxiliary inner ring **190** may have a vertical length less than that of the inner ring **100** to prevent interference with the inner ring **100**.

Also, the auxiliary inner ring **190** may contact the upper end of the container inlet **12** when the container cap **20** is combined with the container inlet **12**.

To this end, the upper end of the container inlet **12** may have a curved cross-section, and an inner circumferential auxiliary protruding part **122** formed as the inner circumferential surface of the upper end of the container inlet **12** protrudes toward the inner circumferential surface of the auxiliary inner ring **190** may be formed.

As a second example illustrated in FIG. **11B**, at least one surface of surfaces of the auxiliary inner ring **190** and the upper end of the container inlet **12**, which contact each other, may form an inclined surface **180** and **241a** in a structure illustrated in FIG. **11a**.

In the first and second examples, the auxiliary inner ring **190** may be disposed between the upper end of the container inlet **12** and the wall part **25** instead of being disposed between the upper end of the container inlet **12** and the inner ring **100**, and a corresponding component may be symmetrically formed.

13

As a third example illustrated in FIG. 1C, the flexibility applying groove 231 may be additionally formed in the structure of FIG. 11B.

Here, an auxiliary projection 122 as in FIG. 11A may be formed on the inner circumferential surface of the container inlet 12.

As a fourth example illustrated in FIG. 11D, the flexibility applying groove 231 may be additionally formed in the structure of FIG. 11A.

As a fifth example illustrated in FIG. 11E, a portion from the upper end of the container inlet 12 to the inner circumferential protruding part 121 in the structure of FIG. 11D may be inclinedly formed.

Specifically, in the container inlet 12, the inner circumferential surface may be inclined from the upper end to the inner circumferential protruding part 121 in a direction in which the internal diameter is decreased.

Due to the above-described structure, the outer circumferential surface of the auxiliary ring 190 may closely contact a portion between the upper end of the container inlet 12 and the inner circumferential protruding part 121 when the container cap 20 is combined with the container inlet 12 or when the container cap 20 is descended to the container inlet 12.

Fifth Embodiment

The present invention may include various embodiments of intentionally increasing the portion closely contacting each other when the container cap 20 is combined with the container inlet 12 to improve the sealing force of the container.

Particularly, when the contact portion in which the upper end of the container inlet 12 contacts the container cap 20 is improved, the sealing force of the container may be improved.

As illustrated in FIGS. 12 to 15, in the container according to the fourth embodiment of the present invention, a recessed groove 220 may be formed between and by an inner upper end 222 and an outer upper end 221 of the container inlet 12, and an inner inclined surface 241 and an outer inclined surface 242 respectively contacting an inner portion of the inner upper end 222 and an outer portion of the outer upper end 221 may be formed in the container cap 20.

Specifically, the recessed groove 220 may be formed at a center of the upper end of the container inlet 12 by the inner upper end 222 and the outer upper end 221.

Here, the inner portion of the inner upper end 222 and the outer portion of the outer upper end 221 may each have a curved surface shape to respectively closely contact the inner inclined surface 241 and the outer inclined surface 242.

Also, sidewalls of the recessed groove 220 may be inclined upward so that upper ends of the inner upper end 222 and the outer upper end 221 are deformed in a direction in which a width of the recessed groove is decreased, i.e., the inner upper end 222 and the outer upper end 221 are contracted toward each other.

In other words, the inner upper end 222 and the outer upper end 221 may have a 'V'-shape.

The container cap 20 has the inner inclined surface 241 and the outer inclined surface 242, which respectively contact an inner portion of the inner upper end 222 and an outer portion of the outer upper end 221.

The inner inclined surface 241 and the outer inclined surface 242 may have various structures to respectively

14

contact the inner portion of the inner upper end 222 and the outer portion of the outer upper end 221 as a portion formed in the container cap 20.

For example, as illustrated in FIGS. 12 to 15, the inner inclined surface 241 may be formed as a portion of the inner ring 100.

Also, the inner inclined surface 241 may be formed as a portion of an auxiliary inner ring 190 when the auxiliary inner ring 190 is provided.

In the embodiments of the present invention in addition to the fifth embodiment, all of the screw-type combining structure and the snap-type combining structure (e.g., the one-piece and two-piece one touch cap and the undercut cap structure) may be applied to the container cap 20 and the container inlet 12.

As illustrated in FIGS. 14 and 15, the container according to the embodiments including the fifth embodiment of the present invention may adopt the container cap in which the body part 26 is rotated around a portion of the wall part 25 by using a hinge.

Specifically, in the container cap 20, a portion of the wall part 25 may be cut in the circumferential direction by a cut part 170 formed on the wall part 25 as illustrated in FIGS. 14 and 15. Here, a portion of the wall part 25, which is not cut, may form a hinge portion 172 disposed at the right side in FIG. 15.

The reference numeral 171 that is not described indicates a handle installed to easily separate the cut part 170 from the wall part 25.

The structure in FIGS. 12 to 15 according to an embodiment may be variously deformed such that, e.g., components corresponding to each other of the container cap 20 and the container inlet 12 may be exchanged.

Specifically, as illustrated in FIGS. 16 and 17, in the container cap 20, an inner contact ring 190 and an outer contact ring 191 may be formed by forming an opposed recessed groove 192 so that at least a portion of the upper end of the container inlet 12 is inserted therebetween.

The inner contact ring 190 and the outer contact ring 191 may have various configurations as components formed by forming the opposed recessed groove 192 so that at least a portion of the upper end of the container inlet 12 is inserted therebetween.

For example, each of the inner contact ring 190 and the outer contact ring 191 may extend downward from the bottom surface of the body part 26.

Here, a portion contacting the upper end of the container inlet 12 of each of the inner contact ring 190 and the outer contact ring 191 may have a curved surface shape.

Also, the sidewalls of the opposed recessed groove 192 may be inclined upward so that lower ends of the inner contact ring 190 and the outer contact ring 191 are deformed in a direction in which a width of the opposed recessed groove 192 is increased, i.e., the inner contact ring 190 and the outer contact ring 191 are spaced apart from each other.

In other words, the inner contact ring 190 and the outer contact ring 191 may have a reverse 'V'-shape.

The upper end of the container inlet 12 may have a sharp shape to be easily inserted to the opposed recessed groove 192 formed by the inner contact ring 190 and the outer contact ring 191.

Also, the upper end of the container inlet 12 may form inclined surfaces 193 and 194 extending inward and outward from an uppermost end so as to be easily inserted to the opposed recessed groove 192.

15

The container in FIG. 16 includes the cut part 170 described above as a modified example of the container in FIG. 15.

Embodiments, particularly the first and second embodiments, may be applied to the fifth embodiment of the present invention except for conflicted portion (the inner circumferential protruding part 121 in FIG. 3 and the structure in FIG. 4B or 5B).

Sixth Embodiment

Referring to the above-described embodiments, as a unit for increasing the sealing force of the container, the contact portion in which the container cap 20 contacts the container inlet 12 may be increased, and particularly the contact portion between the upper end of the container inlet 12 and the container cap 20 may be relatively increased.

Here, the plurality of contact portions may be formed along the path from the outside to the inside of the container instead of being connected to each other.

Particularly, the plurality of contact portions may be formed by structural deformation of the container inlet 12 and the container cap 20 and realized by forming the inclined surfaces having various angles and structures and the flexibility applying groove.

Also, the plurality of contact portions may be deformed in structure by exchanging the structures of the container inlet 12 and the container cap 20.

Hereinafter, various deformation of the container inlet 12 for forming the plurality of contact portions along the path from the outside to the inside of the container will be described.

As illustrated in FIG. 18A, the container inlet 12 may be formed such that an inner upper end 222 and an outer upper end 221 are formed by using a recessed groove 220 as a center in the upper end thereof.

Here, as the recessed groove is formed adjacent to the inner side or the outer side, one of the inner upper end 222 and the outer upper end 221 may have a width greater than that of the other so as to be easily deformed when one of the inner upper end 222 and the outer upper end 221 contacts the container cap 20.

Also, the inner upper end 222 and the outer upper end 221 may have the same height or different heights.

FIG. 18a illustrates a case when the inner upper end 222 has a width less than that of the outer upper end 221.

Also, the recessed groove 220 may have a width that is gradually increased in a direction toward the upper side.

FIG. 18B is a modified example of the container inlet 12 in FIG. 18A.

The container inlet 12 in FIG. 18B may be formed such that the inner upper end 222 has a height less than that of the outer upper end 221.

Furthermore, a minute ring projection 229 may be formed on the inner circumferential surface to increase a contact portion with the container cap 20.

FIG. 18C is a modified example of the container inlet 12 in FIG. 18A.

The container inlet 12 in FIG. 18C may be formed such that a central protruding upper end 225 protruding upward from the recessed groove 220 is additionally formed.

Particularly, the central protruding upper end 225 may have a height greater than that of each of the inner upper end 222 and the outer upper end 221, and the recessed groove 220 may include two divided recessed grooves 220a and 220b.

16

Here, the container cap 20 combined with the container inlet 12 may include a ring-shaped groove to which the central protruding upper end 225 is inserted.

FIG. 18D is a modified example of the container inlet 12 in FIG. 18A.

The container inlet 12 in FIG. 18D may be formed such that an inner upper end 222 is higher than an outer upper end 221 and has a width less than that of the outer upper end 221, and a stepped portion 226 smaller than each of the inner upper end 222 and the outer upper end 221 is formed.

FIG. 18E is a modified example of the container inlet 12 in FIG. 18A.

The container inlet 12 in FIG. 18E may be formed such that an outer upper end 221 has a width less than that of an inner upper end 222 as an opposite case of FIG. 18a.

FIGS. 18A and 18D illustrate a structure in which the inner upper end 222 or the outer upper end 221, which has a relatively small width, has a blocking structure deformed by pressing of the container cap 20 or filling of hot contents, and the container inlet 12 may have a whole or a portion of the structures in FIGS. 18E, 18F, 18G, 21A, 21B, 21C, 26A, and 26B in addition to the structures in FIGS. 8A and 8B of the third embodiment.

That is, as illustrated in FIGS. 18D and 18E, the flexibility applying groove 233 may be formed in the outer circumferential surface, particularly the scut combining projection 13, of the container inlet 12.

Here, the flexibility applying groove 233 pressed by the container cap 20 may be recessed from the upper side to the lower side of the scut combining projection 13.

Here, the flexibility applying groove 233 may be relatively formed by a groove formation projection ring 237 extending upward.

As illustrated in FIGS. 18g and 18h, the flexibility applying groove may be formed in the outer circumferential surface or the inner circumferential surface between the scut combining projection 13 and the upper end of the container inlet 12.

That is, as illustrated in FIGS. 18C and 18G, the flexibility applying groove 232 may be formed in the outer circumferential surface of the container inlet 12.

Here, a ring-shaped flexibility applying projection 230 may be formed on the outer circumferential surface of the container inlet 12 to form the flexibility applying groove 232 pressed by the container cap 20, and the flexibility applying groove 232 may be recessed from the upper side to the lower side of the flexibility applying projection 230.

Here, the flexibility applying groove 232 may be relatively formed by a groove formation projection ring 238 extending upward.

For another example, as illustrated in FIG. 18H, a flexibility applying groove 235 may be formed in the inner circumferential surface of the container inlet 12.

Here, a ring-shaped flexibility applying projection 250 may be formed on the inner circumferential surface of the container inlet 12 to form the flexibility applying groove 250 pressed by the container cap 20, and the flexibility applying groove 232 may be recessed from the upper side to the lower side of the flexibility applying projection 250.

Here, the flexibility applying groove 235 may be relatively formed by a groove formation projection ring 239 extending upward.

The container cap 20 includes a structure such as the inclined surface or the inner ring for pressing the groove formation projection ring 239.

Hereinafter, various deformation of the container cap 20 for forming the plurality of contact portions along the path

17

from the outside to the inside of the container will be described in addition to the container inlet 12.

The container cap 20 in FIG. 19A may have the structure illustrated in FIG. 5C or 8B.

The container cap 20 in FIG. 19B, as a modified example of FIG. 19A, illustrates a case in which a gap between the contact ring 150 and the inner ring 100 is relatively small.

Also, an end of the contact ring 150 may be lower than that of the inner ring 100.

Here, the container inlet 12 may have an inclined surface or a projection ring at a portion contacting the contact ring 150.

In FIG. 19C, as a modified example of FIG. 19A, the contact ring 150 may protrude from the inner circumferential surface of the sidewall part 25 instead of the bottoms surface of the body part 26.

Here, the container inlet 12 may have an inclined surface or a projection ring at a portion contacting the contact ring 150.

FIGS. 20A to 20D illustrate examples in which the container cap and the container inlet, which have various structures, are combined with each other.

FIG. 20A illustrates an example of a structure in which the container cap is combined with the container inlet in FIG. 18g. Since the structure of FIG. 20A is similar to that of FIG. 8C, a detailed description thereof will be omitted.

FIG. 20B illustrates a structure in which the container cap having a structure similar to that in FIG. 10A is combined with the container inlet deformed from the structure of FIG. 18A. Since FIG. 20B has a similar structure to that of FIG. 10A, a detailed description thereof will be omitted.

In the container in FIG. 20C, by comparing with the structure of the container according to other embodiments, an inner upper end 222, an outer upper end 221, and an auxiliary upper end may be formed in the upper end of the container inlet 12, and an insertion projection ring 199 inserted to a recessed groove formed by the inner upper end 222 and the outer upper end 221 may be formed on the container cap 20.

Here, as an auxiliary upper end 220d having a height less than that of each of the inner upper end 222 and the outer upper end 221 is formed inside the inner upper end 222 of the container inlet 12, an additional recessed groove 220c may be formed together with the inner upper end 222.

The container in FIG. 20D is substantially same as or similar to that in FIG. 8B except for the shape of the contact ring 150.

The container in FIG. 21A is characterized in that a plurality of contact portions for blocking external air are formed on a projection formed on the outer circumferential surface of the container inlet 12 such as the scut combining projection 13 and the separation preventing projection 15.

Specifically, at least one recessed groove 271 and 272 is vertically formed in the projection formed on the outer circumferential surface of the container inlet 12 such as the scut combining projection 13 and the separation preventing projection 15, and an insertion projection ring 273 inserted to at least one of the at least one recessed groove 271 and 272 may be formed in the sidewall part 25 of the container cap 20.

As the insertion projection ring 273 is inserted to at least one of the ring-shaped recessed groove 271 and 272 formed in the projection formed on the outer circumferential surface of the container inlet 12 such as the scut combining projection 13 and the separation preventing projection 15, at least one contact portion may be formed.

18

In the container of FIG. 21B, a sealing ring 274 forming a recessed groove 275 defined from the lower side to the upper side may be formed on the inner circumferential surface of the container cap 20. Here, a pressing portion 276 for pressing the sealing ring 274 when the container cap 20 is combined with the container inlet 12 may be formed on the outer circumferential surface of the container inlet 12.

The sealing ring 274 may have various configurations such that, e.g., the sealing ring 274 extends downward while being spaced apart from the inner circumferential surface of the sidewall part 25 of the container cap 20, as a component for forming the recessed groove 275 defined from the lower side to the upper side on the inner circumferential surface of the container cap 20.

Also, the pressing portion 276 may be formed by an inclined surface, as a component formed on the outer circumferential surface of the container inlet 12 to press the sealing ring 274 (strengthen the sealing force) when the container cap 20 is combined with the container inlet 12.

In the container in FIG. 21C, as a modified example of FIG. 21B, at least one protruding part 278 protruding upward from the projection formed on the outer circumferential surface of the container inlet 12 along the circumferential direction in an integrated manner such as the scut combining projection 13 and the separation preventing projection 15 may be formed, and here, an insertion recessed groove ring 277 to which the protruding part 278 is inserted when the container cap 20 is combined with the container inlet 12 may be formed in the container cap 20 at a position corresponding to the protruding part 278.

Seventh Embodiment

The above-described embodiments may have various combinations by combining or replacing each other in a range in which the embodiments are not conflicted.

Also, the container cap may have various structures in addition to the embodiments.

As illustrated in FIGS. 22 to 24, the container according to a seventh embodiment of the present invention may include the double sidewall parts 25 of the container cap 20 and be combined with the above-described embodiments.

Specifically, the container cap 20 may further include an auxiliary sidewall part 25a extending downward from the body part 26 while being spaced apart from the outer circumferential surface of the sidewall part 25 extending downward from the body part 26.

The auxiliary sidewall part 25a may include a scut part 27 combined with the scut combining projection 13 as a portion extending downward from the body part 26 while being spaced apart from the outer circumferential surface of the sidewall part 25 extending downward from the body part 26.

That is, the auxiliary sidewall part 25a may include the scut structure in FIG. 4b and the scut structure in FIG. 5B.

A contact projection ring 430 contacting a contact portion 440 formed on the container inlet 12 when the container cap 20 is combined with the container inlet 12 may be formed on the auxiliary sidewall part 25a.

The contact projection ring 430 may have various structures as a component contacting the contact portion 440 formed on the container inlet 12 when the container cap 20 is combined with the container inlet 12.

For example, the contact projection ring 430 may form a lower end of the auxiliary sidewall part 25a and have a cross-sectional thickness that is relatively small to have flexibility.

19

Furthermore, the contact projection ring **430** may have a sharp end.

The contact portion **440** may be variously formed, e.g., an inclined surface or a projection, as a portion formed on the container inlet **12** to contact the contact projection ring **410** when the container cap **20** is combined with the container inlet **12**.

Particularly, the contact portion **440** may be formed as an inclined surface in the scut combining projection **13**.

A contact projection ring **410** contacting a contact portion **420** formed on the container inlet **12** when the container cap **20** is combined with the container inlet **12** may be formed on the sidewall part **25**.

The contact projection ring **410** may have various structures as a component contacting the contact portion **420** formed on the container inlet **12** when the container cap **20** is combined with the container inlet **12**.

For example, the contact projection ring **410** may form a lower end of the sidewall part **25a** and have a cross-sectional thickness that is relatively small to have flexibility.

Furthermore, the contact projection ring **410** may have a sharp end.

Also, the contact projection ring **410** may have a cross-sectional thickness that is decreased by a plurality of steps at the lower end of the sidewall part

The contact portion **420** may be variously formed, e.g., an inclined surface or a projection, as a portion formed on the container inlet **12** to contact the contact projection ring **410** when the container cap **20** is combined with the container inlet **12**.

The contact portion **420** contacting the contact projection ring **410** formed on the sidewall part may be formed above the scut combining projection **13**.

As illustrated in FIG. **24**, a protruding ring **234** protruding upward may be formed on the upper end of the container inlet **12**, and an inclined surface **244** or a projection contacting the protruding ring **234** may be formed in the container cap **20**.

FIG. **25** shows cross-sectional views illustrating examples of various structures such as projection-projection contact or projection-recessed groove contact when the container cap contacts the container inlet.

The structures in FIG. **25** illustrate examples capable of realizing a blocking function as a flexible portion and a flexible portion or a hard portion and a flexible portion contact each other.

FIGS. **26A** and **26B** are partial cross-sectional views illustrating structures of at least one projection ring **451**, **452**, **453**, and **454** formed on the lower end of the sidewall part of the container cap and the container inlet contacting the same in the container according to the present invention.

At least one projection ring **451**, **452**, **453**, and **454** is formed on the lower end of the sidewall part **25** or the auxiliary sidewall part **25a** and contact a contact portion **461**, **462**, **463**, and **464** formed on the container inlet **12**.

Here, the contact portion **461**, **462**, **463**, and **464** includes an inclined surface or a projection, as a portion contacting the projection ring **451**, **452**, **453**, and **454** when the container cap **20** is combined with the container inlet **12**.

Although the above description merely corresponds to some exemplary embodiments that may be implemented by the present disclosure, as well known, the scope of the present disclosure should not be interpreted as being limited to the above-described embodiments, and all technical spirits having the same basis as that of the above-described

20

technical spirit of the present disclosure are included in the scope of the present disclosure.

The invention claimed is:

1. A container assembly comprising:

a container body containing contents and having a container inlet; and

a container cap combined with the container inlet, wherein a plurality of contact portions are formed on at least one of the container cap and the container inlet, and

the plurality of contact portions form a closed curve enclosing the container inlet so as to block inflow of external air from outside into the container body,

wherein the container cap comprises:

a body part in close contact with an upper end of the container inlet;

a sidewall part which extends downward from the body part and in which a female screw part that is screw-coupled with a male screw part formed on an outer circumferential surface of the container inlet is formed; and

an inner ring protruding from a bottom surface of the body part and inserted into an inner circumferential surface of the container inlet, and

wherein the container inlet includes an inner upper end and an outer upper end, each having a curved outer circumferential surface, a recessed groove being formed between the inner upper end and the outer upper end,

wherein an inclined surface is formed on the container cap, the inclined surface including a first inclined surface and a second inclined surface, the first inclined surface being in contact with the curved outer circumferential surface of the inner upper end and the second inclined surface being in contact with the curved outer circumferential surface of the outer upper end when the container cap is combined with the container inlet,

wherein an inner circumferential protruding part is formed on the inner circumferential surface of the container inlet, and the inner circumferential protruding part protrudes toward the inner ring to contact an outer circumferential surface of the inner ring when the container cap is combined with the container inlet,

wherein the inner circumferential protruding part has a curved shape so that a cross-section in the longitudinal direction of the container inlet is in surface-contact with the outer circumferential surface of the inner ring,

wherein a middle ring is formed on a bottom surface of the body part of the container cap, the middle ring being inserted into the recessed groove formed between the inner upper end and the outer upper end, the middle ring forming at least one close contact portion with the container inlet when the container cap is combined with the container inlet, a bottom of the middle ring being positioned above and spaced apart from a surface defining a bottom of the recessed groove, creating a gap therebetween, and

wherein each of the inner upper end and the outer upper end has a structure inclined at a portion in which the recessed groove is formed and having elasticity.

2. The container assembly of claim 1, wherein the inner upper end of the container inlet is lower than the outer upper end of the container inlet based on the recessed groove in the upper end of the container inlet.

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